



ISLAMIC UNIVERSITY

**DEPARTMENT OF
ELECTRICAL AND ELECTRONIC ENGINEERING**

Faculty of Engineering and Technology

**Bachelor of Science in
Electrical and Electronic Engineering Degree**

First Year Examination	: 2021
Second Year Examination	: 2022
Third Year Examination	: 2023
Fourth Year Examination	: 2024

**CURRICULUM FOR B.SC. ENGINEERING DEGREE
SESSION: 2020 – 2021**

THE DEPARTMENT OF ELECTRICAL AND ELECTRONIC ENGINEERING

Introduction: Islamic University is one of the sixth major public universities in Bangladesh and largest seat of higher education in the south-west part of the country financially aided by the Organization of the Islamic Conference (OIC) and the government of Bangladesh through University Grants Commission, Bangladesh. It was established on 22 November 1979 by the Islamic University Act. Islamic University began its academic operations on 28 June 1986. The university is recognized as the hub of academic and research activities, sports and cultural programs with a very innate and peaceful environment. It is a major international centre for an excellent integration of Islamic Studies with general Studies and Studies of modern science, engineering and technology. The university provides both local and foreign students with the facilities of undergraduate studies, postgraduate research and teaching. It is a campus-oriented university, where all the academic and administrative buildings, central library, student's residential halls, teacher's dormitories, gymnasium, play grounds, tennis court and other facilities are in one self-contained 175-acre land at Shantidanga-Dulalpur, beside the Kushtia-Khulna highway. It is located about 24 km south and 22 km north of the Kushtia and Jhenidah district-towns, respectively.

Electrical and Electronic Engineering (EEE) Department was opened in this university in 1995 as the name of “Electronics and Applied Physics” with Professor Dr. M Ekin Uddin as the founder head of the department. Academic activities were started in the session 1995-96 with 25 undergraduate students. Later, the course curriculum was upgraded and the department was renamed as the “Applied Physics, Electronics and Communication Engineering” in December, 2006. Again to meet the increasing demand of the 21st century the curriculum was updated again and the department was renamed as the “Electrical and Electronic Engineering” in August 2017 under the Faculty of Engineering and Technology.

The courses designed for Bachelor of Science in Electrical and Electronic Engineering (B.Sc. Engg. in EEE) consist of 160 credits (4000 marks) distributed over eight semesters in four academic years. Each academic year is divided into two semesters (Odd and Even), each of duration not less than 13 weeks (65 working days). There shall be final examinations at the end of each semester. The medium of answer in all examinations is in English. The theoretical examination of courses less than or equal to 2 credits shall be of 2 hours duration and courses greater than 2 credits shall be of 3 hours duration. An academic schedule for an academic year shall be announced for general notification before the start of the academic year, on the prior approval of the academic committee.

The department is presently housed in the Applied Science and Technology Building comprising of 2nd and 3rd floor. This department currently runs four years (eight semesters) B.Sc. Engineering program with an annual intake of 50 undergraduate students. It conducts education and research in three major areas- Electronics, Communication and power and energy group with the goal of producing quality graduates who can become leaders in the global arena to serve the society, and to conduct leading-edge research.

EEE graduates are involved in the design and development of modern technological applications such as automation for electromechanical systems, computer systems, embedded systems and electronic control systems applied in process plants, automotive industry, aerospace, and even maintenance.

Presently the Department is offering following programs:

1. Undergraduate Program: B.Sc. Engineering

Postgraduate Programs: M.Sc. Engineering, Master of .Philosophy (M. Phil) and Doctor of Philosophy (Ph. D).

CURRICULUM FOR B.Sc. ENGINEERING



Name of the University: Islamic University, Kushtia-Jhenidah, Bangladesh.

Vision of the University

To provide for instructions in theology and other fields of Islamic Studies and comparative jurisprudence and such other branches of learning at graduate and post-graduate level and to make provision for research including Post-Doctorate research and training for the advancement and dissemination of knowledge.

Mission of the University:

M1. Have a world-class curriculum with talented academicians and conducive academic and research environment for generation and dissemination of knowledge.

M2. To maintain international standards in education and research with a focus on both knowledge and skills, and humanitarian and ethical values to meet the needs of the society and state.

M3. To develop strategic relationships with leading national and international organizations, and universities for academic as well as research collaborations.

Name of the Faculty Offering the Program: Faculty of Engineering and Technology.

Name of Department Offering the Program: Department of Electrical and Electronic Engineering.

Vision of the Program:

To provide highest quality engineering education to prepare an effective and socially responsible graduate to serve the future needs and challenges of the society.

Mission of the Program:

M1. To achieve the highest possible standard graduates to be active, productive, engaged members of society on local, national and international levels.

M2. To develop the graduates as effective researchers on fundamental and applied problems to involve in interdisciplinary research/industries and promoting the development of Scientific and Industrial Revolutions.

M3. To develop the skills to find gainful employment in academics, industry and government organizations.

Objectives of the Program:

PO1. Providing a quality education in electrical and electronic engineering.

PO2. Enhancing research skills of the students through innovative research and developing appropriate technologies contributing to the sustainable development of the nation.

PO3. Establishing linkages with reputed R & D organizations and leading educational institutions.

PO4. Dispensing the practical issues by adapting to technical, economic and social trends.

PO5. Preparing a student to face challenges in the global environment.

Name of the Degree: B. Sc. Engineering in Electrical and Electronic Engineering.

Description of the Program: The B. Sc. Engineering in Electrical and Electronic Engineering will prepare a student to complete for a wide range of job opportunities in home and abroad. While gaining a broad understanding of emerging electrical and electronic engineering and the engineering business environment, a student will also develop essential leadership, management and communication skills and the global perspective he needs to excel in a carrier in engineering. The program provides with the necessary skills and experience that will enable a student to take his place in the first-paced world of electrical and electronic engineering as a creator of next generation, intelligent, connected devices and systems. A senior student will undertake an individual Project/Thesis that is either linked to research within the department or potentially linked to industry relevant design activities. He can also pursue his own ideas and undertake a personalized project.

Program Educational Objectives (PEOs)

- PEO1.** Achieve basic knowledge, theories, principles, process and procedure the main areas of electrical and electronic engineering.
- PEO2.** Demonstrate high level analytical and critical thinking skills to solve conceptual and quantitative engineering problems.
- PEO3.** Develop skill to demonstrate safe and acceptable research individually and in team.
- PEO4.** Integrate technical and communication knowledge with professional and industry based education to build up successful professional careers in industry, government and academia nationally and globally.
- PEO5.** Enhance the skill for communicating scientific results effectively in written, oral and in interactive presentation.
- PEO6.** Enhance the spirit to contribute the educational, cultural, social, technological and economic development of society with moral and ethical values and development patriotism.

Program Learning Outcomes (PLOs)

PLO1	Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals to the solution of complex problems of electrical and electronic engineering.
PLO2	Problem analysis: Identify, formulate, research and analyze complex electrical and electronic engineering problems.
PLO3	Design/development of solutions: Design solutions for complex Electrical and Electronic Engineering problems and design system components or processes that meet the specified needs with technological concerns.
PLO4	Investigation: Conduct investigations of complex electrical circuits and electronic devices problems, considering experimental design, data analysis and interpretation.
PLO5	Modern tool usage: Create, select and apply appropriate techniques, resources and modern electrical and electronic tools necessary for engineering practice.
PLO6	The engineer and society: Apply reasoning informed by contextual knowledge to develop the electrical and electronic technologies for society benefits.
PLO7	Environment and sustainability: Understand the impact of professional electrical and electronic engineering solutions in societal and environmental contexts and demonstrate the knowledge for sustainable development.
PLO8	Ethics: Apply ethical principles and commit to the professional ethics, responsibilities and the norms of the engineering practice.
PLO9	Individual work and team work: Function effectively as an individual and as a member or leader of diverse teams and in multidisciplinary settings.
PLO10	Communication: Communicate effectively about complex electrical and electronic engineering activities with the engineering community and with society at large in both oral and written.
PLO11	Project management and finance: Demonstrate knowledge and understanding of electrical and electronic engineering and management principles and apply these to one's work as a team member or a leader to manage projects in multidisciplinary environments.
PLO12	Life-long learning: Recognize the need for and have the preparation and ability to engage in independent, life-long learning in the broadest context of technological change.

Mapping Program Mission with PEOs

PEOs	PEO1	PEO2	PEO3	PEO4	PEO5	PEO6
M1	√	√				
M2			√		√	
M3		√		√		√

Program Objectives (PEO) to Program Learning Outcome (PLO) Mapping

PLO/PEO	PEO1	PEO2	PEO3	PEO4	PEO5	PEO6
PLO1	√					
PLO2		√				
PLO3			√	√		
PLO4			√	√		
PLO5		√		√	√	
PLO6				√		√
PLO7						√
PLO8			√	√		√
PLO9			√	√		√
PLO10				√	√	√
PLO11				√		√
PLO12			√			√

Mapping Courses with the PLOs

Program Learning Outcome (PLO)		PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
Humanities													
Sl. No.	Course Code												
1.	HUM-1111						√			√	√		√
2.	HUM-1211										√	√	√
3.	HUM-2211						√	√	√	√		√	√
4.	NCC 1(BS)							√				√	
5.	NCC 2(IS)							√	√	√	√	√	√
Basic Science													
1.	PHY-1103	√	√	√	√	√							
2.	PHY-1104	√	√	√	√	√							
3.	PHY-1203	√	√	√	√								
4.	PHY-1204	√		√	√								
5.	CHEM-1105	√	√	√	√	√	√						
6.	CHEM-1106	√	√	√									
7.	MATH-1107	√	√	√	√								
8.	MATH-1207	√	√	√	√								
9.	STAT-2105	√	√	√	√	√							√
10.	MATH-2107	√	√	√	√								
11.	MATH-2207	√	√	√	√	√							
12.	MATH-2208	√	√	√									
Allied Engineering													
1.	ICT-1109	√	√	√	√		√						√
2.	CSE-1110	√	√	√	√								√
3.	CSE-1209	√	√	√	√	√							√
4.	CSE-2109	√	√	√	√	√							√
5.	CSE-2110	√	√	√	√						√		
6.	ME-1205	√	√	√	√	√		√	√				√
7.	ME-1206		√	√	√						√		
8.	IPE-2111	√	√	√	√		√			√	√	√	√

Core Engineering													
1.	EEE-1101	√	√	√	√	√							
2.	EEE-1102	√	√	√	√	√							
3.	EEE-1200	√	√								√		√
4.	EEE-1201	√	√	√	√	√							
5.	EEE-1202		√	√	√						√		
6.	EEE-2101	√	√	√	√	√							√
7.	EEE-2102		√	√	√	√							√
8.	EEE-2103	√	√	√	√	√							
9.	EEE-2104		√	√	√	√							
10.	EEE-2200	√	√								√		√
11.	EEE-2201	√	√	√	√	√							√
12.	EEE-2202		√	√	√	√							√
13.	EEE-2203	√	√	√	√	√							√
14.	EEE-2204	√	√	√	√	√					√		√
15.	EEE-2205	√	√	√	√	√							
16.	EEE-2206	√	√	√	√	√							
17.	EEE-3101	√	√	√	√	√							
18.	EEE-3102		√	√	√	√							
19.	EEE-3103	√	√	√	√	√							
20.	EEE-3104	√	√	√	√	√	√						
21.	EEE-3105	√	√	√	√	√							
22.	EEE-3106	√	√	√	√	√							
23.	EEE-3107	√	√	√	√	√	√						
24.	EEE-3108	√	√	√	√	√							
25.	EEE-3110	√		√	√		√						
26.	EEE-3112	√	√	√	√	√							
27.	EEE-3200	√	√								√		√
28.	EEE-3201	√	√	√	√	√							
29.	EEE-3203	√	√	√	√	√							
30.	EEE-3204	√	√	√	√	√							
31.	EEE-3205	√	√	√	√	√							
32.	EEE-3206	√	√	√	√	√							
33.	EEE-3207	√	√	√	√	√	√						
34.	EEE-3209	√	√	√	√	√							√
35.	EEE-3212	√		√	√	√	√						
36.	EEE-4000	√	√	√	√	√				√	√		√
37.	EEE-4100	√	√							√	√		
38.	EEE-4101	√	√	√	√	√							
39.	EEE-4102	√	√	√	√								√
40.	EEE-4103	√	√	√	√	√	√						
41.	EEE-4104	√	√	√		√							
42.	EEE-4105	√	√	√	√	√							
43.	EEE-4106	√	√	√	√	√							
EEE-41** Elective I													
44.	EEE-4111	√	√	√	√	√							√
45.	EEE-4112	√	√			√							
46.	EEE-4121	√	√	√	√	√							
47.	EEE-4122	√	√			√							
48.	EEE-4131	√	√	√	√	√							
49.	EEE-4132	√	√	√	√		√						
50.	EEE-4200	√	√								√		√
51.	EEE-4201	√	√	√	√	√							
52.	EEE-4202	√	√	√									
EEE-42** Elective II													
53.	EEE-4213	√	√	√	√	√							
54.	EEE-4215	√	√	√	√	√							√

55.	EEE-4217	√	√	√	√	√							
56.	EEE-4219	√	√	√	√								
EEE-42** Elective III													
57.	EEE-4223	√	√	√	√	√	√	√					√
58.	EEE-4225	√	√	√		√	√					√	√
59.	EEE-4227	√	√	√	√	√						√	√
60.	EEE-4229	√	√	√	√	√							
EEE-42** Elective IV													
61.	EEE-4233	√	√	√	√	√							
62.	EEE-4235	√	√	√	√								
63.	EEE-4237	√	√	√	√	√							
64.	EEE-4239	√	√	√	√	√							

RULES AND REGULATIONS

1. Duration of Course and Course Structure (Ref. Academic Ordinance Faculty of Engineering (AOFE) article no-9.1)

- The B. Sc. Engineering degree program shall extend over a period of 04 (four) academic years, each of a normal duration of one calendar year, divided into 2 semesters. Thus, B. Sc. Engineering degree will consist of 08 (eight) consecutive semesters.
- The curricula of the B. Sc. Engineering degree in the different Departments shall be proposed by the Committee of Courses and approved by the Syndicate on the recommendation of the Academic Council through the Faculty.
- The Committee of Courses shall review the curricula at least once in three Academic Year and recommend changes and revision (if any) to the Faculty. The Faculty will recommend to the Academic Council for final approval by the syndicate.
- Teaching for the courses is reckoned in credits and the credits allotted to various courses will be determined by the Academic committee with the following guidelines:

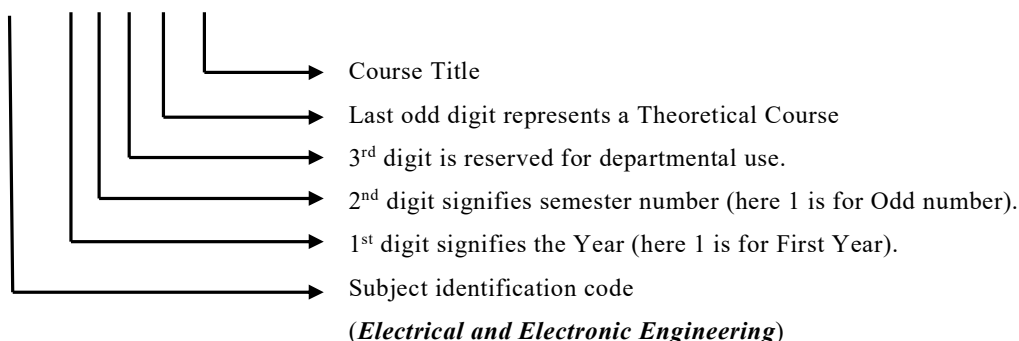
Type of Course	Contact Hour (in a semester)	No. of Credit
(i) Theory Lecture	: 1 hour/week	1.0
(ii) Tutorial	: 1 hour/week	1.0
(iii) Lab/Practical/ Sessional/Design	: 2 hours/week	1.0
	: 3 hours/week	1.5
(iv) Project/Thesis	: 3 hours/week	1.5
	: 6 hours/week	3.0
(v) Field work	: 2 weeks of field work	1.0
(vi) Seminar/Special Studies: Preparation of technical paper and its presentation.		1.0

2. COURSE IDENTIFICATION SYSTEM (Ref. Academic Ordinance Faculty of Engineering (AOFE) article no-9.1)

Each course shall be designated by maximum 05 (five) letters for the concerned Department (generally short name of the Department). This will be followed by a hyphen and then by four-digit number for the course of which the first digit will correspond to the year, second digit corresponds to the semester and last two digit of odd number denotes the Theoretical courses and even number denotes the Practical/ Sessional/ Lab/ Project/Industrial tour or Training/ Field work/ Viva-voce courses. For Example, EEE-1101 will denote the First year First semester Theoretical course of the Department of Electrical and Electronic Engineering, CSE-2202 will denote the Second year Second semester Practical/ Sessional/ Lab/ Project/Industrial tour or Training/ Field work/ Viva-voce

courses of the Department of Computer Science and Engineering, ACCE-3101 will denote the Third year First semester Theoretical course of the Department of Applied Chemistry and Chemical Engineering, ICT-4202 will denote the Fourth year Second semester Practical/ Sessional/Lab/ Project/Industrial tour or Training/ Field work/ Viva-voce courses of the Department of Information and Communication Technology. The course designation system is illustrated by the following example:

EEE 1 1 0 1 Basic Electrical Engineering



3. Duration of Examination [Ref. AOFE article no- 13 (p)]

Duration of Theoretical and Practical Examinations:

- The duration of theoretical course final examination will be 04 (four) hours.
- The duration of practical course examination will be defined by the Committee of Courses of the respective Department.

4. Academic Calendar [Ref. AOFE article no- 8.2]

- The academic year shall ordinarily be divided into two regular semesters each ordinarily having duration of not less than 13 weeks of classes.
- There shall be a final examination at the end of each semester and the examination will be conducted as per Academic regulations.
- The academic committee of the Department will announce the academic schedule for each semester ordinarily before the start of the class subject to the approval of the Academic Council.

Academic schedule may be prepared according to the following guidelines based on two regular Semesters:

Semester 1		No. of weeks
Classes		13
Recess before examination		2
Semester end final examination		3
Result publication		3
Total:		21
Semester-2		No. of weeks
Classes		13
Recess before examination		2
Semester end final examination		3
Result publication		3
Total:		21
Vacation after even semester		10
Total:		52 weeks

5. CREDIT REQUIREMENT AND DURATION OF STUDY [Ref. AOFE article no- 9.2]

- a) For theoretical courses, 13 (thirteen) class-hour of 60 (sixty) minutes is defined as one credit.
- b) For practical/Sessional or lab courses, 26 (twenty-six) lab hours' work is defined as one credit. One lab hour is equal to $3 \times 60 = 180$ minutes
- c) Course load: A maximum of 22 credits with not more than 12 courses (theory and practical) may be allowed in a semester.
- d) Based on course pattern, content and weightage, individual course may have different credit hour.
- e) **The total credits required to be earned or fulfilled for obtaining a Bachelor of Science Engineering (B.Sc. Engineering) degree will be 160 out of 160 credits. However, the distribution of credits and courses must be designed in the Curricula by the committee of courses of respective Department.**
- f) **Non-Credit/University Course:** There shall be a Non-credit course namely Islamic Studies/Bangladesh Studies carrying 100 marks to be completed within the Second semester. However, if any student failed to pass the Non-credit course (obtained less than 50% marks) in the second semester he/ she must complete the Non- credit course within following academic year. Students coming from Madrasa stream (both Dakhil and Alim), Non-Muslim and foreign students shall take the Non-credit course Bangladesh Studies. Those coming from college stream (both S.S.C. & H.S.C./ Dakhil & H.S.C.) shall take the Non-credit Course Islamic Studies.

Assignment of Credit:

- a) Credit for each Theoretical Course = 2.0 to 4.0
- b) Credit for each Practical/ Sessional Course = 1.0 to 3.0
- c) Credit for Industrial Tour/ Field work/ In-plant Training = 1.0 to 3.0
- d) Credit for Project/ Internship Dissertation = 4.0 (Maximum)
- e) Credit for Viva-voce = 1.0 (Maximum)

NB: The step of credit increment should not be less than 0.5

6. DISTRIBUTION OF MARKS: [Ref. AOFE article no- 10]

- a) **For Theoretical Courses:**
 - i) Class attendance : 10%
 - ii) Quiz/presentation : 05%
 - iii) In-course/tutorial exam/ assignment : 15%
 - iv) Semester end final examination : 70%
- b) **For Practical/Laboratory/Sessional/ Courses:**
 - i) Laboratory performance during lab hours : 05%
 - ii) Lab attendance : 10 %
 - iii) Lab note book on experiment : 05%
 - iv) Viva-voce on experiment during practical exam : 10%
 - v) Semester end final exam : 70%
- c) **For Industrial Tour/Field work /In-plant Training Courses:**
 - i) Report of the Industrial Tour/Field work /In-plant Training : 50%
 - ii) Overall performance in the Tour/ Field work /Training : 30%
 - iii) Viva-voce/Quiz/Presentation : 20%
- d) **For Project /Internship Courses:**
 - i) Presentation and Dissertation of the Project / Internship work : 70%
 - ii) Overall Performance during the work (evaluated by supervisor) : 30%

7. EVALUATION OF THE COURSES [Ref. AOFE article no- 11]

The performance of a student in a theoretical course will be evaluated in the following ways:

- a) For a theory course, the evaluation will be made by the summation of class attendance, quiz/ presentation, tutorial/ in-course exam/ assignment and Semester final exam marks.
- b) For any course attendance, quiz/ presentation, tutorial/ in-course exam/ assignment will be evaluated by the course teacher/s and the marks must be submitted to the Chairman of the examination committee before commencement of the semester final examination.
- c) Respective examination committee will conduct the semester final examination.
- d) The attendance sheet must be submitted to the Chairman of the Department before commencement of the semester final examination.

- e) Quiz/ presentation, tutorial/ in-course exam/ assignment marks must be shown in the notice board before the end of the class.
- f) Minimum 04 (four) tutorial/in-course exam/ assignment will be taken for each theory course. Average of all the tutorial/ in-course exam/ assignment marks will be considered for final calculation.
- g) For theoretical course final examination, generally there will be two examiners: course teacher will be the first examiner and the second examiner will be within the department or from a relevant department of Islamic University or from the same/allied departments of the other universities appointed by the examination committee. **Examiner will be evaluated maximum two theory courses in each year.**
- h) The answer scripts of final examination will be evaluated by two examiners. The arithmetic mean of the marks given by two examiners in a course will be considered as the obtained marks of that course. Total marks of a student in each course should be rounded up.
- i) If the difference of marks between the two examiners varied equal to or above 20% for any answer script of a course that will be evaluated by a third examiner. The examination committee will appoint the neither third examiner if the members of the corresponding examination committee nor previously appointed examiner (first & second examiner) are nominated as third examiner.
 - a) The nearest two marks among the examiners (1st, 2nd and 3rd) shall be taken into account.
 - b) If the differences of marks of third examiner from the first and second examiner become equal, then obtained mark will be calculated from average of third examiner mark and nearest higher mark of the first or second examiner.
- j) The evaluation of practical/laboratory/ Sessional/ course will be done as follows-
 - i. The practical/ lab course teacher will evaluate lab attendance, lab performance and lab note book marks on experiment. Lab attendance mark will give in accordance with the percentage of attendance in the lab class and works.
 - ii. Examiners appointed by the respective examination committee will evaluate experiments and viva- voce on experiment during examination.
- k) The percentage and marks of attendance of students for all courses (theoretical and practical) will be given as follows:

Attendance	Marks
95% and above	10
90% to 94%	09
85% to 89%	08
80% to 84%	07
75% to 79%	06
70% to 74%	05
65% to 69%	04
60% to 64%	03
Less than 60%	00

- l) Industrial Tour/Field work/In-plant Training course will be evaluated as: The report, overall performance in the Industrial Tour/Field work/In-plant Training and Quiz/Viva-voce /Presentation will be evaluated by the examination committee.
- m) Project/Internship course will be evaluated by the following ways-
 - i) Overall performance during the work of the Project/Internship course will be evaluated by the teacher who supervises the Project/ Internship work.
 - ii) Presentation and dissertation of the Project/ Internship will be evaluated by the members of the respective examination committee.
- n) Viva-voce course evaluation- Viva-voce course will be conducted in even semester and evaluated by the examination committee of the respective semester.

16. GRADING SYSTEM

A. Grades and Grade Points:

M	Letter Grade	Grade Point	Interpretation
80% and above	A+ (A plus)	4.00	Outstanding
75% to less than 80%	A (A regular)	3.75	Excellent
70% to less than 75%	A- (A minus)	3.50	Very Good
65% to less than 70%	B+ (B plus)	3.25	Good
60% to less than 65%	B (B regular)	3.00	Satisfactory
55% to less than 60%	B- (B minus)	2.75	Below Satisfactory
50% to less than 55%	C+ (C plus)	2.50	Average
45% to less than 50%	C (C regular)	2.25	Below Average
40% to less than 45%	D	2.00	Poor
less than 40%	F	0.00	Fail
	I		Incomplete

In case of Theory course-

a) Only "D" or higher grade will be counted as credits earned by a student.

b) Absence from the examination will be considered as incomplete.

In case of Industrial Tour/ Field work/ In-plant Training, Project / Internship, Practical and Viva-voce courses-

Obtained Grade point less than C+ will not be counted as credit earned by a student.

17. CREDIT CALCULATION:

Grade Point Average (GPA) is calculated as total points secured divided by credits offered. No credits will be granted for grade less than D. Grade Point Average (GPA) will be calculated as follows:

$$\text{GPA} = \frac{\sum \text{PS}}{\sum \text{CO}}$$

where, C = Credit Hour (s) in any course, GP = Grade point acquired in the corresponding course, PS = Points secured, CO= Credits Offered (PS=GP×CO)

Computation of Cumulative Grade Point Average (CGPA) is shown below:

$$\text{CGPA} = \frac{\sum \text{TPS}}{\sum \text{TEC}}$$

where C = Credit Hour (s) in any course, GP = Grade point acquired in the corresponding course, TPS = Points secured

TEC= Total Earned Credits

Semester-Wise Distribution of Credits

Sl. No.	Year/ Semester	Theory		Sessional/Training/ Project/Thesis		Oral/Technical Seminar		Non-Credit Course s	Total Credits
		No. of Courses	Credits	No. of Courses	Credits	No. of Courses	Credits		
1.	1 st Year 1 st Semester	6	15.5	4	4.5				20
2.	1 st Year 2 nd Semester	6	15.5	3	3.5	1	1.0	IS/BS	20
3.	2 nd Year 1 st Semester	6	16.0	3	4.0				20
4.	2 nd Year 2 nd Semester	5	14.0	4	5.0	1	1.0		20
5.	3 rd Year 1 st Semester	4	12.0	6	8.0				20
6.	3 rd Year 2 nd Semester	5	15.0	3	4.0	1	1.0		20
7.	4 th Year 1 st Semester	4	12.0	6	8.0				20
8.	4 th Year 2 nd Semester	5	15.0	2	4.0	1	1.0		20
Total		41	115.0	31	41.0	4	4.0		160

Distribution of Courses:

Sl. No.	Course Type		Marks	Credits	Weight (%)
1.	Departmental Courses				68.12
	Core Courses	Theory	2500	75.0	
		Lab	1050	29.0	
	Capstone Courses	Training/ Internship	50	1.0	
		Thesis/ Project	150	4.0	
	Sub Total		3750	109	
2.	Related Courses				15.00
	Basic Science	Theory	800	20.0	
		Lab	200	4.0	
	Sub Total		1000	24.0	
	Humanities	Theory	300	7.5	4.69
	Islamic Studies/Bangladesh Studies	Theory	100	NCC	
	Sub Total		400	7.5	
	Allied Engineering	Theory	500	12.5	9.69
		Lab	150	3.0	
		Sub Total		650	
3.	Oral	Viva-voce	150	3.0	2.50
		Technical Seminar	50	1.0	
	Sub Total		200	4.0	
Total			6000	160	100%

Course Outline

B.Sc. Engineering First Year, First Semester, Examination 2021

Sl. No.	Course Codes	Course Title	Marks	Contact Hrs/Wk	Credits
1.	EEE-1101	Electrical Circuits I	100	3	3.0
2.	EEE-1102	Electrical Circuits I Sessional	50	3	1.5
3.	PHY-1103	Physics I	100	3	2.5
4.	PHY-1104	Physics I Sessional	50	2	1.0
5.	CHEM-1105	Chemistry	100	3	2.5
6.	CHEM -1106	Chemistry Sessional	50	2	1.0
7.	MATH-1107	Mathematics I	100	3	2.5
8.	ICT-1109	Information and Communication Systems	100	3	2.5
9.	CSE-1110	Computer Programming	50	2	1.0
10.	HUM -1111	Communicative English	100	3	2.5
Total			800	27	20.0

B.Sc. Engineering First Year, Second Semester, Examination 2021

Sl. No.	Course Codes	Course Title	Marks	Contact Hrs/Wk	Credits
1.	EEE-1201	Electrical Circuits II	100	3	3.0
2.	EEE-1202	Electrical Circuits II Sessional	50	3	1.5
3.	PHY-1203	Physics II	100	3	2.5
4.	PHY-1204	Physics II Sessional	50	2	1.0
5.	ME-1205	Mechanical Engineering	100	3	2.5
6.	ME-1206	Mechanical Engineering Sessional	50	2	1.0
7.	MATH-1207	Mathematics II	100	3	2.5
8.	CSE-1209	Computer Organization and Architecture	100	3	2.5
9.	HUM-1211	Economics	100	3	2.5
10.	NCC1/NCC2	Bangladesh Studies/ Islamic Studies	100	3	00
11.	EEE-1200	Viva-voce	50		1.0
Total			900	28	20.0

B.Sc. Engineering Second Year, First Semester, Examination 2022

Sl. No.	Course Codes	Course Title	Marks	Contact Hrs/Wk	Credits
1.	EEE-2101	Electronics I	100	3	3.0
2.	EEE-2102	Electronics I Sessional	50	3	1.5
3.	EEE-2103	Electrical Machines I	100	3	3.0
4.	EEE-2104	Electrical Machines I Sessional	50	3	1.5
5.	STAT-2105	Statistics for Engineers	100	3	2.5
6.	MATH-2107	Mathematics III	100	3	2.5
7.	CES-2109	Data Structures and Algorithm	100	3	2.5
8.	CSE-2110	Data Structures and Algorithm Sessional	50	2	1.0
9.	IPE-2111	Industrial Management and Production Engineering	100	3	2.5
Total			750	26	20.0

B.Sc. Engineering Second Year, Second Semester, Examination 2022

Sl. No.	Course Codes	Course Title	Marks	Lab Hrs/Wk	Credits
1.	EEE-2201	Electronics II	100	3	3.0
2.	EEE-2202	Electronics II Sessional	50	3	1.5
3.	EEE-2203	Electrical Machines II	100	3	3.0
4.	EEE-2204	Electrical Machines II Sessional	50	2	1.0

5.	EEE-2205	Digital Electronics	100	3	3.0
6.	EEE-2206	Digital Electronics Sessional	50	3	1.5
7.	MATH-2207	Numerical Methods for Engineers	100	3	2.5
8.	MATH-2208	Numerical Methods for Engineers Sessional	50	2	1.0
9.	HUM-2211	Professional Ethics and Law	100	3	2.5
10.	EEE-2200	Viva-voce	50		1.0
Total			750	25	20.0

B.Sc. Engineering Third Year, First Semester, Examination 2023

Sl. No.	Course Code	Course Title	Marks	Contact Hrs/Wk	Credits
1.	EEE-3101	Electronics III	100	3	3.0
2.	EEE-3102	Electronics III Sessional	50	3	1.5
3.	EEE-3103	Measurements and Instrumentation	100	3	3.0
4.	EEE-3104	Measurements and Instrumentation Sessional	50	3	1.5
5.	EEE-3105	Electromagnetic Fields and Waves	100	3	3.0
5.	EEE-3106	Electromagnetic Fields and Waves Sessional	50	3	1.5
8.	EEE-3107	Microprocessor, Microcontrollers and Peripherals	100	3	3.0
9.	EEE-3108	Microprocessor, Microcontrollers and Peripherals Sessional	50	3	1.5
10.	EEE-3110	Electrical and Electronic Shop Practice	50	2	1.0
11.	EEE-3112	Electrical and Electronic Circuit Simulation	50	2	1.0
Total			700	28	20.0

B.Sc. Engineering Third Year, Second Semester Examination 2023

Sl. No.	Course codes	Course Title	Marks	Contact Hrs/Wk	Credits
1.	EEE-3201	Power System Analysis I	100	3	3.0
2.	EEE-3203	Control System Engineering	100	3	3.0
3.	EEE-3204	Control System Engineering Sessional	50	3	1.5
4.	EEE-3205	Communication Engineering I	100	3	3.0
5.	EEE-3206	Communication Engineering I Sessional	50	3	1.5
6.	EEE-3207	Electrical Properties of Materials	100	3	3.0
7.	EEE-3209	Signals and Linear Systems	100	3	3.0
8.	EEE-3212	Electrical Service Design	50	3	1.0
9.	EEE-3200	Viva-voce	50		1.0
Total			700	24	20.0

B.Sc. Engineering Fourth Year, First Semester Examination 2024

Sl. No.	Course Codes	Course Title	Marks	Contact Hrs/Wk	Credits
1	EEE-4000	Thesis/ Project	50	2	1.0
2.	EEE-4100	Industrial Attachment/ Training/Internship	50	-	1.0
3.	EEE-4101	Power System Analysis II	100	3	3.0
4.	EEE-4102	Power System Analysis II Sessional	50	3	1.5
5.	EEE-4103	Communication Engineering II	100	3	3.0
6.	EEE-4104	Communication Engineering II Sessional	50	3	1.5
7.	EEE-4105	Digital Signal Processing	100	3	3.0
8	EEE-4106	Digital Signal Processing Sessional	50	3	1.5
9.	EEE-41**	Elective I	100	3	3.0
10.	EEE-41**	Elective I Sessional	50	3	1.5
Total			700	25	20.0

B.Sc. Engineering Fourth Year, Second Semester, Examination 2024

Sl. No.	Course Codes	Course Title	Marks	Contact Hrs/Wk	Credits
1.	EEE-4000	Thesis/Project	100	4	3.0
2.	EEE-4201	Switchgear and Protection	100	3	3.0
3.	EEE4202	Switchgear and Protection Sessional	50	2	1.0
4.	EEE-42**	Elective II / Elective III / Elective IV	400	12	12.0
5.	EEE-4200	Technical Seminar	50		1.0
Total			700	21	20.0

List of Elective Courses

Elective I					
Sl. No.	Course Codes	Course Title	Marks	Contact Hrs/Wk	Credits
1.	EEE-4111	Power Electronics	100	3	3.0
2.	EEE-4112	Power Electronics Sessional	50	3	1.5
3.	EEE-4121	Cellular Mobile Communication	100	3	3.0
4.	EEE-4122	Cellular Mobile Communication Sessional	50	3	1.5
5.	EEE-4131	High Voltage Engineering	100	3	3.0
6.	EEE-4132	High Voltage Engineering Sessional	50	3	1.5
Total			450	18	13.5

Elective II (Electronics Group)					
Sl. No.	Course Codes	Course Title	Marks	Contact Hrs/Wk	Credits
1.	EEE-4213	Optoelectronics	100	3	3.0
2.	EEE-4215	VLSI Design	100	3	3.0
3.	EEE-4217	Micro-electro Mechanical Systems (MEMs)	100	3	3.0
4.	EEE-4219	Bioelectronics	100	3	3.0
Total			400	12	12.0

Elective III (Power and Energy Group)					
Sl. No.	Course Codes	Course Title	Marks	Contact Hrs/Wk	Credits
1.	EEE-4223	Renewable Energy for Sustainable Development	100	3	3.0
2.	EEE-4225	Power Plant Engineering	100	3	3.0
3.	EEE-4227	Nuclear Power Engineering	100	3	3.0
4.	EEE-4229	Power System Operation and Control	100	3	3.0
Total			400	12	12.0

Elective IV (Communication Group)					
Sl. No.	Course Codes	Course Title	Marks	Contact Hrs/Wk	Credits
1.	EEE-4233	Optical Fiber Communication	100	3	3.0
2.	EEE-4235	Microwave and Satellite Communication	100	3	3.0
3.	EEE-4237	Data Communication and Networking	100	3	3.0
4.	EEE-4239	Telecommunication Engineering	100	3	3.0
Total			400	12	12.0

Course Profile

B.Sc. Engineering First Year, First Semester Examination 2021

Course Code:	EEE-1101
Course Title:	Electrical Circuits I
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	N/A

Rationale of the Course: Electrical circuits are found in many complicated system and have wide range of applications. Hence proper circuit design and analysis is an ultimate requirement from different engineering fields. This course is intended to enable the students to develop the sound understanding and ability to design and analyze the basic electrical/magnetic circuits through the fundamental tools of linear circuit analysis.

Course Objectives:

- To provides an introduction to fundamental electrical concepts as well as the technical skills and knowledge to analyze the simple circuits.
- To facilitate students to understand the basics of AC fundamentals.
- To help learners to comprehend laws of magnetic circuits.
- To build and analyze some practical, useful devices in real world.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Explain the fundamental concepts of linear electrical circuit elements and magnetic quantities.	Cognitive (Understand)
CLO2	Apply different circuit analysis techniques and circuit theorems to solve circuits for unknown quantities.	Cognitive (Apply)
CLO3	Interpret magnetic circuits for understanding advance courses.	Cognitive(Understand, Apply)
CLO4	Solve RL, RC and RLC alternating current (AC) circuits with phasor algebra.	Cognitive (Analyze)
CLO5	Design different types of DC and AC circuits.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Electrostatics: Matter and charge, Coulomb's law, Electric field lines, electric field and torque on a dipole, Electric flux, Gauss's law and its applications, electric potential.
2.	Circuit Variables and Parameters: Voltage, current, resistance and resistivity, Ohm's law, power and energy, Simple circuits: Series-parallel circuits, voltage and current divider rules, Kirchhoff's current and voltage laws, Open and short circuits.
3.	Techniques of Circuit Analysis: Current-voltage sources and Source conversion, Node voltage method and mesh current method including super node and super mesh, DC Network theorems: Thevenin's, Norton's and superposition theorems with applications in circuits having independent and dependent sources, maximum power transfer condition and reciprocity theorem, Wye-delta transformation.
4.	Capacitors: Electric field and capacitance of capacitor, Charging and discharging of capacitors, Dielectrics in capacitors, Capacitors in series and parallel, Energy storage.
5.	Alternating Current Circuits: Introduction to alternating current circuits, instantaneous, average and RMS values, complex number in AC circuit, phasor diagram, AC through resistor, capacitor and inductor, Frequency response of RL, RC and RLC circuits, Power relations in AC circuits: Real, reactive and apparent power, power factor, power factor improvement, Q-factor and band width.
6.	Magnetic Quantities and Variables: Magnetic flux, permeability and reluctance, magnetic field strength, magnetic potential, magnetization curve, Laws in magnetic circuits: Biot-Savart's law and Ampere's circuital law, Magnetic circuits: Series, parallel and series-parallel circuits, Inductors and inductance, Faraday's law of of electromagnetic induction.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√								
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Problem solution ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Problem solution ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Charles K. Alexander : Fundamentals of Electric Circuits
2. B. L. Theraja : A Text Book of Electrical Technology-I
3. Resnic & Halliday : Physics, part-II
4. R. M. Kerchner & G. F. Corcoran : Alternating Current Circuits

Course Code:	EEE-1102
Course Title:	Electrical Circuits I Sessional
Course Credits:	1.5
Course Type:	Core Course
Co-requisite:	EEE-1101

Rationale of the Course: This course provides the student a hands-on experience of verifying the theorems and laws they are taught in EEE-1101 course.

Course Objectives:

- To show how the theorems and laws work in ideal and non-ideal environment.
- To provide hands on experience of measuring the AC and DC quantities of a circuit.
- To show the process of designing and constructing an electrical circuit and then verify it with necessary measuring instrument.
- To provide knowledge about common errors that may occur during circuit design.

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand operation of a practical electrical circuit comprising with passive elements for both AC and DC.	Psychomotor(Imitation, Manipulation)
CLO2	Identify the error in the circuit if there is any.	Psychomotor(Imitation, Manipulation)
CLO3	Design and construct a simple electrical circuit applying network theorems and laws	Psychomotor(Precision, Articulation)

Course Content

Sl. No.	List of the Experiments
1.	Familiarization with laboratory equipment's and electric meters.
2.	Study and verification of Ohm's Law with series and parallel combination of resistances
3.	Study and verification of voltage and current divider rules.
4.	To convert a resistive T-network into equivalent Π -network and vice-versa.
5.	Study of series circuit and verification of Kirchhoff's Voltage Law (KVL).
6.	Study of parallel circuit and verification of Kirchhoff's Current Law (KCL).
7.	Verification of Thevenin's and Norton's Theorem
8.	Verification of Superposition Theorem
9.	Verification of Maximum Power Transfer Theorem
10.	Verification of Reciprocity Theorem
11.	Three phase power measurement using two wattmeter method
12.	Determination of coupling coefficient of a magnetically coupled circuit.
13.	Determination of coupling coefficient of a conductively coupled circuit.
14.	Understanding and application of the characteristics of B-H curve

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√										
CLO3			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO2	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO3	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	PHY-1103
Course Title:	Physics I
Course Credit:	2.5
Course Type:	Basic Science
Prerequisite:	N/A

Rationale of the Course: Use of acoustical device such as microphone and loudspeaker is very common in this modern world. Highly skilled manpower is needed for Building acoustics and recording sectors. Boom of optical sectors urgently needs huge manpower. Hence, this course is designed to cover introduction to microphone, loudspeaker, recording, architectural acoustics, optical instruments in detail.

Course Objectives:

- To facilitate necessary knowledge of oscillation, acoustic and optics
- To acquaint students with the basic tools of acoustic and optical instruments.
- To help the students to become expertise in microphone, loudspeaker, recording reproduction, optical instruments with their construction, their functions, characteristics with applications.
- To gain designing skills and real time applications of microphones and loudspeakers.

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to

CLOs	Descriptions	Learning domain with level
CLO1	Understand the characteristics and functional concept of various types of waves.	Cognitive(Understand)
CLO2	Design, construct and test microphones and loudspeakers.	Cognitive(Apply)
CLO3	Calculate whether the room is live or dead.	Cognitive(Analyze)
CLO4	Produce and apply ultrasonic instruments.	Cognitive(Apply)
CLO5	Develop recording and reproduction skills.	Cognitive(Create)

Course Contents

Sl. No.	Course Content (as Summary)
1.	Harmonic Oscillation and Wave: Simple harmonic motion, Fundamental wave equation, Graphical representation, Kinetic energy, Total Energy, Plane wave, Spherical wave, Standing wave, Wave motion, Characteristics of wave motion, Particle velocity, wave velocity, wave energy.
2.	Microphones: General characteristics of microphones, pressure mic, gradient and combination mic. Carbon mic. Capacitor, moving coil, crystal mic. Ribbon mic
3.	Loudspeakers: Basic theory of direct radiator loudspeaker, loudspeaker enclosure, Dynamic loudspeaker, the horn loudspeaker, Elementary horn speaker.
4.	Architectural Acoustics: Reverberation, Reverberation time, Live room, Optimum reverberation time, Sabine's theory, Eyring's theory, Room Design, Effects of audience on reverberation, Measurement of absorption coefficient.
5.	Recording and Reproduction: Magnetic recording media; Properties of magnetic recording materials, Head structure and working; Recording loss; Reproduction loss; Disk recording; Tape recording; Film recording and reproduction.
6.	Nature of light: Corpuscular theory; Huygen's Principle.
7.	Interference: Young's experiment; Coherent sources; Fringe Analysis; Bi-prism; Newton's ring; Michelson Interferometer.
8.	Diffraction: Rectilinear propagation; Fresnel Assumptions; Fresnel diffraction at single slit; Fraunhofer diffraction at single slit; Resolving power of Telescope and microscope.
9.	Polarization: Light as transverse wave; Plane of polarization; Polarizing Angle; Polarization by reflection and refraction; Brewster's Law, Malus's Law; Double refraction; Nicol Prism; Polariscopes.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√								
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion	- Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Presentation - Discussion	- Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Group Assignment - Semester end final examination - Participation

Learning Materials:

1. Brijlal & Ghosh : A textbook of Sound
2. Brijlal and Subramonium : A Textbook of Optics
3. Beranak : Applied Acoustics
4. Kinsler & Frey : Fundamentals of Acoustics
5. Stewart : Magnetic Recording Process
6. J. L. Hunter : Acoustics

Course Code:	PHY-1104
Course Title:	Physics I Sessional
Course Credit:	1.0
Course Type:	Basic Science Course
Co-requisite:	PHY-1103

Rationale of the Course: The sessional course will give opportunity to verify practically the fundamental characteristics of electrostatic signal, acoustic devices, magnetic instrument and optical instruments taught in the theoretical course PHY-1103

Course Objectives:

- To know how to use different instruments used to measure electrical, magnetic and acoustics quantities.
- To provide hands on experience of measuring magnetic induction.
- To know the limitations and errors of measurements in practical.
- To recognize the process of generation of electricity with magnetism.

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to –

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand how to generate waves.	Psychomotor(Imitation, Manipulation)
CLO2	Apply the knowledge to produce microphone and loudspeakers.	Psychomotor(Manipulation, Precision)
CLO3	Utilize diffraction grating to measure wavelength of light.	Psychomotor(Manipulation, Precision)
CLO4	Employ polarimeter for the measurement of concentration of a solution.	Psychomotor(Manipulation, Precision)
CLO5	Design acoustic auditorium.	Psychomotor(Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Familiarization with laboratory equipment's and instrument
2.	To determine the radius of curvature of a lens by Newton's ring method
3.	To determine the angle of prism, minimum deviation and refractive index of the materials of a prism.
4.	Draw and testing of different types of microphones and Loud speakers
5.	Conversion of Loudspeaker into Dynamic Microphone.
6.	Determination of directivity of a Loudspeaker.
7.	Determination of directivity and sensitivity of a Microphone.
8.	Determination of wavelength of a monochromatic light by plane diffraction grating
9.	Calibration of a polarimeter and hence determination specific rotation of a sugar solution
10.	To study the acoustic beats displaying on the oscilloscope

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√	√							
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Participation
CLO2	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO3	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO4	- Demonstration - Experiment, - Report writing	- Lab performance - Semester end final examination - Participation
CLO5	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals
2. Md. Sekul Islam : Practical Applied Physics.

Course Code:	CHEM-1105
Course Title:	Chemistry
Course Credit:	2.5
Course Type:	Basic Science
Pre-requisite:	N/A

Rationale of the Course: Basic knowledge of chemicals and their properties are very important in many branches of engineering. This course aims to give basic ideas about various aspects of inorganic and physical chemistry including basic theories, chemical calculations and their applications in different industries and in various fields of our everyday life.

Course Objectives:

- To identify the electronic configuration and structure of atoms for different elements and their periodic properties and atomic interactions to each other's
- To explain different concepts of acids and bases and their relative strengths
- To prepare different types of solutions and perform chemical calculations related to the concentration of solution
- To know about chemical kinetics and chemical equilibrium and also learn how energy changes, occur during chemical reaction
- To explain inter-conversion of energy in the chemical, thermal and electrochemical process and their energy calculation.

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand atomic structure, periodic table, chemical bonding, modern concept of acids and bases, chemical kinetics, chemical equilibrium, p^H and buffer solution, and electrochemistry.	Cognitive (Understand)
CLO2	Explain periodic properties of element, chemical bonding's and chemical equilibrium.	Cognitive (Understand)
CLO3	Apply modern concept of acids and bases, chemical kinetics and different laws of electrochemistry in different chemical environments.	Cognitive (Apply)
CLO4	Prepare various solutions and make necessary dilutions.	Cognitive (Apply)
CLO5	Determine the order, half-life, spontaneity etc. of chemical reactions and their equilibrium.	Cognitive (Analyze)
CLO6	Calculate the energy changes occur during a chemical reaction.	Cognitive (Analyze)

Course Content:

Sl. No.	Course Content (as Summary)
1	Atomic Structure: Modern concept of atomic structure (Rutherford's and Bohr's atomic model), quantum numbers, distribution of electrons in atoms, Aufbau principle, Pauli exclusion principle, Hund's rule of maximum multiplicity, wave nature of electron, de Broglie relation, Heisenberg uncertainty principle, preliminary idea of orbital's, physical significance of s , p and d orbitals
2	Periodic Table: Periodic law, classification of elements, modern periodic table in the light of electronic configurations of elements, different types of elements, periodic properties, atomic, covalent and ionic radii, ionization energy, electro-negativity, electron affinity, effective nuclear charge. Important uses of noble gases and lanthanide elements.
3	Chemical Bonding: Different types of chemical bonds (ionic, covalent, co-ordinate, hydrogen and metallic), valence bond theory (VBT), hybridization of orbital's, molecular orbital theory (MOT), electronic configurations of simple molecules in terms of MO concept (He_2 , N_2 , O_2 , F_2 , Cl_2).
4	Modern Concept of Acids and Bases: Arrhenius, Bronsted-Lowry, Lewis, Lux-Flood and Usanovich concepts of acids and bases, strength of acids and bases.
5	Chemical kinetics: Rate of reactions, rate equations, order and molecularity, zero order, first order and second order reactions, determination of order of reaction, temperature effect on reaction rate.
6	Chemical Equilibrium: Equilibrium and equilibrium constants, Le Chatelier principle and its applications.
7	pH and Buffer Solution: Ionization of water, pH and pH scale, buffer solutions, mechanism of buffer action, Henderson-Hasselbalch equation.
8	Electrochemistry: Electrolytes and electrolysis, Faradays laws of electrolysis and their significance, ionic mobility, Kohlrausch's law, transference number, electrochemical cells, electrode reactions and potentials, reference electrodes, storage batteries.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3			√	√								
CLO4				√								
CLO5					√							
CLO6						√						

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion - Online Resource	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO6	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation

Learning Materials:

1. R. D. Madan : Modern Inorganic Chemistry
2. W.U. Malik, G.D. Tuli and R. D. Madan : Selected Topics in Inorganic Chemistry
3. Darrel D. Ebbing : General Chemistry
4. Raymond Chang : General Chemistry
5. M. M. Haque and M. Y. A. Mollah : Principles of Physical Chemistry
6. B. S. Bahl and G. D. Tuli and ArunBahl : Essentials of Physical Chemistry

Course Code:	CHEM-1106
Course Title:	Chemistry Sessional
Course Credit:	1.0
Course Type:	Basic Science
Co-requisite:	CHEM-1105

Rationale of the Course: The course will provide the student some basic experimental tests of acid and bases, preparing different kinds of standard chemical solutions based on the course CHEM-1105.

Course Objectives:

- To prepare chemical solution and standardize the solutions of acid, base, oxidizing and reducing agent via titration.
- To learn the metal ions (such as $\text{Fe}^{2+}/\text{Fe}^{3+}$ and Cu^{2+} ions) in a solution as well as some understanding of the basic laws of electrochemistry.

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to

CLOs	Descriptions	Learning Domain with Level
CLO1	Prepare different kinds of standard chemical solution.	Psychomotor (Imitation)
CLO2	Standardization of some solution through titration.	Psychomotor (Manipulation, Precision)
CLO3	Analyze and estimate different kinds of metal ions in a solution.	Psychomotor (Manipulation, Precision)

Course Content:

Sl. No.	List of the Experiments
1.	Identify the acid radical of the supplied samples qualitatively (a) Preparation of Mohr's salt (b) Preparation of ammonium alum
2.	Determination of the strength of HCl by Na_2CO_3 .
3.	Determination of the strength of NaOH by Oxalic acid.
4.	Determination the Integral Heat of solution of supplied samples calorimetrically
5.	Determination of pH of the supplies samples and identify these samples acid or basic

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3			√									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO2	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO3	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	MATH-1107
Course Title:	Mathematics I
Credit Value:	2.5
Course Type:	Basic Science
Pre-requisite:	NA

Rationale of the Course: Modern science, technology and engineering would have not been possible without extensive use of mathematics. Mathematics contributes to the developments and understanding in many disciplines and provides the foundation for many of today's innovations and tomorrow's solutions. This course has been introduced to give learners intermediate ideas of set theory, matrices, complex numbers and trigonometry with some applications.

Course Objectives:

- To understand basic ideas and principles of algebra and trigonometry.
- To facilitate necessary knowledge about set theory, matrices, complex numbers and complex operators.
- To apply the knowledge of functions of complex arguments, statistical curve fitting and trigonometry.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand basics of set theory and its applications, matrices, complex numbers, complex operators and inequities, trigonometry, Indeterminate forms and Curve Fitting of different functions.	Cognitive (Understand)
CLO2	Apply complex numbers, complex operators, inequities in circuit and signal analysis.	Cognitive (Apply)
CLO3	Utilize trigonometric function in signal and systems in communication engineering.	Cognitive (Apply)
CLO4	Analyze circuits of electrical and electronic engineering using matrices.	Cognitive (Analyze)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Elements of Set Theory: Introduction to sets, Set algebra, Types of sets: Finite, Infinite, Countable and Uncountable sets, Relations and Functions.
2.	Matrices: Definition and algebra of matrices, Special types of matrices, Determinant of square matrix, Properties, Cramer's rule, Rank of a matrix, Elementary transformation and normal form. The Adjoint and Inverse of a square matrix, Matrix inversion by partitioning, Solution of system of linear equations, Gauss-Jordan method.
3.	Complex Numbers, Complex Operators and Inequities : Complex j operator, Complex numbers, Basic operations with complex numbers, Graphical representation of complex number, Polar and exponential forms of complex numbers, Powers and Roots of Exponential and Trigonometric Functions, Circular functions of complex number and Hyperbolic functions, Inverse Circular and Inverse Hyperbolic functions, Logarithmic functions, Separation of real and Imaginary parts of all types of Functions, Properties of Inequities, Graphical solution of inequities, Algebraic solution of inequities, Inequities involving absolute values, Graphical solution of inequities with two variables.
4.	Trigonometry: Trigonometric functions, Graphs of Trigonometric functions, Complex number, De-Moivre's theorem and it's applications, Functions of complex arguments, Gregory's series, Summation of trigonometric series, hyperbolic functions.
5.	Indeterminate Forms and Curve Fitting: Fitting of curves by least square method for linear, parabolic, and exponential. Regression Analysis.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3		√	√									
CLO4			√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
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Learning Materials:

1. P.N. Wartikar and J.N. Wartikar : A text book of Applied Mathematics ,Vol – I and –II
2. Shahidullah & Bhattachajee : A text book on higher algebra
3. M.L Khanna : Matrices
4. B.C Das & N. Mukhejee : Higher Trigonometry

Course Code:	ICT-1109
Course Title:	Information and Communication Systems
Course Credit:	2.5
Course Type:	Allied Engineering
Pre-requisite:	NA

Rationale of the Course: In this modern world progress and prospects of Information and Communication system are remarkable and became heart of nearly every company where a team of savvy computer experts are needed to keep its systems up and running. This course gives a fundamental overview of Information and Communication systems with some practical real-life experience.

Course Objectives:

- To facilitate basic understanding of information communication and computer system.
- To select appropriate communication system for a particular purpose.
- To discuss the fundamentals of IOT.
- To utilize IT in real-life purpose.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the basics of information technology, computer fundamentals, communication systems, IoT and real life IT experience.	Cognitive (Understand)
CLO2	Solve the basic problems of communication systems	Cognitive (Evaluate)
CLO3	Apply the IoT and IT in real life scenario.	Cognitive (Apply)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction: IT Concepts and Applications, The role of IT in the present era, Types of Computers, The Elements of a Computer & Communication System, Data/Information, Hardware, Software.
2.	Computer fundamentals: The Computer Generations, Basic Computers, Input and Output Devices, Memory Systems, CPU, Bus Lines, Ports, Application Software, System Software, Integrated Software, Desktop Accessories, Project Management Software, Multimedia Presentation.
3.	Basic Communication Systems: Communication Systems and Networking, Types of Computer Networking, www, E-mail, Voice-mail, Video-Conferencing, Tele-conferencing, Bulletin Board System, Public Information Services, Online Information Services, Internet Relay Chat, FTP, Video-Text, Facsimile, Electronic Meeting Systems, Usenet, Telnet, Electronic Commerce.
4.	IoT: Introduction to IoT, Basic ideas of IoT, Types of sensors, and application areas.
5.	Real Life IT Applications: Office Automation Systems, Electronic Publishing Systems, Desktop Publishing Systems, E-Learning, E-Education, E-Training, E-Agriculture, E-Medicine, E-Health, E-Commerce, E-Government, E-Entertainment.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										√
CLO2		√	√			√						
CLO3		√	√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Tutorial ▪ Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Information Technology – The Breaking Wave : Dennis P. Curtin et al., Irwin
2. Using Information Technology-
A Practical Introduction to Computers & Communications : William, Sawyer, Hutchinson
3. Information Technology Today,
Encyclopedia of Information Technology : S. Jaiswal
4. Fundamentals of Computers : V Rajaraman
5. Computer Fundamentals : Pradeep K. Sinha

Course Code:	CSE-1110
Course Title:	Computer Programming
Course Credit:	1.0
Course Type:	Allied Engineering
Co-requisite:	ICT-1109

Rationale of the Course: Software is written and hardware is programmed by different computer programming languages. So, skilled computer programmers are required in different branches of engineering. This course will begin with a brief description of basic programming languages and their classifications. The language of Python programming will be taught in detail. Students will be able to solve some real world problems.

Course Objectives:

- To introduce the students with different types of programming languages.
- To use the control statements to write primary computer programs.
- To handle arrays, pointers, and functions to write advanced programs.
- To use structure, string operations and union along with other tool to solve real life problems using computer programming.
- To use primary object oriented programming when needed.

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the basic tokens of Python programming.	Psychomotor (Imitation, Manipulation)
CLO2	Apply the structure of a typical Python program and I/O statements.	Psychomotor (Imitation, Manipulation)
CLO3	Solve the mathematical and engineering problems using Python.	Psychomotor (Manipulation, Precision)
CLO4	Use functions to import data, analyze data, and produce output for solving real life problems.	Psychomotor (Manipulation, Precision, Articulation)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction to C Programming: Programming language and their classification, variables and Constants, Operators-arithmetic and logical and bitwise, and expression.
2.	Control Statements: Control flow statements and blocks: if-else, Loop statements.
3.	Array and Pointers: Arrays-single dimensional and multi-dimensional, Strings as array of characters, string Library functions Solving problems using Arrays.
4.	Functions: Library functions, User-defined functions, Arguments passing among functions, Variable scope, Recursion.
5.	String Operations: Declaring and Initializing string variables, string I/O operations, Standard library string functions, Problem solving using string operations and arrays.
6.	Solve Real World Problems: Introduction to solve real world problem using Python Machine Learning Concept.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√								√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resource	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Web based reading and consulting materials will be used.

Course Code:	HUM-1111
Course Title:	Communicative English
Course Credit:	2.5
Course Type:	Humanities
Pre-requisite:	NA

Rationale of the Course: Communication skills can help many aspects of our life, from our professional career, to social gatherings, to our family life. This course is designed to enable students to develop their competence in reading, writing, speaking, listening and grammar for academic purposes. The students will be encouraged to acquire skills and strategies for using language appropriately and effectively in various situations.

Course Objectives:

- To facilitate necessary knowledge about the communication model, structure of sentences and vocabulary enrichment
- To understand native speakers, professionals and students talking about their work and study.
- To develop necessary skills to communicate fluently with others verbally
- To enhancing the skills for apprehension of wide variety of written texts and other materials
- To apply the students' acquired knowledge in writing descriptions and explanations on different things/objects/places/peoples etc. related to student's field of study and to write study- and work-related letters/reports.

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to -

CLOs	Descriptions	Learning Domain with Level
CLO1	Identify the basic building blocks of communication process and structure of English sentences.	Cognitive (Understand) Affective (Receiving, Responding)
CLO2	Understand how reading texts can increase vocabulary and improve fluency in English.	Cognitive (Understand) Affective (Receiving, Responding)
CLO3	Use English Language correctly in spoken and written forms (have a basic knowledge of English pronunciation and grammar)	Cognitive (Apply) Affective (Receiving, Responding and Valuing)
CLO4	Communicate (orally and in written) confidently in various contexts and situations	Cognitive (Apply) Affective (Responding, Valuing)
CLO5	Acquire proficiency in listening comprehension, distinguish between British and American pronunciation and spelling and usages of words in different contexts.	Cognitive (Apply) Affective (Receiving)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction to Communication: Definition, Principles, Nature, Forms and Types. A Model of Communication Process.
2.	Grammar and Usage: Basic sentence patterns (introduction to subject, predicate, object, complement and adverbial), sentence types (declarative, imperative and exclamatory etc.), simple sentence and its structures; tenses and their use in context, introduction to present, past and future tenses.
3.	Speaking and Listening Skills: Listening Process; Classification of Listening; Purpose of Listening; Common Barriers to the Listening Process; Measures to Improve Listening; Listening as an Important Skill in Work Place, Socializing (starting a conversation, introducing oneself, greetings, giving and responding to good wishes etc.), expressing moral attitude (apologizing, granting forgiveness, expressing appreciation etc.), expressing intellectual skill (agreement/disagreement, denying something, accepting offer/invitation etc.)
4.	Reading and Writing: Reading different types of texts including books, newspapers, diagrams, charts/tables, graphs, course brochures and job advertisements. Writing short and long paragraphs, academic letters/applications, writing official letters (CV/resume, memo, notice, job application, order letter, quotation) and electronic mail (E-Mail); descriptions and explanations on different things/objects/places/peoples etc.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1						√			√			
CLO2						√			√			
CLO3						√			√	√		√
CLO4									√	√		√
CLO5						√			√	√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture Note, ▪ Text Book, ▪ Web Material	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture Note, ▪ Text Book, ▪ Web Material, ▪ Audio/video	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture Note, ▪ Text Book, ▪ Web Material	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture Note, ▪ Text Book, ▪ Web Material	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture Note, ▪ Text Book, ▪ Web Material	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Jeya Santhi V. and R Selvam : Advanced Skills for Communication in English: Book I
2. Ahmed Sadruddin : Learning English the Easy Way
3. Lesikar R.V. and Petit J.D. : Business Communication- Theory and Application

B.Sc. Engineering First Year, Second Semester Examination 2021

Course Code:	EEE-1201
Course Title:	Electrical Circuits II
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-1101

Rationale of the Course: Electrical circuits are heart of many electrical systems. It has very wide applications in different engineering branches. So, this core course has been introduced to give learner in depth ideas of electrical circuits and their applications.

Course Objectives:

- To learn the ideas and principles of transient
- To understand working of ac networks, poly-phase circuits
- To analyze the different types of ac circuits
- To design of different types of filters.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the transient behavior of first and second order electrical circuits.	Cognitive (Understand)
CLO2	Apply the node and loop analysis in standard series and parallel AC circuits.	Cognitive (Apply)
CLO3	Analyze the electrical power and its conversion in poly-phase circuits.	Cognitive (Analyze)
CLO4	Solve the circuit parameters using AC networks theorem.	Cognitive (Evaluate)
CLO5	Design different types of filter circuits.	Cognitive (Create)

Course Content:

Sl. No.	Course Contents (as Summary)
1.	Transients: Introduction, Types of transients, Transients in R–L circuits (DC), Transients in R–C circuits (DC), Double energy transients for undamped, under damped, over damped and critical damping
2.	Series and Parallel AC Circuits: Voltage divider rule, Current divider rule, AC through in series R–L circuits, Power factor, Q–factor of a coil, AC through in series R–C circuits, Impedance and phasor diagram, AC through in series R–L–C circuits, Admittance, conductance and susceptance, Series equivalent of a parallel circuit.
3.	Poly-phase Circuits: Generation of poly-phase voltages, Phase sequence, Phase sequence at load, Numbering of phases, Interconnection of three phases, Star or Wye (Y) connection, Voltages, currents and power in Y–connections, Delta or Mesh (Δ) connection, Voltages, currents and power in Δ –connections, Balanced Y/ Δ and Δ /Y conversions.
4.	Power–AC Networks: Power measurement in 3-phase circuits, Average, Apparent and reactive power, The power triangle, The total P, Q and S, Power factor correction.
5.	AC Network Analysis: Determination of voltage sign, Mesh and Nodal Analysis, Delta–Star and Star–Delta conversions, Kirchhoff’s laws, Superposition theorem, Thevenin’s theorem, Norton’s theorem, Maximum power transfer theorem, and Millman’s theorem.
6.	A.C. Bridges: AC Bridge, Maxwell’s inductance bridge, Maxwell’s wien bridge, Anderson Bridge, Hay’s bridge, De–Sauty Bridge
7.	Filters: Conventional filter design and operation, Elementary filter sections, Fundamental equations of an ideal filter, Low–pass RC filter, Low–pass RL filter, High–pass RC filter, High–pass RL filter, RC Pass band filter, RC Band stop filter, Filter performance.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3		√	√									
CLO4		√	√									
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion	▪ Participation; Presentation ▪ Tutorial ▪ Final exam
CLO2	▪ Lecture ▪ Presentation ▪ Discussion	▪ Participation; Presentation ▪ Tutorial ▪ Final exam
CLO3	▪ Lecture ▪ Presentation ▪ Discussion	▪ Participation; Presentation ▪ Tutorial ▪ Final exam
CLO4	▪ Lecture ▪ Presentation ▪ Discussion	▪ Participation; Presentation ▪ Tutorial ▪ Final exam
CLO5	▪ Lecture ▪ Presentation ▪ Discussion	▪ Participation; Presentation ▪ Tutorial ▪ Final exam

Learning Materials

- | | |
|--|-------------------------------------|
| 1. Charles Alexander, Matthew Sadiku | : Fundamentals of Electric Circuits |
| 2. M. Russell Kerchner, F. George Corcoran | : Alternating-current Circuits |
| 3. B.L. Theraja | : Textbook of Electrical Technology |

Course Code:	EEE-1202
Course Title:	Electrical Circuits II Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE - 1201

Rationale of the Course: This laboratory course is designed to be the experimental companion to EEE 1201- Electrical Circuit and Filter Design. The course is intended to enhance the students' understanding of important analytical principles developed in EEE 1201 by engaging them in developing practical ac systems in the laboratory.

Course Objectives:

- To build and examine and alternating current circuits based on AC concepts.
- To identify circuit faults and provide corresponding solutions.
- To demonstrate the reason behind real time deviations from theoretical analysis.
- To learn safety measurement rules in handling ac circuits.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Apply the electrical measuring equipment's used for modern industry.	Psychomotor (Imitation, Manipulation, Precision)
CLO2	Test and evaluate alternating current circuits.	Psychomotor (Manipulation, Precision)
CLO3	Write technical reports using collected experiment data.	Psychomotor (Manipulation, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Familiarization with electrical circuits, equipment and meters.
2.	To study the transient response of RC, RL and RLC Circuits.
3.	Study the characteristics of an AC series R-C circuit and find out the phase angle between $V_s \sim V_R$ and $V_s \sim V_C$.
4.	Study the characteristics of an AC series R-C circuit and find out the phase angle between $V_s \sim V_R$ and $V_s \sim V_C$.
5.	Study the characteristics of an AC series R-L circuit and find out the phase angle between $V_s \sim V_R$ and $V_s \sim V_L$.
6.	. Study the characteristics of an AC series R-L-C circuit and find out resonance frequency, upper cutoff frequency, lower cutoff frequency and bandwidth and quality factor.
7.	Three phase power measurement using two wattmeter method
8.	Determination of phase sequence of a three phase system.
9.	Determination of coupling coefficient of a magnetically coupled circuit.
10.	Determination of coupling coefficient of a conductively coupled circuit
11.	Determination of line voltage, phase voltage, line current and phase current of a 3-phase Y-connected and Δ -connected load.
12.	Determination of iterative and image impedance of a symmetrical and asymmetrical T and π network.
13.	Determination of different parameters of a two port network.
14.	Observation of frequency response and determination of cut off frequency of low pass and high pass filter.
15.	Observation of frequency resonance and determination of cut off frequency of band pass and band stop filter.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√	√									
CLO2			√	√								
CLO3			√							√		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO2	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO3	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	PHY-1203
Course Title:	Physics II
Course Credit:	2.5
Course Type:	Basic Science
Pre-requisite:	N/A

Rationale of the Course: It is essential for engineering/science students to develop strong understanding in the topics of physics. This course introduces students to the fundamental aspects of modern Physics, relating them to everyday applications. It aims to introduce students to frontiers of modern Physics and illustrate the benefits that scientific investigation brings to society

Course Objectives:

- To demonstrate the understanding of the scientific method of work and the evolution of physics from the classical to its modern era.
- To enhance the understanding and problem solving skills on related basic topics in modern physics
- To develop the basic concepts of atomic and nuclear physics, relativity, quantum mechanics and statistical physics in their relevant scope.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the structure of atoms, properties of nucleus, nuclear energies, special theory of relativity	Cognitive (Understand)
CLO2	Explain the interaction of various types of radiations of matter, principle of radioactivity and nuclear reactors.	Cognitive (Understand)
CLO3	Illustrate classical and quantum mechanical systems.	Cognitive (Understand)
CLO4	Apply the special theory of relativity and Fermi-Dirac statistics in physical sciences.	Cognitive (Apply, Analyze)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Atomic Physics: Thomson's plum pudding model, Rutherford's nuclear atom model, Bohr's atomic model, Sommerfeld's relativistic model, Vector atom model, Different quantum numbers, Electron configuration, Pauli exclusion principle, Stark effect and Zeeman effect.
2.	Relativistic Physics: Postulates of Einstein special theory, Michelson-Morley experiment, Lorentz transformation, Length contraction and Time dilation, Twin paradox, Relativity of mass, Mass-energy equivalence, Velocity addition, Introduction to General relativity.
3.	Nuclear Physics: Properties of a nucleus, Mass defect and Binding energy, Nuclear force, Meson theory, Characteristics of nuclear force. Nuclear fission and fusion, Chain reaction and Nuclear reactor, Natural and artificial radioactivity, Radioactive decay laws, Half-life and mean life, Secular and Transient equilibrium, radioactive dating, Uses of radioactivity.
4.	Wave-Particle Duality: Blackbody radiation, Photoelectric effect, Quantum theory of light, The Compton effect, Pair production, Pair Annihilation, De Broglie matter waves, Phase and group velocities, Heisenberg's uncertainty principle and its applications.
5.	Quantum Mechanics: Limitations of Classical mechanics, Postulates of quantum mechanics, Eigen values and eigen function, Time dependent and time independent Schrodinger equations, Expectation values, Particle in a box, Harmonic oscillator, Schrodinger's equation for the hydrogen atom.
6.	Statistical distribution laws: Phase space, the probability of distribution, the most probable distribution, Maxwell-Boltzmann distribution, Evaluation of constants, Molecular energies in an ideal gas, Bose-Einstein distribution, Fermi-Dirac distribution, Transition between states, Comparison of results.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3		√	√									
CLO4			√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion - Online Resources	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Presentation - Discussion - Online Resources	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion - Online Resources	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Discussion/Group Discussion - Online Resources	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation

Learning Materials

1. A Beiser : Concept of Modern Physics
2. B. L. Theraja : Modern Physics
3. Murugesan : Modern Physics

Course Code:	PHY-1204
Course Title:	Physics II Sessional
Course Credit:	1.0
Course Type:	Basic Science Course
Co-requisite:	PHY-1203

Rationale of the Course: The Sessional course will give opportunity to observe the basic laws of atomic physics, statistical physics, quantum mechanics and radioactivity phenomena taught in the theoretical course PHY-1203

Course Objectives:

- To develop a foundation on conceptual approach to physical sciences by uncovering the deep laws behind physical phenomenon/effects.
- To demonstrate their understanding of the fundamental postulates and principles of special relativity and quantum mechanics.
- To understand about phenomena of radioactive decay, nuclear fissions and fusion.
- To enhance analytical reasoning and problem solving skills

Course Learning Outcomes (CLOs): By the end of the course, the student will able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the atomic structure and spectral series of hydrogen.	Psychomotor(Imitation, Manipulation)
CLO2	Observe the Compton effect	Psychomotor(Imitation, Manipulation)
CLO3	Detect and determine different nuclear phenomenon.	Psychomotor(Imitation, Manipulation)

Course Content:

Sl. No.	List of the Experiments
1.	Photoelectric effects (a) Determination of Plank's constant(b) Determination of work function (c) Study of photoelectric current intensity (i) as a function of bias voltage (ii) at different frequencies of irradiation (iii) at different intensities (at constant frequency)
2.	To determine e/m of an electron using a Helmholtz coil.
3.	To determine the charge of an electron by Millikan oil-drop experiment.
4.	To determine the value of Bohr magneton (μ_B) from Zeeman Effect experiment.

5.	To verify the Rutherford's atomic model, α -scattering experiment, using Gold and Aluminum foil.
6.	To verify Heisenberg's uncertainty principle using single slit diffraction experiment.
7.	To obtain the speed of light using Fizeau focault method.
8.	To study radioactivity of artificial Ag^{108} and Ag^{110} and determination of their half-lives.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2			√									
CLO3			√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO2	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO3	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course No.	ME-1205
Course Title:	Mechanical Engineering
Course Credit:	2.5
Course Type:	Allied Engineering
Pre-requisite:	N/A

Rationale of the Course: Mechanical engineering plays a critical role in manufactured technologies from cars to airplanes to refrigerators. It is specialized and essential field in the daily activities of modern industrialized societies where need highly skilled manpower. Hence, this course is designed to cover introduction to thermal and mechanical system as well as heat engine, steam generator, refrigeration and air condition systems.

Course Objectives:

- To develop basic ideas in the study of steam generation, steam turbines and internal combustion engines.
- To understand various refrigeration and air conditioning cycles and enhance their ability in calculating cooling loads in commercial and residential systems.
- To demonstrate engineering competency in the field of fluid machines, turbines, pumps and different modes of heat transfers.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the basic concepts of energy sources, refrigeration, psychrometry and air conditions systems.	Cognitive (Understand)
CLO2	Explain the basic construction and working principles of heat engine, IC engine and steam generators.	Cognitive (Understand)
CLO3	Identify, formulate and solve the related mathematical problems with basic mechanical engineering.	Cognitive (Understand, Apply, Analyze)

CLO4	Design and realize a physical system, component to meet desired needs within realistic constraints such as economic, environmental, social, political, ethical, health and safety, manufacturability, and sustainability.	Cognitive (Apply, Analyze, Evaluate)
CLO5	Develop and design refrigerator and air conditioner units considering environmental issues.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction to the Sources of Heat Energy: Renewable and non-renewable sources and their potential, Classification of heat engines, working principle, applications, Fuel, Lubrication and cooling systems of I.C. engines, Introduction to internal combustion engines and their cycles, Study of SI and CI engines and gas turbines with their accessories.
2.	Introduction to Steam Generation: Boilers and their classification, Working principle of few common and modern boiler, Boiler mountings and accessories, Performance of boiler
3.	Thermodynamics: First law, Systems and surroundings, Conservation of energy, Different laws of thermodynamics processes, Energy transfer as heat for a control volume. Entropy and Second law: Reversibility and irreversibility; Definition and corollaries of second law of thermodynamics. Entropy: its transfer and change. Characteristics of some thermodynamic cycles: Analysis of different thermodynamic cycles, vapor power cycles, Representation of various cycles on PV & TS planes.
4.	Basic Concepts of Refrigeration Systems: Vapor compression refrigeration, Absorption refrigeration, cop, refrigerants and their classifications and properties.
5.	Basic Concepts of Psychrometry: Introduction, psychrometric terms, enthalpy, psychrometric chart, psychrometric process, sensible cooling and heating, humidification and dehumidification.
6.	Basic Concepts of Air Conditioning Systems: Introduction, objectives and major components of air conditioning systems; Humidity; Dew point.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√	√							
CLO4			√	√	√		√	√				√
CLO5			√	√	√		√					

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion - Online Resource	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Presentation - Discussion/Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination
CLO3	- Lecture - Presentation - Discussion/Group Discussion	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination
CLO4	- Lecture - Presentation - Discussion	- Tutorial/Assignment - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Discussion - Online Resource	- Quiz - Tutorial/Assignment/ - Semester end final examination - Participation

Learning Materials:

- | | |
|----------------------------|---|
| 1. R.S. Khurmi & J.K Gupta | : A Text Book of Thermal Engineering |
| 2. R. K Rajput | : Engineering Thermodynamics |
| 3. R. S Khurmi & J.K Gupta | : Mechanical Engineering. |
| 4. R. S Khurmi & J.K Gupta | : A Text Book of Refrigeration and Air Conditioning |

Course No.	ME-1206
Course Title:	Mechanical Engineering Sessional
Course Credit:	1.0
Course Type:	Allied Engineering
Co-requisite:	ME-1205

Rationale of the Course: This laboratory course is designed to be the experimental companion to ME 1209-Mechanical Engineering. The course is intended to enhance the students' understanding of important analytical principles developed in ME 1209 by engaging them in measuring and analyzing various parameters of engines, turbines, refrigerators, air conditioners and pumps.

Course Objectives:

- To identify machinery faults and provide corresponding solutions.
- To demonstrate the reason behind real time deviations (if any) from theoretical analysis in mechanical systems.
- To learn dealing with industrial tools and machineries.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Develop and conduct appropriate experimentation, analyze and interpret data, and use engineering judgment to draw conclusions.	Psychomotor(Imitation, Manipulation, Precision)
CLO2	Write technical reports using collected experiment data in mechanical systems.	Psychomotor (Manipulation, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	To study a Boiler mountings and accessories and calculate its efficiency
2.	To study a Petrol Engine and its accessories.
3.	To study a Diesel Engine accessories and calculate its efficiency
4.	To study a Gas Turbine accessories and calculate its efficiency
5.	To study the Refrigeration System.
6.	To study the Air Conditioning System.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√	√	√								
CLO2			√							√		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Participation
CLO2	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Lab performance ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	MATH-1207
Course Title:	Mathematics II
Course Credit:	2.5
Course Type:	Basic Science
Pre-requisite:	NA

Rationale of the Course: It is essential for the engineering/science students to develop strong analytic skills which cannot be achieved without learning calculus of one and multiple variables. This course is intended to build up strong base in calculus as well as to include the experience of applications of its contents in the relevant courses of science/engineering.

Course Objectives:

- To enhance the skill of basic mathematical topics required for learning ordinary and partial differentiation.
- To develop the basic concepts of differentiation, integration and ordinary differential equations with its wide scope.
- To apply the knowledge of problem solving using calculus in the related areas of science/engineering.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the basics of differential and integral calculus.	Cognitive (Understand)
CLO2	Calculate derivatives and apply them to compute error propagation.	Cognitive (Apply)
CLO3	Apply the differential equations to solve real world problems.	Cognitive (Apply, Analyze)
CLO4	Evaluate series expansion of functions and apply series concept in engineering problems.	Cognitive (Evaluate)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Differential Calculus: Function: Domain, Range, Inverse function and graphs functions, Limits, Continuity and properties of continuous functions.
2.	Ordinary Differentiation: Differentiability, Differentiation of real-valued functions, Successive differentiation, Leibnitz theorem.
3.	Expansion of functions: Rolle's Theorem, Lagrange's means value theorem, Taylor's and Maclaurin's formula, Maxima and Minima.
4.	Partial Differentiation: Tangent and Normal, Applications of the derivatives.
5.	Integral Calculus: Indefinite Integral: Method of substitution, Integration by parts, Integration of special trigonometric and rational functions, Fundamental theorem of calculus.
6.	Definite Integral: General properties of definite integral, Evaluation of definite integrals and reduction formula, Applications of integration.
7.	Multiple Integral: Determination of Length area and volume.
8.	Mathematical Functions: Gamma function and Beta function, Relation between beta and gamma function, Bessel's differential equation, Bessel's functions, Legendre differential equations and Legendre polynomials.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√										
CLO3		√	√									
CLO4			√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Das and Mukherjee : Differential Calculus
2. Das and Mukherjee : Integral Calculus
3. S. L. Ross : Differential Equations
4. Edwards : Differential Calculus
5. F. Ayres : Calculus
6. H. T. H. Piaggio : An Elementary Treatise on Differential

Course Code:	CSE-1209
Course Title:	Computer Organization and Architecture
Course Credit:	2.5
Course Type:	Allied Engineering
Pre-requisite (If any):	ICT - 1109

Rationale of the Course: The aim of the course is to develop a fundamental understanding of internal structuring, working and implementation of a computer system including the CPU control and data path, memory hierarchy, tradeoff between execution cycles, software hardware tradeoff.

Course Objectives:

- To understand the structure, function and characteristics of computer systems.
- To understand the design of the various functional units and components of computers.
- To identify the elements of modern instructions sets and their impact on processor design.
- To explain the function of each element of a memory hierarchy.
- To identify and compare different methods for computer I/O.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand basic computer systems, memory, I/O device, CPU and control unit,	Cognitive(Understand)
CLO2	Apply data transfer techniques in digital computer and the I/O interfaces.	Cognitive(Apply)
CLO3	Analyze the performance of memories and ALU.	Cognitive(Analyze)
CLO4	Apply micro-program to control the CPU, pipelined architectures, hazards and superscalar operations.	Cognitive(Apply)
CLO5	Evaluate tradeoffs between hardware and software techniques to achieve a performance goal	Cognitive(Evaluate)

Course Content:

Sl. No.	Course Content (as Summary)
1.	The Computer system: Computer interconnection structures, Components and functions, Bus interconnection.
2.	Memory: Computer memory system overview, Internal memory, External memory. CD Driver R/W technique, DVD, Cache memory and its mapping functions, L1 & L2 cache. Virtual memory management.
3.	I/O: Input/output devices, I/O modules, Programmed and interrupt driven I/O, DMA, Some applications.
4.	CPU: Integer and Floating point arithmetic, Signed operand multiplication. Fast multipliers, Instruction sets, Types of operand, Types of operations, Addressing modes, CPU structure and functions. Processor organization, Register organization, Instruction cycle, Instruction pipelining.
5.	Control Unit: Hardwired control, Microprogrammed Control, Microinstruction, Grouping of Signals, Microprogram Sequencing.
6.	Advanced Technologies: RISC and CISC machines, parallel processing, multi-programming.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3		√	√									
CLO4		√	√									
CLO5			√	√	√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion	▪ Quiz ▪ Tutorial/Assignment ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | |
|----------------------------------|--|
| 1. Stallings | : Computer Organization |
| 2. D.V. Hall | : Microprocessors and Interfacing |
| 3. Y. Yu, C, Marut. | : Assembly Language Programming and Organization of the IBM PC |
| 4. Cormen. et.al. 2nd Edition | : Algorithm |
| 5. Schaum's series, S. Lipschutz | : Data structures |
| 6. Robert, Sedgewick, Addison | : Algorithms in C, Wesley, 2000 |

Course Code:	HUM-1211
Course Title:	Economics
Course Credit:	2.5
Course Type:	Humanities
Pre-requisite:	NA

Rationale of the Course: Economics plays role in our everyday life. Studying economics enable students to understand past, future and current models and apply them to societies, governments, business and individuals. Hence, this course is designed to cover a broad background in economic theory and concepts and various model to play major role in investment by making decisions based on economic analysis and design considerations.

Course Objectives:

- Make fundamentally strong base for decision making skills by applying the concepts of economics.
- Educate the students on how to systematically evaluate the various cost elements of a typical manufactured product, an engineering project or service, with a view to determining the price offer.
- Prepare engineering students to analyze profit/revenue data and carry out make economic analysis in the decision making process to justify or reject alternatives/projects.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand major principles of economic analysis for decision making among alternative courses of action in engineering.	Cognitive(Understanding) Affective(Receiving, Valuing)
CLO2	Apply economic principles to prices and quantities in competitive supply and demand for goods and for money.	Cognitive(Apply) Affective(Receiving, Valuing)
CLO3	Solve economic problems involving comparison and selection of alternatives by using analytical techniques.	Cognitive(Analyze, Evaluate) Affective(Valuing)
CLO4	Analyze the performance and functioning of government, markets and institutions in the context of social and economic problems.	Cognitive(Analyze, Evaluate) Affective(Valuing)
CLO5	Use data to describe the relationships among variables in order to analyze economic issues.	Cognitive(Apply, Analyze, Evaluate) Affective(Valuing)
CLO6	Communicate economic thought and analysis in both written and oral context to varied audiences.	Cognitive(Analyze, Evaluate) Affective(Receiving, Responding, Valuing)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction to Economics: Definition and scope; subject matter of economics; central economic problems, microeconomics vs. Macroeconomics, different economic systems; Concept of engineering economics, Engineering efficiency and Economic efficiency.
2.	Demand and Supply: Concepts of demand; law of demand; demand schedule and demand curve; concepts of supply; supply schedule and supply curve; shift in demand and supply curves; equilibrium in the market; elasticity of demand and supply.
3.	Production, Cost, and Revenue: Meaning of production; factors of production; concepts of total, average and marginal costs; fixed and variable costs; total, average and marginal revenue.
4.	Theory of the Firm: Concept of market; perfect competition and monopoly; total, average, and marginal revenue of a firm under perfect competition and monopoly; equilibrium of a firm under perfect competition and monopoly.
5.	Basic Concepts of Macroeconomics: Aggregate demand; aggregate supply; aggregate economic behavior; macroeconomic objective.
6.	Measuring Macroeconomic Variables: Output growth; unemployment; price level; consumption function; price indexes; inflation; Philips curve; business cycle; circular flow of economy; two, three and four sectors economy.
7.	National Income Accounting and Determination: Concepts of GDP, GNP, GNI and national income; methods of national income accounting; Keynesian model of national income determination; the expenditure multiplier; the multiplier effects of determining equilibrium income.
8.	Budgets of Bangladesh: Revenue and capital budget; balanced and unbalanced budget; income and expenditure of the government

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1											√	
CLO2												√
CLO3											√	
CLO4											√	
CLO5												√
CLO6										√		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO6	- Lecture - Presentation - Demonstration	- Tutorial/Assignment/group Assignment - Semester end final examination - Participation

Learning Materials:

- Hubbard, G., Garnett, A. M., Lewis, P., & O'Brien, A. P. : Microeconomics
- Krugman, P., & Wells, R : Essentials of economics
- Mankiw, N. G : Principles of Economics
- Samuelson, P. A., & Nordhaus, W. D. : Economics
- Government of Bangladesh : Budget Document/Five Year Plans

Course Code:	NCC 1
Course Title:	Bangladesh Studies
Course Credit:	0
Course Type:	Non Credit Course
Pre-requisite:	Foreign Student Bangladeshi students with Madrasa background

Rationale of the Course: Bangladesh is the only nation on earth who had fought and died for our mother language Bangla. It has glorious history and culture. This course provides students with an introduction to acquire knowledge about the Bangladesh.

Course Objectives:

- To introduce the students to the Historical Background of the Bangladesh.
- To study about the Liberation, Location, Environment, Economics, Culture and so on relating to the Bangladesh.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Learn about the historical background of Bangladesh	Cognitive(Understanding)
CLO2	Know about the geography and inhabitants of Bangladesh	Cognitive(Understanding)
CLO3	Describe about the economics and politics of Bangladesh	Cognitive(Understanding)

Course Content:

Sl. No.	Course Content (as Summary)
1.	বাংলাদেশ অধ্যয়ঃ সামাজিক, রাজনৈতিক, অর্থনৈতিক ও সাংস্কৃতিক প্রেক্ষাপট; বায়ান্নর ভাষা আন্দোলন, চুয়ান্নর যুক্ত ফ্রন্ট নির্বাচন ও চৌষট্টির গণতান্ত্রিক আন্দোলন, ছেষট্টির ছয় দফা আন্দোলন, আগরতলা ষড়যন্ত্র মামলা ও বঙ্গবন্ধু, উনসত্তরের গণঅভ্যুত্থান, সত্তরের সাধারণ নির্বাচন, ৭ই মার্চের ভাষন ও স্বাধীনতার ঘোষণা এবং একাত্তরের মুক্তিযুদ্ধ ।
2.	বাংলাদেশের ভৌগলিক পরিচিতিঃ বাংলাদেশের ভৌগলিক অবস্থান ও সীমারেখা; নদ-নদী, পাহাড়-পর্বত, সমুদ্র উপকূল, বনাঞ্চল, জলবায়ু ও বৃষ্টিপাত, বন্যা ও অন্যান্য প্রাকৃতিক দুর্যোগ, প্রাকৃতিক সম্পদ (বনজ, প্রাণীজ, মৎস্য ও খনিজ) জনসংখ্যা, নগর ও নগরায়ন, পরিবেশ, ও পরিবেশ ব্যবস্থাপনা ।
3.	বাংলাদেশের ভূ-প্রকৃতি ও অধিবাসীঃ ভূমি(প্রকৃতি, প্রকারভেদ ও ব্যবহার), বাঙালি, উপজাতি ও অভিবাসী এবং তাদের নৃতাত্ত্বিক ও সাংস্কৃতিক পরিচিতি ।
4.	বাংলাদেশের অর্থনীতিঃ প্রধান উৎপাদন খাতসমূহ ও তাদের অর্থনৈতিক গুরুত্ব (কৃষি, শিল্প, বাণিজ্য, যোগাযোগ, ব্যাংকিং ও রাজস্ব); উৎপাদন ব্যবস্থা ও আর্থ-সামাজিক অবকাঠামো, জনসংখ্যা ও বেকার সমস্যা, জাতীয় অর্থনীতিতে নারী ও নিলবর্গীয়দের অবস্থান, জাতীয় অর্থনৈতিক পরিকল্পনা ও বাস্তবায়ন ।
5.	বাংলাদেশের রাজনীতিঃ বাংলাদেশের সংবিধান ও তার বিবর্তন, বাহাত্তরের সংবিধানের সন্নিবেশিত জাতীয় মূলনীতি ও মৌলিক অধিকার সমূহ, সংবিধানের সংশোধনী ও অধ্যাদেশ সমূহ, শাসন ও বিচারব্যবস্থা, রাজনৈতিক দল ও তাদের অনুসৃত মতবাদসমূহ ।

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1							√					
CLO2							√					
CLO3											√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation

Learning Materials:

১।	এম.আর. আখতার মুকুল	:	আমি বিজয় দেখেছি ।
২।	এম. আ. আখতার মুকুল	:	বাহান্নর জবানবন্দী ।
৩।	এম. আ. আখতার মুকুল	:	বাহান্নর ভাষা আন্দোলন ।
৪।	মেজর রফিকুল ইসলাম	:	একাত্তরের মুক্তিযুদ্ধ ।
৫।	ডঃ মোহাম্মদ হান্নান	:	হাজার বছরের ইতিহাস ।
৬।	ডঃ মোহাম্মদ হান্নান	:	বাংলাদেশের মুক্তিযুদ্ধের ইতিহাস ।
৭।	মোস্তফা কামাল	:	বাঙালী বাংলাদেশ ও বঙ্গবন্ধু ।
৮।	শৈলেন্দ্রকুমার বন্দোপাধ্যায়	:	জিন্মা/পাকিস্তান ।

৯।	ইমন সরকার	:	স্বাধীনতার ঘোষক কে?
১০।	আনু মোহাম্মদ	:	ধর্মরাত্রি ও গণতান্ত্রিক আন্দোলন ।
১১।	এ্যান্টনী ম্যাসকারেনহাস	:	বাংলাদেশ রক্তের খন ।
১২।	মোঃ সিরাজুল ইসলাম	:	বাংলাদেশের ইতিহাসঃ ৩য় খন্ড ।
১৩।	নীহাররঞ্জন রায়	:	বাঙালীর ইতিহাস : আদিপর্ব
১৪।	আব্দুল্লাহ ফারুক	:	বাংলাদেশের অর্থনৈতিক ইতিহাস
১৫।	Dr. AMA Rahim	:	Current Issues of Bangladesh Economy.

Course Code:	NCC 2
Course Title:	Islamic Studies
Course Credit:	0
Course Type:	Non Credit Course
Pre-requisite:	Only for Bangladeshi Muslim students with School, College or both background

Rationale of the Course: Islam is the complete code of life bestowed by Almighty Allah for mankind. It covers all aspects of human rights. This course provides students with an introduction to acquire knowledge about the main aspects of Islam in terms of the Holy Quran and Hadith.

Course Objectives:

- To understand main aspects of Islam in terms of rules provided by the Holy Quran and Hadith.
- To study about the lifestyle of Prophet Mohammad (SW) and Islamic Culture, Islamic Economics and Islamic State and so on relating to the Islam.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand main concepts of Islam, its history, holy books, Islamic Culture, Islamic Economics and Islamic State.	Cognitive(Understanding)
CLO2	Apply learnings of Islam in daily life.	Cognitive(Apply)

Course Content:

Sl. No.	Course Content (as Summary)
1.	ইসলাম ও অন্যান্য প্রসিদ্ধ ধর্মের সংক্ষিপ্ত পরিচিতি ।
2.	তাওহীদ, ঈমান ও ইসলাম; শিরক এবং ইহার প্রকার ভেদ; বিদ'আত ও ইহার প্রকারভেদ ।
3.	ইসলামী শরীয়াতের উৎস সমূহ (আল-কুরআন, আল-হাদীস, ইজমা ও কিয়াস) ।
4.	ইসলামে হালাল ও হারামের বিবরণ এবং অপরাধ দমনের বিধান ।
5.	ইবাদতঃ- (ক) ফরয সালাত, যাকাত, সওম ও হজ্জ (খ) ওয়াজিব সমূহ; (গ) সুন্নত সমূহ; (ঘ) মুস্তাহাব সমূহ ।
6.	ইসলামী আচার আচরণঃ- (ক)মু'আসালাত-ইসলামের দৃষ্টিতে পারস্পারিক লেনদেন, শ্রমিক ও মালিক সম্পর্ক, (খ) মু'আবারাত খনদাতা ও গ্রহীতা, বিক্রোতা, মাতা-পিতা, ছোটবড়, ছাত্র-শিক্ষক এর মধ্যে এবং প্রতিবেশীর সহিত সম্পর্ক; (গ) ইসলামে ভাতৃত্ব ও জনকল্যানমূলক কাজ ।
7.	ইসলামী আখলাকঃ-(ক) তাকওয়া, সব্র, আদল, ইহসান, সিদ্ক, লজ্জা, শিষ্টাচার, ধুমপান, মাদকদ্রব্যপান, পর্দা, আমানত ও খিয়ানত ।
8.	অনুবাদঃ-(ক) সূরা আল মু'মিনুন এর ১ম রুকু; (খ) সূরা ফীল থেকে সূরা নাস; (গ) বিশটি হাদীস (ইবি কর্তৃক ১৯৯৫ও৯৬ সালের প্রকাশিত সিলেবাস উলে-খ মোতাবেক)
9.	ইসলামী শিক্ষা ও সংস্কৃতিঃ স্বরূপবৈশিষ্ট্য গুরুত্ব ও প্রভাব । অর্থনীতি :- দারিদ্র বিমোচনে পুজিবাদ, সমাজবাদ ও ইসলামের ভূমিকা । ইসলামী রাষ্ট্রঃ সংজ্ঞা, স্বরূপ, বৈদেশিক নীতি ও অধিকার । ইসলামে বিচার ব্যবস্থা :ঃ ইসলামী আইনের উৎস সমূহ, আধুনিক ও ইসলামী বিচার ব্যবস্থা, বিচারকের যোগ্যতা ও গুনাবলী, ইসলামী বিচার পদ্ধতি, সাম্যের ভিত্তিতে ন্যায় বিচার প্রতিষ্ঠা, মহানবী (সাঃ) ও খলিফাগণের বিচার ব্যবস্থা । ইসলামে সামরিক ব্যবস্থা :ঃ জিহাদের সংজ্ঞা, প্রকারভেদ ও গুরুত্ব; ইসলামে সমরনীতি, আধুনিক সভ্যতায় যুদ্ধ ।
10.	ইসলাম প্রচারঃ- ইসলামে তাবলীগের প্রয়োজনীয়তা ও গুরুত্ব; ইসলাম প্রচার এর পদ্ধতি ও প্রচারের গুনাবলী; বাংলাদেশে ইসলাম প্রচার এন.জি.ও. এবং খৃষ্টান মিশনারী তৎপরতা; ইসলাম প্রচারের মাধ্যম সমূহের ভূমিকা ।
11.	মানবাধিকারঃ ইসলাম ও পাশ্চাত্য ।

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1							√	√				√
CLO2								√	√	√	√	√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Participation
CLO2	- Lecture - Discussion - Demonstration	- Tutorial/Assignment - Semester end final examination - Participation

Learning Materials:

1. Islamic Studies : Prof. Dr. Muhd. Abdur Rahman Anwari et. Al.
2. The Holy Quran :
3. Hadith :

Course Code:	EEE-1200
Course Title:	Viva-voce
Course Credit:	1.0
Course Type:	Core Course
Pre-requisite:	Knowledge of All 1 st and 2 nd semester courses

Rationale of the Course: Communication plays very vital role in aspect of life. Viva-voce is such an opportunity where a learner is judged how he/she can communicate with others from his/her gained knowledge. The objective of this course is to verify the students how well they can express their knowledge orally.

Course Objective:

- To assess the students ability to communicate with another person.
- To supplement the information obtained through other evaluation techniques.
- To identify and analyze the students presence of mind.
- To evaluate the students spontaneity and mannerism.

Course Learning Outcomes (CLOs): By the end of the all courses in First Year, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Convince the audience orally about their knowledge from the courses in First Year	Affective(Receiving, Responding, Valuing)

Course Content:

Sl. No.	Course Contents
1.	1st Year Course Curriculum

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√								√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation/Discussion	- Participation; - Presentation

Learning Materials: Knowledge from 1st Year Curriculum

B.Sc. Engineering Second Year, First Semester Examination 2022

Course Code:	EEE-2101
Course Title:	Electronics I
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	N/A

Rationale of the Course: Electronics have wide range of applications in science and engineering. So, knowledge and understanding of electronics is compulsory for engineering student. This course is introduced to facilitate learners the basic concepts and principles of operation of various electronics devices related to their use and working in electronic systems and applications.

Course Objectives:

- To familiarize the students with construction and electrical characteristics of semiconductor diodes, transistors and FETs.
- To explain their use and behavior in fundamental application like rectification and amplification.
- To find the circuit parameters and use them to solve real world problems.

Course Learning Outcomes (CLOs): at the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the semiconductors, diode, BJT and FET.	Cognitive (Understand)
CLO2	Explain the operation of diode, BJT and FET.	Cognitive (Understand)
CLO3	Identify particular parameters/electronic component in a circuit like diode, BJT and FET.	Cognitive (Analyze)
CLO4	Determine and choose an electronic component necessary for using in a circuit.	Cognitive (Analyze and Evaluate)
CLO5	Design rectifiers, filter, clippers and clamper circuit using diodes.	Cognitive (Analyze, Evaluate and Create)
CLO6	Design amplifier circuit using BJTs.	Cognitive (Analyze, Evaluate and Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Semiconductor: Semiconductor, Bonds in semiconductor, commonly used semiconductors, Energy band description of semiconductors, Types of semiconductors.
2.	Diode: Junction Diode, P-N Junction, Properties of P-N junction, Applying voltage of P-N junction, P-N junction diode.
3.	Rectifiers and Filters: Rectifiers, Half-wave rectifier, Full-wave rectifier, Bridge rectifier, Rectifier design, Voltage regulation, Use of Zener diode as voltage regulator, Voltage multipliers, Clippers and Clamper Design, Filters, Shunt capacitor filter, Series inductor filter, L-C Filter, The C-L-C Filter, filter design.
4.	The Bipolar Junction Transistor (BJT): Introduction, Transistor construction, Transistor configuration CE, CB, CC Transistor static characteristics.
5.	Transistor Bias Stabilization: Cut off and Saturation points, DC load line, AC load line, Factors contributing to thermal instability, Effect of temperature, Stability factor, Stabilization techniques, Transistor biasing, Fixed bias, Emitter stabilized bias, Voltage divider bias, DC-bias with voltage feedback, Bias compensation, Basic design of a CE transistor amplifier; Amplifying action.
6.	Small Signal Low-Frequency BJT Models: Transistor equivalent circuits, Hybrid parameters, Analysis of CE, CB and CC amplifiers using h-Parameters, CE amplifier with emitter resistor.
7.	Field-Effect Transistors (FETs): FET, Junction FET, Construction of JFET, Static characteristics of FET, Parameters of FET, DC-biasing of a JFET, FET applications, MOSFET, Construction and characteristics of MOSFET, Depletion-type of MOSFET, Enhancement-type MOSFET.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2	√	√										
CLO3		√	√									
CLO4		√	√									
CLO5				√	√							√
CLO6				√	√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Participation ▪ Semester end final examination
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Audio/Video ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Participation ▪ Semester end final examination
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Tutorial ▪ Assignment/Group Assignment ▪ Participation ▪ Semester end final examination
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Audio/Video ▪ Online Resources	▪ Tutorial ▪ Assignment/Group Assignment ▪ Participation ▪ Semester end final examination
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Audio/Video ▪ Online Resources	▪ Tutorial ▪ Assignment/group Assignment ▪ Mini Project ▪ Participation ▪ Semester end final examination
CLO6	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Audio/Video ▪ Online Resources	▪ Tutorial ▪ Assignment/group Assignment ▪ Mini Project ▪ Participation ▪ Semester end final examination

Learning Materials:

1. Boylestad and Nashelsky : Electronic Devices & circuit Theory, Prentice-Hall of India Pub. 1996
2. Millman & Tub : Principles of Electronics
3. B.L Theraza : A textbook of Electrical Technology (Vol. IV)
4. Malvino : Principles of Electronics
5. J. Brophy : Basic Electronics for Scientists. McGraw-Hill Pub. 1983
6. Bell : Electronic Devices Circuit
7. Millman and Halkias : Electronic Devices Circuit

Course Code:	EEE-2102
Course Title:	Electronics I Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-2101

Rationale of the Course: This course provides the student hands on experience of verifying the theorems and laws they are taught in EEE-2101 course.

Course Objectives:

- To introduce the students with the electronic devices like diodes and transistors along with their use in the circuit.
- To explain the students how to choose a particular diode or transistor using its electrical parameters.
- To help students gain hands-on experience of using such electronic devices in the rectifier and amplifier circuits.

Course Learning Outcomes (CLOs): at the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Identify diodes and transistors.	Psychomotor(Imitation, Manipulation)
CLO2	Choose diodes and transistors for a particular use in circuit.	Psychomotor(Manipulation, Precision)
CLO3	Design and construct low power rectifier and amplifier circuits using diodes and transistors.	Psychomotor(Manipulation, Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Familiarization with basic electronic devices and components
2.	V-I characteristic of a rectifier diode.
3.	Study and demonstration of Zener diode as a voltage regulator
4.	Study and demonstration of regulated dc power supply
5.	To study the waveform of a half-wave rectifier circuit and find its ripple factor without and with filter.
6.	Study and demonstration of clipper circuit
7.	Study and demonstration of clamper circuit
8.	To study the V-I characteristics of a Bipolar Junction Transistor (BJT).
9.	Study and demonstration of common collector amplifier circuit
10.	Study and demonstration of common base amplifier circuit
11.	Study and demonstration of common emitter amplifier circuit
12.	To study the input/output characteristics of a Field Effect Transistor (FET).

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√										
CLO2		√	√	√								
CLO3			√	√	√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Participation
CLO2	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Participation
CLO3	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Lab performance ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-2103
Course Title:	Electrical Machines I
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	N/A

Rationale of The Course: Modern world revolution would have not been possible without invention and extensive uses of electrical machineries. Basic knowledge of electrical machines are important requirements for engineering students. This course is intended to provide basic conception of electrical machines.

Course Objectives:

- To understand the basic principle and construction of a transformer.
- To accumulate the basic ideas about the efficiency and regulation of a transformer
- To test and identify faults in transformer and induction motor

Course Learning Outcomes (CLOs): By the end of the course student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the basic principle, construction and equivalent circuit of a transformer and inductor.	Cognitive (Understand)
CLO2	Identify the physical components of transformer and inductor.	Cognitive (Understand)
CLO3	Determine and interpret efficiency and regulation of a single phase transformer and inductor.	Cognitive (Apply, Analyze)
CLO4	Conduct open circuit test and short circuit test of a transformer	Cognitive (Analyze, Evaluate)
CLO5	Test and identify faults in single phase and three phase induction motor	Cognitive (Apply, Analyze, Evaluate)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Single Phase Transformer: Principles, Types, Equivalent circuits, Performance and testing, Regulation, Losses and efficiency, Parallel operation, Auto-transformer, Instrument transformers.
2.	Poly Phase Transformer: Poly phase transformer construction, Poly phase transformer connections, Harmonics in polyphase transformer, transformer cooling.
3.	Single Phase Induction Motor: Principle, Construction and types, Performance, Double revolving field theory, Cross field theory, Equivalent circuits.
4.	Polyphase Induction Motor: Principle of operation, Constructional details, Classifications, Equivalent circuits, starting torque and maximum torque, Speed-torque relations, Losses and efficiency, Circle diagram, Starters, Methods of speed control, Methods of braking and plugging, Induction generator.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√								
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion	▪ Quiz ▪ Tutorial/Assignment ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion/Demonstration	▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation

CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial ▪ Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial ▪ Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Stephen J. Chapman : Electric Machinery and Power System Fundamentals
2. A Fitzgerald : Electric Machinery
3. A. Puchstein, T E Loyd and AG : Alternating Current Machines
Conard
4. J. Rosenblatt & M. H. Friedman : Direct and Alternating Current Machinery
5. Charles I Hubert : Electric Machines: Theory, Operating Applications, and Controls
6. B.L. Thereja, A.K. Thereja : A Textbook of Electrical Technology, Volume II
7. Er. R.K. Rajput : Electrical Machines in S.I. UNITS

Course Code:	EEE-2104
Course Title:	Electrical Machines I Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-2103

Rationale of the Course: This course provides the student a hand on experience of testing the generators, motors and transformers and its measure their efficiency.

Course Objectives:

- Getting idea about the faults of generator, motor and transformer and induction motor
- To develop skills to test the generator, motor, transformer and induction motor to determine their efficiency

Course Learning Outcomes (CLOs): At the end of the course student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Identify faults in single phase and three phase transformer, generator and motor.	Psychomotor(Imitation, Manipulation)
CLO2	Conduct open circuit test and short circuit test of a transformer	Psychomotor(Manipulation, Precision)
CLO3	Determine the line voltage & phase voltage and line current & phase current relation in primary and secondary winding of DC generator and DC motor.	Psychomotor(Manipulation, Precision, Articulation)
CLO4	Measure circuit parameter of single phase and three phase induction motor and determine the efficiency	Psychomotor(Manipulation, Precision)

Course Content:

Sl. No.	List of the Experiments
1.	To study of a single phase transformer To perform (a) open circuit tests (b) short circuit tests (c) carry out load test and (d) voltage regulation and efficiency
2.	Construction of three-phase transformer using three single-phase transformer and observation of line voltage & phase voltage and line current & phase current relation in primary and secondary winding with balance and unbalanced loading.
3.	Open circuit test of a 3- ϕ induction motor
4.	No-load Test of a 3-phase Induction Motor
5.	No load test blocked rotor test of three phase induction motor.
6.	Measurement of circuit parameters of single phase induction motor and determination of efficiency.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√	√									
CLO2			√	√								
CLO3			√	√								
CLO4				√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO2	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO3	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO4	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	STAT-2105
Course Title:	Statistics for Engineers
Course Credit:	2.5
Course Type:	Basic Science
Pre-requisite:	N/A

Rationale of the Course: Statistics is a crucial process behind how we make discoveries science, make decisions based on data and make predictions. Probability and statistics are the critical tools for robustness analysis, measurement system error analysis, test data analysis, probabilistic risk assessment and many other fields in science and engineering world. Hence, this course is designed to cover a broad background in statistical concepts and probability for dealing with variability, uncertainty and complexity in engineering and use a variety of software packages in solving practical problems in reference to electrical engineering.

Course Objectives:

- Practice statistical methodology and tools in the engineering problem-solving process
- Develop probabilistic and statistical models for some applications, and apply statistical methods to a range of problems in science and engineering
- Perform the basic tests.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to–

CLOs	Descriptions	Learning Domain with Level
CLO1	Develop conceptual understanding of some major ideas in statistics and probability theory including moments, sample, event, probability, random variables and provide the elementary background for regression analysis, correlation coefficients	Cognitive (Understand)
CLO2	Explain and apply major distributions of random variables to various engineering problem.	Cognitive (Understand and Apply)
CLO3	Analyze the statistical data using numerical and graphical techniques.	Cognitive (Apply and Analyze)
CLO4	Develop probabilistic and statistical models for some applications, and apply statistical methods to a range of problems in science and engineering.	Cognitive (Apply, Analyze and Evaluate)
CLO5	Test different hypothesis.	Cognitive (Analyze and Evaluate)
CLO6	Use and compare a wide variety of software packages to perform statistical analysis and inference.	Cognitive (Apply, Analyze and Evaluate)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Analysis of Statistical Data: Collecting and presenting data, Location, Dispersion and their measures, Skewness, Kurtosis and their measures, Moment and Cumulants and Practical examples.
2.	Probability: Concept of probability, Sample Space, Events union and Intersection of Events. Probability of events, Loss of probability, Conditional probabilities. Bay's Theorem, Chebysec's Inequality and Practical examples
3.	Probability Distributions: Random Variables, The Binomial Distribution, The Hyper geometric distribution. The Mean and Variance of a Probability Distributions, Chebyshev's theorem, the Poisson approximation to the Binomial Distribution, Poisson Processes, The Geometric Distribution, The Multinomial Distribution. The Weibull Distribution, Joint Probability Densities.
4.	Linear Regression and Correlation: Correlation, Rank correlation. Partial and Multiple Correlations Linear Regression for two Variables, Principle of Least Squares Method, Lines of best fit, Residual Analysis and examples, Multiple regressions.
5.	Test of Significance: Basic ideas of Null hypothesis, Alternative hypothesis, Type-I error Type-II error level of significance Degree of freedom, Rejection region and Acceptance region. Test of Single mean, Single variance, two sample means and Variances. Test for 2×2 contingency tables, Independence test and practical examples. Application in quality control, failure pattern, depreciation calculation.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√	√							
CLO5				√	√							
CLO6			√	√	√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO6	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | | |
|----------------------|---|--|
| 1. P.G. Hoel | : | Introductory Statistics |
| 2. S.G. Gupta | : | Fundamentals of Statistics |
| 3. A. J. B. Anderson | : | Interpreting Data |
| 4. H. Cramer | : | The Elements of Probability Theory |
| 5. D. V. Lindley | : | Introduction to Probability and Statistics |
| 6. S. Lipschutz | : | Probability |

Course Code:	MATH-2107
Course Title:	Mathematics III
Course Credit:	2.5
Course Type:	Basic Science
Pre-requisite:	N/A

Rationale of the Course: Strong analytical skills and problem solving are basic requirements for engineering students which can be achieved by mathematical knowledge. This course is intended to broaden strong base in vector analysis, co-ordinate geometry and to build up a strong base in ordinary differential equations as well as to include the experience of applications of its contents in the relevant courses of science and engineering.

Course Objectives:

- To enhance the understanding and problem solving skills on related/necessary basic mathematical topics.
- To develop the basic concepts of vectors, co-ordinate geometry and ordinary differential equations in their wide scopes.
- To build up problem solving experience in each topic along with the addition of application familiarity in the related areas of science and engineering.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the fundamental aspects of vectors, co-ordinate geometry and ordinary differential equations.	Cognitive (Understand)
CLO2	Apply the concepts of vector analysis in electromagnetic field equations and the important quantities associated with vector fields.	Cognitive (Apply)
CLO3	Apply geometric properties and relationships to solve multistep problems in two and three dimensions.	Cognitive (Apply)
CLO4	Demonstrate differential equations to model natural phenomena, engineering systems and many other situations in connection with numerous problems that are encountered in the various branches of science and engineering.	Cognitive (Apply)
CLO5	Analyze real world scenarios to recognize when ordinary differential equations are appropriate in order to solve problems using multiple approaches.	Cognitive (Analyze)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Vector Analysis: Definitions and examples of vectors, Theorems on vector and scalar product of vectors, Vector Differentiation, Differential operators: Gradient, Divergence and Curl of vector function, Vector integration: line, surface and volume integrals; Integral theorem: Gauss's divergence theorem, Stoke's theorem, Green's theorem and Related integral theorem.
2.	Co-ordinate Geometry of Two Dimensions: change of axes, system of circle, straight line and ellipse.
3.	Co-ordinate Geometry of Three Dimensions: Straight line, plane and sphere.
4.	First-order Differential Equations: Classification of Differential Equation, Degree and Order of Ordinary Differential Equation, Initial Value Problem, Boundary Value Problem, First-order ordinary differential equations, exact differential equations and integrating factors, Separable equations and equations reducible to this form, Linear equations and Bernolli equations.
5.	Higher Order Differential Equations: Explicit solution of Higher-order linear differential equations, Homogeneous linear equation with constant coefficients, Method of undetermined coefficients, Variation of parameters, Cauchy-Euler equation.
6.	Series Solutions: Series solutions of linear differential equations, Power series solutions about an ordinary point, Solutions about singular points, the method of Frobenious.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3		√										
CLO4		√	√									
CLO5			√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Problem solution - Discussion - Online Resources	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Problem solution - Discussion - Online Resources	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Problem solution - Discussion - Online Resources	- Quiz - Tutorial - Assignment/group Assignment - Semester end final examination

CLO4	▪ Lecture ▪ Problem solution ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial ▪ Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | |
|---------------------------|---|
| 1. M. R Spiegel | : Vector Analysis and Tensor Calculus |
| 2. S. L. Ross | : Differential Equations |
| 3. Rahman & Bhattacharjee | : Analytical Geometry of Conic Sections |
| 4. M. A. Sattar | : Vector Analysis |

Course Code:	CSE-2109
Course Title:	Data Structures and Algorithm
Course Credit:	2.5
Course Type:	Allied Engineering
Pre-requisite:	CSE-1110

Rationale of the Course: The main purpose of this course is to provide the students with solid foundations in the basic concepts of data structures and algorithms. This course offers the students a mixture of theoretical knowledge and practical experience by covering the topics analysis of algorithms, arrays, trees, binary search trees, multi-way search trees, dictionaries, hash tables, graphs, graph traversals, graph algorithms, sorting.

Course Objectives:

- To facilitate necessary knowledge about data structures such as linear lists, stacks, queues, hash tables, binary trees, heaps, binary search trees, and graphs and various data algorithms such as searching, sorting, hashing.
- To choose the appropriate data structure and algorithm design method for a specified application.
- To solve problems using data structures such as linear lists, stacks, queues, hash tables, binary trees, heaps, binary search trees, and graphs and writing programs for these solutions.
- To efficiently implement the different data structures and solutions for specific problems.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understanding of data structures such as linear lists, stacks, queues, hash tables, binary trees, heaps, binary search trees, and graphs and various data algorithms such as searching, sorting, hashing.	Cognitive(Understand)
CLO2	Identify the usage of various data structures including arrays, lists, stacks, queues, trees and graphs and the usage of various data algorithms such as searching, sorting, hashing.	Cognitive(Understand)
CLO3	Analyze the performance characteristics of algorithms using mathematical and measurement techniques.	Cognitive(Analyze)
CLO4	Interpret and summarize the advantages and disadvantages of various data structure implementations.	Cognitive(Evaluate)
CLO5	Design and apply appropriate data structures for solving computing problems.	Cognitive(Apply)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction: Basic Terminology, Elementary data organization, Data structures, Algorithms, Complexity of Algorithms.
2.	Arrays, Records and Pointers: Introduction, Linear arrays, Representing of Linear Arrays in Memory, Traversing Linear Arrays, Inserting & Deleting, Sorting, Searching, Multidimensional Arrays, Pointers Array, Record Structures, Parallel Arrays.
3.	Linked Lists: Introduction, Linked List, Representation of Linked Lists in Memory, Traversing a Linked List, Searching a Linked List, Memory Allocation, Insertion into a Linked List, Deletion from a Linked List, Header Linked Lists, Two way Lists.
4.	Stacks, Queues & Recursion: Introduction, Stacks, Array Representation of stacks, Arithmetic Expressions, Application of stacks, Recursion, Implementation of Recursive Procedures by Stacks, Queues, Deques, Polarity Queues.
5.	Trees: Introduction, Binary trees, Representing Binary Trees in Memory, Traversing Binary Trees, Traversal Algorithms using Stacks, Binary Search Trees, HAECE, Path Lengths, General Trees.
6.	Graphs and Their Application: Introduction, Graph Theory Terminology, Sequential Representation of Graphs, Warshall's Algorithm, Linked Representation of a graph, Operations on Graphs.
7.	String Processing, Sorting and Searching: Introduction, Basic Terminology, Storing Strings, String Operations, Word Processing, Pattern matching Algorithms, Sorting, Insertion Sort, Selection Sort, Merging, Merge-Sort, Radix-Sort, Searching and Data modification, Hashing.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√								
CLO5			√		√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Journal Paper	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | |
|------------------------|--------------------------------------|
| 1. Robert L. Kruse. | : Data Structures and Program Design |
| 2. Taunenbaum, Langsam | : Data Structure using C |
| 3. Seymour Lipschutz | : Data Structures |
| 4. Reingold and Hansen | : Data Structure |

Course Code:	CSE-2110
Course Title:	Data Structures and Algorithm Sessional
Course Credit:	1.0
Course Type:	Allied Engineering
Co-requisite:	CSE 2109

Rationale of the Course: The course introduces the basics of computational complexity analysis and various algorithm design paradigms. The goal is to provide students with solid foundations to deal with a wide variety of computational problems, and to provide a thorough knowledge of the most common algorithms and data structures.

Course Objectives:

- Demonstrate a familiarity with major algorithms and data structures.
- Apply important algorithmic design paradigms and methods of analysis.
- Analyze the performance of algorithms.
- Prove rigorous correctness for algorithms.
- Synthesize efficient algorithms in common engineering design situations.

Course Learning Outcomes (CLOs): By the end of the course student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Identify the appropriate data structures.	Psychomotor(Imitation, Manipulation)
CLO2	Apply the appropriate algorithms in different situations	Psychomotor (Manipulation and Precision)
CLO3	Test efficiency of algorithms	Psychomotor(Manipulation and Precision)
CLO4	Evaluate the correctness of algorithms.	Psychomotor(Manipulation, Precision, Articulation)
CLO5	Develop improved communicative and collaborative skills.	Affective(Receiving, Responding)

Course Content:

Sl. No.	List of the Experiments
1.	To apply sorting algorithms on array, lists, and trees.
2.	To employ searching algorithms on various types of data structures.
3.	To test insertion and deletion algorithms on different types of data structures.
4.	To compare the complexity of algorithms on different types of data structures.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√								
CLO5										√		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO2	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO3	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Lab performance ▪ Semester end final examination ▪ Participation

CLO4	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO5	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	IPE-2111
Course Title:	Industrial Management and Production Engineering
Course Credit:	2.5
Course Type:	Allied Engineering
Pre-requisite:	N/A

Rationale of the Course: The knowledge of this course is required for all engineering graduates who wish to choose industry as their career. This course is designed to develop understanding of various functions of management, human resources, marketing, finance along with providing knowledge about safety and security, industrial laws and management in different areas. Knowledge of these issues will help engineers to be successful in their career pursued in manufacturing sector.

Course Objectives:

- To provide an opportunity for students to understand management skill in industry.
- To analyze the financial accounting data for the management

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the role and functions of management, marketing, marketing mix, market segmentation, product life cycle, pricing, advertising etc.	Cognitive (Understand)
CLO2	Recognize and apply the steps of decision making process,	Cognitive (Understand, Apply)
CLO3	Outline meaning and functions of HRM, Job analysis, recruitment, selection and HRM related laws,	Cognitive (Apply, Analyze)
CLO4	Describe production facilities - location and layout, plant maintenance, industrial safety, production planning and control; Production Process - Job, Batch, Repetitive and Continuous,	Cognitive (Apply, Evaluation)
CLO5	State the meaning of inventory and inventory management, reasons for holding inventory, steps in material procurement,	Cognitive (Apply, Analyze, Evaluate)
CLO6	Apply EOQ and ABC analysis to minimize costs and control inventory,	Cognitive (Apply, Analyze)
CLO7	Identify sources of finance, financing organizations, type of costs.	Cognitive (Apply, Analyze)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Principles of Management: Definition, functions, roles and skills of management; Business organizations, Taylor's scientific management principles and Fayol's 14 principles of management; Planning, Organizing, Motivating, controlling and Coordination.
2.	Human Resource Management (HRM): Meaning, Importance and Functions of HRM; HR planning, Job analysis, Job description, Job specification, Recruitment, Selection, Employee training and HR related laws: The Factories Act 1947, The Boiler's Act 1923 and The Industrial Relations Ordinance 1969.
3.	Plant Management: Facilities - Location and Layout; Plant maintenance - Types and objectives; Industrial safety; Causes and costs of accidents; Production planning and control; Production Process - Job, Batch, Repetitive and Continuous.

4.	Principles of Marketing: Definition and Concepts of marketing, marketing mix, market segmentation, new product development, product life cycle, pricing, supply chain network, advertising, and sales promotion.
5.	Inventory Management: Inventory, Inventory Management, Reasons for holding inventory, Steps in material management and procurement, EOQ, and ABC analysis.
6.	Financial Management: Sources of finance, financing organization, Cost concept and classification, Break even analysis, budgets and budgetary control, make or buy analysis, income statement, balance sheet, and ratio analysis.
7.	Strategic Management: Meaning of Strategic Management, Key Concepts: Mission, Goals and Objectives, Analyzing the External Environment of the Firm, Business Level Strategy, and Corporate Level Strategy.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√							√		√	
CLO2		√	√									√
CLO3		√	√						√		√	√
CLO4			√		√		√					
CLO5									√	√		
CLO6					√							
CLO7					√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resource	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO6	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO7	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | |
|-------------------------------|---|
| 1. Koontz, O Daniell | : Principles of management |
| 2. Kotler P., Stauton William | : Principles of marketing management |
| 3. S.C. Kuchal | : Financial Management |
| 4. T.R. Sharma | : Industrial organization & engineering Economics |

B.Sc. Engineering Second Year, Second Semester Examination 2022

Course Code:	EEE-2201
Course Title:	Electronics II
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-2101

Rationale of the Course: This course comes under the core engineering course and will enable the students to comprehend the theory, concepts, characteristics and working principles of electronic circuits and software and their applications. The knowledge of various devices acquired by the students will help them to design, test, troubleshoot and repair electronic circuits.

Course Objectives:

- To understand and implement the advanced electronic circuits and software.
- To explain their use and behavior in fundamental applications.
- To design and plan to solve the problems of electronic circuits.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Explain the frequency effect on an amplifier, feedback and Oscillator Circuits, Power amplifiers	Cognitive (Understand)
CLO2	Understand the circuit analysis using MATLAB and programing concepts for automated testing of electronic device.	Cognitive (Understand)
CLO3	Investigate the amplifier circuit behavior with frequency, Oscillator Circuit for feedbacks, power gain of power amplifiers, and diode, transistor characteristics by MATLAB	Cognitive (Apply, Analyze)
CLO4	Justify the frequency effect on an amplifier design, feedback effect on oscillator circuit design and the application of power amplifier class	Cognitive (Analyze, Evaluate)
CLO5	Design the Oscillator and power amplifier circuits	Cognitive (Apply, Analyze)
CLO6	Plan to solve the identified problem of electronic devices	Cognitive (Evaluate)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Amplifier Frequency Response of BJT and FET: Basic Concepts, The Decibel, Low-Frequency Amplifier Response, High-Frequency Amplifier Response, Total Amplifier Frequency Response, Frequency Response of Multistage Amplifiers, Frequency Response Measurements, Application Activity
2.	Feedback and Oscillator Circuits: Feedback Concepts, Feedback Connection Types, Practical Feedback Circuits, Feedback Amplifier—Phase and Frequency Considerations, Oscillator Operation, The Nyquist criterion for the stability of a feedback amplifier, Phase-Shift Oscillator, Wien Bridge Oscillator, Tuned Oscillator Circuit, Crystal Oscillator, Unijunction Oscillator
3.	Power Amplifiers: The class A, class, AB and B amplifiers, tuned Class B and C amplifiers, Neutralization, Class B and C Push-pull amplifiers, Design of push- pull amplifiers, Complementary and symmetry amplifiers.
4.	Electronics And Circuit Analysis Using MATLAB: MATLAB Fundamentals, Plotting Commands, Control Statements, Dc Analysis, Diode Analysis, Transistor Analysis
5.	Basic programing concepts for automated testing of electronic devices: Programming Basics, Automated Testing Basics, The Simple Sequential Program, Conditional Execution, Program Loops, Branching and Subroutines, Troubleshooting

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√								
CLO5				√	√							
CLO6				√	√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resource	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resource	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO6	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resource	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | | |
|---|---|--|
| 1. Robert Boylestad and Louis Nashelsky | : | Electronic Devices and Circuits |
| 2. Thomas L. Floyd and David M. Buchla | : | Electronics Fundamentals: Circuits, Devices & Applications |
| 3. Adel S. Sedra and Kenneth C. Smith | : | Microelectronic Circuits: Theory and Applications |
| 4. J. Millman and C.C. Halkias | : | Electronic Devices and Circuits |
| 5. Donald A. Neamen | : | Semiconductor Physics and Devices |
| 6. Albert P. Malvino and David J. Bates | : | Electronic Principles |

Course Code:	EEE-2202
Course Title:	Electronics II Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-2101

Rationale of the Course: In this course students will perform experiments to practically verify the theories learned in the theory course EEE 2201.

Course Objectives:

- To give hands-on experience of electronic circuits studied in course EEE 2201.
- To familiarize the students about the design principles of feedback amplifiers and oscillators.
- To show them how the power amplifiers are designed.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Identify MOS devices and use them in the amplifier circuits	Psychomotor(Imitation, Manipulation)
CLO2	Design feedback amplifiers using Op Amps	Psychomotor(Imitation, Manipulation, Precision)
CLO3	Design oscillator circuits using feedback principle	Psychomotor(Imitation, Manipulation, Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	To design a R-C coupled two-stage amplifier and show its frequency response and bandwidth
2.	Study and demonstration of direct coupling amplifier circuit.
3.	To design a R-F tuned amplifier and show its frequency response and bandwidth
4.	To design the linear operation and frequency response of a Class-B Push-Pull complementary symmetry power amplifier.
5.	Study and demonstration of a JFET as a constant current circuit.
6.	To find the characteristics of JFET and hence design a single stage amplifier (V_m curves, f_{vs} A_v curve, bandwidth)
7.	Design, construction and testing of Hartley oscillator using BJT.
8.	Design, construction and testing of Colpitt's oscillator using BJT
9.	Design, construction and testing of Phase-shift oscillator using BJT
10.	Design, construction and testing of an astable multivibrator using BJT
11.	Design, construction and testing of a monostable multivibrator using BJT
12.	Design, construction and testing of a bistable multivibrator using BJT

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√	√									
CLO2			√	√								√
CLO3			√	√	√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO2	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO3	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-2203
Course Title:	Electrical Machines II
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-2103

Rationale of The Course: Modern world revolution would have not been possible without invention and extensive uses of electrical machineries. Basic knowledge of electrical machines are important requirements for engineering students. This course is intended to provide basic conception of electrical machines.

Course Objectives:

- To understand the basic concepts of DC and synchronous machines with particular focus on their fundamentals and operating characteristics.
- To understand the construction and operating principles of special machines and their applications in special purposes.
- To develop skills about the testing of DC and synchronous machines.
- To accumulate basic ideas about many techniques to analyze the performance characteristics of DC and synchronous machines and solve related machinery problems.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand elementary concepts of DC machines and apply concepts of DC machines to analyze the performance characteristics of DC generators and motors and solve DC machinery problems.	Cognitive (Understand, Apply, Analyze)
CLO2	Understand construction and principles of synchronous machines and apply concepts of these machines to analyze the performance characteristics of synchronous generators and motors and solve related machinery problems.	Cognitive (Understand, Apply, Analyze)
CLO3	Apply concepts of special machines to analyze and solve machinery problems.	Cognitive (Understand, Apply, Analyze)

Course Content:

Sl. No.	Course Content (as Summary)
1.	D.C. Generator: Principles, Construction, Classification, Armature windings, Voltage build up, Armature reactions and Commutation, Performance and testing, Compounding of d.c. generator, Generator characteristics, Voltage regulation, Losses and efficiency, Parallel operation.
2.	D.C. Motor: Operation, Types, Back e.m.f, Torque equations, Motor characteristics, Speed-Torque Characteristics, Speed regulation, Losses and efficiency, Methods of speed control, Methods of braking, Starters, Amplidyne and Metadyne.
3.	Special Machines: Welding machine, Brushless machines, universal motor, stepper motor, reluctance motor, repulsion motor, servomotor, Hysteresis motor, permanent magnet motor and electrostatic motor.
4.	Synchronous Generator: Alternator construction. Armature winding, air gap flux and voltage compensation. Determination of machine parameters. Vector diagram and alternator regulation by different methods. Parallel operation: necessary condition, synchronizing, circulating current. Transient condition of alternator, transient and sub-transient reactances, Blondel's two reaction analysis. Power balance, loss and efficiency.
5.	Synchronous Motor: Characteristics operation and vector diagram. Effect of excitation on power factor and motor performance. Application and testing of synchronous motor. Synchronous capacitor and power factor improvements.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√	√	√								√
CLO2	√	√	√	√								√
CLO3			√	√	√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Journal Paper	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion/Demonstration	▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Stephen J. Chapman : Electric Machinery and Power System Fundamentals
2. A Fitzgerald : Electric Machinery
3. A. Puchstein, T E Loyd and AG Conard : Alternating Current Machines
4. J. Rosenblatt & M. H. Friedman : Direct and Alternating Current Machinery

Course Code:	EEE-2204
Course Title:	Electrical Machines II Sessional
Course Credit:	1.0
Course Type:	Core Course
Co-requisite:	EEE-2203

Rationale of the Course: This course intends to provide the student hands on experience of testing and controlling of both AC and DC machines they are taught in EEE 2203 course.

Course Objectives:

- To demonstrate the details construction of AC and DC machines.
- To develop skills for analyzing the performance characteristics of AC and DC machines.
- To achieve practical experience about the testing of AC and DC machines.
- To gain practical experience about the controlling of AC and DC machines.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand elementary concepts and construction of AC and DC machines, and practically find out the efficiency of these machines.	Psychomotor(Imitation, Manipulation)
CLO2	Analyze the performance characteristics of AC and DC machines.	Psychomotor(Manipulation, Precision)
CLO3	Control the speed of AC and DC machines.	Psychomotor(Manipulation, Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Determination of Open Circuit Characteristic Curve of a Separately Excited DC Generator
2.	Determination of Variation of Terminal Voltage with Speed of a Separately Excited DC Generator at Constant Excitation
3.	Determination of External Characteristics of a DC series Generator
4.	Study of Loading Characteristics of a DC Shunt Generator
5.	Speed Control of a DC Shunt Motor by Varying Armature Voltage, Keeping the Field Current Constant (Armature Control Method)
6.	Speed Control of a DC Shunt Motor by Varying Field Current with Armature Voltage Kept Constant (Flux Control Method)

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√		√									√
CLO2		√	√		√				√			√
CLO3		√		√	√				√			√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO2	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO3	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-2205
Course Title:	Digital Electronics
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-2101, EEE-2201

Rationale of the Course: Digital electronics plays vital roles in modern technologies of electrical and electronic system designs and automations. This course is designed to cover the design techniques, circuit implementation and operation of various digital logic devices and circuits and along with their practical applications.

Course Objectives:

- To introduce the concept of digital and binary systems to the students
- To enable the students to understand, differentiate and analyze combinational and sequential logic circuits
- To build up students capacity to design and construct various combinational and sequential logic circuits
- To develop students idea about the application and utilization of digital systems

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the fundamental concepts and techniques used in digital electronics.	Cognitive (Understand)
CLO2	Examine the structure of various number systems and its application in digital design.	Cognitive (Apply, Analyze)
CLO3	Investigate and justify the basic requirements of digital logic design.	Cognitive (Evaluate)
CLO4	Design the combinational and sequential logic circuits.	Cognitive (Apply, Analyze)
CLO5	Develop skills to solve the identified problems of digital electronics and devices.	Cognitive (Create)

Course Content:

Sl. No.	Course Content
1.	Introduction to Number Systems and Logic Gates: number systems, number base conversions, complements, binary codes, BCD numbers, Boolean algebra, postulates and theorems, basic logic functions, digital logic gates, logic families (DL, DTL, TTL, ECL, BiCMOS)
2.	Minimization and Implementation of Boolean Functions: Forms of Boolean functions, canonical and standard forms, Shannon's theorem, Minimization of Boolean functions using Karnaugh map, Quine-Mcclusky method, Iterative consensus method, Implementation of switching functions (Using various gates: NOR, NAND, AND-OR-INVERTER).
3.	Modular Combinational Circuit Design; pass transistor, pass gate, Half adder, Full adder, multiplexer, De-multiplexer, decoder, encoder, binary arithmetic elements and ALU design.
4.	Converters: Digital to analog conversion, D/A converter circuitry, DAC specifications and applications, Analog to digital conversion, Digital ramp ADC. A/D resolution and accuracy
5.	Sequential Logic: Difference between combinational circuits and sequential circuits, Types of sequential circuit, Flip-Flops (Basic flip-flop circuit, clocked RS flip-flop, D flip-flop, JK flip-flop, T flip-flop), Triggering of Flip-flop, Analysis of clocked sequential circuits (state table, state diagram, state equations), state reduction, state assignment.
6.	Register and Memory Unit: Basic shift register, Serial In/Serial out shift registers, Serial In/Parallel out shift register, Parallel In/Serial out shift register, Bidirectional shift register, Memory: Integrated circuit memory, Classification and architecture, RAM memory cells, Read only memory, Magnetic core memory.
7.	MOSFET Digital Circuits: NMOS inverter, CMOS inverter, CMOS logic circuits, Clocked CMOS logic circuits, transmission gates.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√	√									
CLO3			√	√								
CLO4		√	√	√								
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion - Online Resources	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Presentation - Discussion - Online Resources	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Online Resources	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Discussion/Group Discussion - Online Resources	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Discussion/Group Discussion - Online Resources	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation

Learning Materials:

- | | |
|-------------------------|--|
| 1. Thomas L. Floyd | : Digital Fundamentals |
| 2. A.P.Godse, D.A.Godse | : Digital Electronics And Logic Design |
| 3. M. Morris Manno | : Digital and Computer Design |
| 4. Tocci and Widmer | : Digital Systems |
| 5. V.K. Jain | : Switching Theory and Digital Electronics |
| 6. S.C. Lee | : Digital Circuit and Logic Design |

Course Code:	EEE-2206
Course Title:	Digital Electronics Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-2205

Rationale of the Course: In this course students will perform experiments to practically verify the theories, concepts, design techniques, circuit implementation and operation learned in the course EEE 2205.

Course Objectives:

- To demonstrate the concept of digital and binary systems through practical experiments.
- To enable the students to understand, differentiate and analyze combinational and sequential logic circuits
- To develop students idea about the practical use and limitations of digital circuits

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Learn the design techniques, implementation and operation of digital devices and circuits	Psychomotor(Imitation, Manipulation, Precision)
CLO2	Construct and explain the operation of various combinational and sequential logic circuits such as adders, A/D or D/A converters, counters, memory devices and displays	Psychomotor(Imitation, Manipulation, Precision)
CLO3	Design and construct digital circuits using MOS	Psychomotor(Imitation, Manipulation, Precision)

Course Content:

Sl. No.	List of the Experiments
1.	Introduction to IC logic families, logic gate ICs and testing of logic gates.
2.	Verification of universal gates.
3.	Design and implementation of half adder, full adder, half subtractor and full subtractor circuit.
4.	Design and implementation of binary to gray and gray to binary code converter circuit.
5.	Design and implementation of BCD to Ex-03 code converter and display of the output in 7 segment display.
6.	Design and implementation of 7 segment display.
7.	Study of 2 to 4 line decoder and 4 to 1 line multiplexer.
8.	Study of 3 bit shift register and 3 bit ripple counter.
9.	Design and implementation of 3 bit binary counter using T flip flop.
10.	Study of Johnson counter

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√	√									
CLO2		√	√	√								
CLO3			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Participation
CLO2	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO3	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-2207
Course Title:	Numerical Methods for Engineers
Course Credit:	2.5
Course Type:	Core Course
Pre-requisite (If any):	MATH-I

Rationale of the Course: The numerical methods are used for deeper understanding to predict the anomalies which are not possible in the analytic methods. Moreover, numerical analysis investigates and provides accurate solution to real-life problems from the field of science, engineering, biology, astrophysics and finance etc. Hence, this course is designed to cover a broad background in numerical methods and use appropriate software like MATLAB in solving practical problems in reference to electrical engineering.

Course Objectives:

- To understand the idea about the basics of numerical methods for analysis of experimental results.
- To be aware of the basic methods for solving linear and nonlinear problems of algebra, differentiation, integration and the solution of differential equations.
- To develop practical skills in the use of the numerical methods using software.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the different types of errors and approximation theory introduced by numerical analysis, algorithms and develop programming skills in MATLAB.	Cognitive(Understand)
CLO2	Solve different types of algebraic, transcendental equations, various linear and nonlinear systems of equations, integration, differential equations.	Cognitive(Apply, Analyze)
CLO3	Apply the major techniques of using computational methods and software resources in solving various problems of professional activity.	Cognitive(Apply, Analyze)
CLO4	Develop the algorithms of numerical methods and implement them in practice by means of software products (MATLAB).	Cognitive(Apply, Analyze)
CLO5	Formulate the problem and select the appropriate numerical method to solve it.	Cognitive(Evaluate)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Computer Application to Numerical Methods: Classification of numerical methods, Errors in numerical calculations, Solution of algebraic and transcendental equations, The bisection method, The iteration method, Newton-Raphson method, The Secant method, Muller's method.
2.	Numerical Solution of Simultaneous Equations: Elementary operations of matrices, Gauss-Jordan elimination method, Necessity of normalization, Zero diagonal elements and positioning of size, Matrix inversion, Jacobi's method, Gauss-Seidel iterative method.
3.	Polynomials: Introduction, Errors in polynomial Interpolation, Difference operators, Expressing a polynomial in factorial notation, Finite difference interpolation, The Lagrange polynomials, Lagrange's interpolation formula for unequally spaced data.
4.	Numerical Differentiation & Integration: Numerical differentiation, Differentiation based on equal interval interpolation, Differentiation based on Lagrange's interpolation formula, Numerical integration, General quadrature formula, Trapezoidal rules, Simpson's and Romberg's formula, Legendre polynomials, Gaussian quadrature.
5.	Numerical Solution of Ordinary Differential Equations: Introduction, Solution of Taylor's series, Euler's method, Runge-Kutta method, Predictor-Corrector method, Boundary-value problems.
6.	Solution of Partial Differential Equations: Introduction, Examples of Partial differential equation, the approximation of derivatives of finite differences, Laplace equation, Iterative Method, Successive Over-relaxation and Alternating and Direction Methods, Parabolic differential equation, Derivation of the elliptic differences equation.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3		√	√									
CLO4			√	√								
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Participation
CLO2	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation

Learning Materials:

- | | |
|-------------------------------------|--|
| 1. S. S. Sastry | : Introduction Methods of Numerical Analysis |
| 2. Gerald and Wheatley Addison | : Applied Numerical Analysis |
| 3. Chapra and Canale | : Numerical Methods and Engineering |
| 4. M.R. Spiegel | : Theory and problems of Probability and statistics |
| 5. C K Santha and S Balachandra Rao | : Numerical Methods with Programs in BASIC, FORTRAN and PASCAL |

Course Code:	EEE-2208
Course Title:	Numerical Methods for Engineers Sessional
Course Credit:	1.0
Course Type:	Core Course
Co-requisite:	EEE-2207

Rationale of the Course: In this course students will perform experiments to practically verify the fundamentals of numerical solution of simultaneous equation, differentiation, integration, polynomials, ordinary & partial differential equations and operation learned in the course EEE-2207.

Course Objectives:

- To study of various numerical methods.
- To learn numerical methods and their applications in the field of science and engineering

Course Learning Outcomes (CLOs): at the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Solve engineering problems with fundamental of numerical solution of simultaneous equation, differentiation, integration, polynomials, ordinary & partial differential equations	Psychomotor (Imitation, Manipulation)
CLO2	Apply the concepts numerical methods to real world phenomena such as electrical networks, traffic flow, archeological dating, economic interdependencies, population movement, communication networks, and weather prediction	Psychomotor (Imitation, Manipulation)
CLO3	Improve their ability to communicate via the language of mathematics with the abstract world	Psychomotor (Imitation, Manipulation, Precision)

Course Content:

Sl. No.	List of the Experiments
1.	Statistical Analysis of Data
2.	Numerical solution of algebraic equations
3.	Numerical solution of differential equations
4.	Numerical solution of partial differential equations
5.	Numerical solution of differentiation and integration
6.	Find the Roots of Nonlinear Equations
7.	Curve Fitting and Correlation

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3		√	√									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Online video demonstration for high voltage safety requirements	▪ Quiz ▪ Lab performance ▪ Participation
CLO2	▪ Practical implementation of electrical wiring using wires and light, Fans and switch	▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO3	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Lab performance ▪ Semester end final examination ▪ Participation

Learning Materials:

2. Lab guide/Manuals

Course Code:	HUM-2211
Course Title:	Professional Ethics and Law
Course Credit:	2.5
Course Type:	Humanities
Pre-requisite (If any):	N/A

Rationale of the Course: The professional ethics is one of the most valued criteria in the labor market. Having good conduct in the workplace can be the passport to a successful career. Professional competence is the habitual and judicious use of communication, knowledge, technical skills, clinical reasoning, emotions, values, and reflection in daily practice for the benefit of the individual and community being served. Hence, this course will familiarize students with the ethical and legal issues concerning the profession.

Course Objectives:

- To understand the moral values that ought to guide the Engineering profession
- To resolve the moral issues in the profession
- To justify the moral judgment concerning the profession.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand basic purpose of profession, professional ethics and various moral and social issues.	Affective(Receiving, Responding)
CLO2	Create awareness of professional rights and responsibilities of an Engineer, safety and risk benefit analysis of an engineer.	Affective(Receiving, Responding, Valuing)
CLO3	Develop strategies for identifying and dealing with typical ethical issues, both personal and organizational.	Affective(Valuing, Organizing)
CLO4	Excel in competitive and challenging environment to contribute to industrial growth.	Affective(Valuing, Organizing, Characterizing)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction to Professional Ethics: Basic Concepts, Governing Ethics, Personal & Professional Ethics, Ethical Dilemmas, Life Skills, Emotional Intelligence, Thoughts of Ethics, Value Education, Dimensions of Ethics, Profession and professionalism, Professional Associations, Professional Risks, Professional Accountabilities, Professional Success, Ethics and Profession.
2.	Basic Theories: Basic Ethical Principles, Moral Developments, Deontology, Utilitarianism, Virtue Theory, Rights Theory, Casuist Theory, Moral Absolution, Moral Rationalism, Moral Pluralism, Ethical Egoism, Feminist Consequentialism, Moral Issues, Moral Dilemmas, Moral Autonomy.
3.	Professional Practices in Engineering: Professions and Norms of Professional Conduct, Norms of Professional Conduct vs. Profession; Responsibilities, Obligations and Moral Values in Professional Ethics, Professional codes of ethics, the limits of predictability and responsibilities of the engineering profession, Central Responsibilities of Engineers.
4.	Work Place Rights & Responsibilities: Ethics in changing domains of Research, Engineers and Managers; Organizational Complaint Procedure, difference of Professional Judgment within the Nuclear Regulatory Commission (NRC), the Hanford Nuclear Reservation. Ethics in changing domains of research - The US government wide definition of research misconduct, research misconduct distinguished from mistakes and errors, recent history of attention to research misconduct, the emerging emphasis on understanding and fostering responsible conduct, responsible authorship, reviewing & editing.
5.	Global issues in Professional Ethics: Introduction – Current Scenario, Technology Globalization of MNCs, International Trade, World Summits, Issues, Business Ethics and Corporate Governance, Sustainable Development Ecosystem, Energy Concerns, Ozone Deflection, Pollution, Ethics in Manufacturing and Marketing, Media Ethics; War Ethics; Bio Ethics, Intellectual Property Rights

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1							√	√				
CLO2						√		√				
CLO3								√	√		√	√
CLO4								√			√	

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion	▪ Quiz ▪ Tutorial/Assignment ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion ▪ Journal Paper	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | |
|-------------------|---|
| 1. J. D. Mabboth | : Professional Ethics and Civics Morals |
| 2. A. R. Khan | : An Introduction to Ethics |
| 3. M. Radar | : Ethics and the Human Community |
| 4. Emile Durkheim | : Business Ethics |

Course Code:	EEE-2200
Course Title:	Viva-voce
Course Credit:	1.0
Course Type:	Core Course
Pre-requisite:	Knowledge of 3rd to 4th semester courses

Rationale of the Course: Communication plays very vital role in aspect of life. Viva-voce is such an opportunity where a learner is judged how he/she can communicate with others from his/her gained knowledge. The objective of this course is to verify the students how well they can express their knowledge orally.

Course Objective:

- To assess the students ability to communicate with another person.
- To supplement the information obtained through other evaluation techniques.
- To identify and analyze the students presence of mind.
- To evaluate the students spontaneity and mannerism.

Course Learning Outcomes (CLOs): By the end of the all courses in First Year, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Convince the audience orally about their knowledge from the courses in Second Year	Affective(Receiving, Responding, Valuing)

Course Content:

Sl. No.	Course Contents
1.	1st Year and 2nd Year Course Curriculum

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√								√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Presentation ▪ Discussion	▪ Participation ▪ Presentation

Learning Materials

Knowledge from 1st Year and 2nd Year Curriculum

B.Sc. Engineering Third Year, First Semester Examination 2023

Course Code:	EEE-3101
Course Title:	Electronics III
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-2101, EEE-2201

Rationale of the Course: Modern technology would have not been possible without advancement of electronics. Proper knowledge, understanding, application and troubleshooting are urgent for core engineering students. This course will enable learners to have comprehensive understanding of the theory, concepts, characteristics and working principles of advanced electronic devices and their applications in modern electronics.

Course Objectives:

- To familiarize the students with operational amplifier applications
- To introduce the concept of signal wave-shaping
- To enable the students to understand and analyze timer circuits
- To improve students skill of analysis, selection, design and troubleshooting of circuits
- To enable the students to learn about the design, features and applications of various ICs

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the constructions and basic operation of differential amplifiers, operational amplifier, active filters, pulse and switching, phase locked loop, switch capacitor circuits and power supply.	Cognitive (Understand)
CLO2	Solve problems related to operational amplifiers, active filters and power systems.	Cognitive (Understand)
CLO3	Identify and apply proper pulse and switching circuit and PLL in the field of communication engineering.	Cognitive (Apply, Analyze)
CLO4	Relate and compare between different power supply sources.	Cognitive (Evaluate)
CLO5	Design & construct different active filters and power supply.	Cognitive (Create)

Course Contents:

Sl. No.	Course Content (as Summary)
1.	Differential Amplifiers: Differential amplifier, DC analysis, AC analysis, CMMR, Constant current bias, Current mirror.
2.	Operational Amplifiers: Introduction, Block diagram, Internal circuit, Characteristics, Basic parameters of Op-Amp, Open-loop of OP-Amp, Applications; Inverting amplifier, Non-inverting amplifier, Summing amplifier, Difference amplifier, Integrator, Differentiator, Instrumentation amplifier, Logarithmic amplifiers, Antilog amplifier, Multiplier, Divider, Voltage to current converter, Current to voltage converter.
3.	Active Filters: Types, Advantages, Filter design Frequency scaling, First order and second order high pass and low pass filters, Band pass filters, Notch filters, Comparators.
4.	Pulse and Switching: Pulse characteristics, Bistable multivibrators, Monostable multivibrators, Astable multivibrators, Waveform generators, Square wave, Triangular wave, Sawtooth wave, Voltage controlled oscillator (VCO), Schmitt trigger, Blocking oscillator, UJT, Varactor diode, 555 timer circuit and applications.
5.	Phase Locked Loops (PLL): Operating principles, Major building blocks, Application of PLL: FM demodulation, FSK demodulation, AM detector, Frequency synthesizer.
6.	Switch Capacitor Circuits: Theory of operation, Switched capacitor integrator, Universal filters.
7.	Power Supplies: Voltage regulators, Fixed and variable voltage regulators, switching regulators, Transistor voltage regulators, IC voltage regulators, Shunt and series voltage regulators, voltage dividers, Protection of equipment from low and high voltage, stabilized power supply with ICs.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√										
CLO3			√	√								
CLO4		√		√								
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO3	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO4	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO5	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam

Learning Materials

1. Gaykwad : Operational Amplifier
2. Millman & Tub : Electronic Devices and Circuits
3. Coughlin & Driscoll : Operational Amplifier
4. Mithal & Vanswasi : Pulse and Digital Electronics
5. Allen. Motershed : Electronic Devices & Circuits
6. Millman and Halkias : Electronic Devices & Circuits
7. Gupta & Kumar : Handbook of Electronics

Course Code:	EEE-3102
Course Title:	Electronics III Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-3101

Rationale of the Course: In this course students will perform experiments to practically verify the theories, concepts, design techniques, circuit implementation and operation learned in the course EEE-3101.

Course Objectives:

- To demonstrate the concept of experimental knowledge on electronics.
- To enable the students to design, construct and analyze various electronics circuits.
- To develop students idea about the practical uses and limitations of ICs.

Course Learning Outcomes (CLOs): By the end of the course students will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Analyze the operation of Op-Amp and construct various circuits using Op-Amp.	Psychomotor (Imitation, Manipulation)
CLO2	Design and construct active filter circuits.	Psychomotor (Manipulation, Precision)
CLO3	Design and construct the PLL circuits.	Psychomotor (Manipulation, Precision)
CLO4	Design and construct the power supplies	Psychomotor (Precision, Articulation)

Course Contents:

Sl. No.	List of the Experiments
1.	To design and testing of an inverting and non-inverting amplifier to measure voltage gain.
2.	To study the characteristics of an Op-Amp- (a) Input offset voltage (b) Slew rate (c) CMMR.
3.	To determine the operation of an Op-Amp as- (a) Summing amplifier (b) Difference amplifier.
4.	To design and construct a first order low pass and high pass active filter using Op-Amp
5.	To design and construct a second order low pass and high pass active filter using Op-Amp.
6.	To design and testing voltage to current converter using Op-Amp.
7.	To design and construct a PLL circuits.
8.	To design and construct a power supplies.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1		√		√								
CLO2			√		√							
CLO3			√		√							
CLO4			√		√							

Mapping of Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Demonstration - Online materials - Experimental works - Report writing	- Participation - Lab performance - Note book - Final exam
CLO2	- Demonstration - Online materials - Experimental works - Report writing	- Participation - Lab performance - Note book - Final exam
CLO3	- Demonstration - Online materials - Experimental works - Report writing	- Participation - Lab performance - Note book - Final exam
CLO4	- Demonstration - Online materials - Experimental works - Report writing	- Participation - Lab performance - Note book - Final exam

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-3103
Course Title:	Measurements and Instrumentation
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	N/A

Rationale of the Course: In this modern age of science and Technology, the use of measurements of any parameter and Instrumentation is common in each and every field. Measurements of any parameter and Instrumentation is essential field in the modern industry where need highly skilled manpower. Hence, this course is designed to cover introduction to measurements of parameters and instrumentation in detail.

Course Objectives:

- To understand the necessities of physical variables as encountered in everyday life and during scientific experimentation along with their significance of measurements.
- To understand the properties (e.g. mechanical, hydraulic, electrical, magnetic, optical, pneumatic and so on) of the physical variables of interests.
- To understand how to convert physical variables into measurable forms through proper selection of devices along with their properties and selection criteria.
- To understand how to choose the physical variable sensing devices, write and solve governing equations to control, design and implement circuits of interests for measurements.

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Describe the functional fundamental concept on various types of instruments and realize nature of transducer, its characteristics, working principles and applications.	Cognitive (Understand, Apply)
CLO2	Design, construct and test electrical instruments.	Cognitive (Analyze)
CLO3	Design, construct and test electronic instruments.	Cognitive (Apply Analyze)
CLO4	Produce and apply display monitoring instruments.	Cognitive (Apply Analyze)
CLO5	Analyze and develop wave generator.	Cognitive (Analyze, Evaluate)

Course Contents:

Sl. No.	Course Contents(as Summary)
1.	Measuring Instruments: Absolute and secondary instrument, Principle of operation, Essentials of indicating instruments. Classification, Indicating instruments, operation principle of ammeters, voltmeters, watt- meters and watt-hour meter.
2.	Introduction of Instrumentation Error: Classification of error, normal law of error, guarantee of error.
3.	Electrical Instruments: Classifications, Moving Iron, Attraction and repulsion type, Errors in moving iron, Moving coil meter, Permanent magnet instrument, torque equation, Error dynamic type instrument, Operating principle, Dynamometer, Voltmeter and ammeters, Errors in dynamometer instruments, Ohmmeters, series and shunt type meter, Multi-meter, Thermal instrument, Induction instruments, Rectifier instrument, Extension of range of voltmeter and ammeter, Ammeter shunt, Voltmeter multipliers, Multi-range meters, Instrument transformer, Current and voltages transformer
4.	Electronic Instruments: Measurement fundamentals, Types of error, Voltmeter sensitivity, Loading effect, Amplified DC meter, Chopper type, AC voltmeter, RMS voltmeter, selecting factors of meter, Digital voltmeter, Component measurement, Q Meter, Frequency Meter.
5.	Oscilloscopes: Block diagram, CRT deflections, Screens, CRT circuits, Vertical deflection system, Delay line, Multiple trace Horizontal deflection system, Probes and transducers, Storage Oscilloscopes, digital storage Oscilloscopes. Sampling Oscilloscopes
6.	Signal Generation and Analysis: Sine wave generators, Frequency synthesized generator, Sweep frequency generator, Wave analyzers, Harmonic distortion analyzer, Spectrum analyzer, Applications.
7.	Transducers: Classifications, Selection of transducer, Strain gages, Displacement transducers, LVDT, Temperature measurements, Thermometer, thermocouples, Flow meter, Liquid level detector.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√		√									
CLO2			√	√								
CLO3			√	√								
CLO4			√									
CLO5		√		√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Journal Paper	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials

- | | |
|------------------|--|
| 1. B L Theraja | : Electro Technology. |
| 2. W.D. Cooper | : Electronic Instrumentation and Measurement Techniques. |
| 3. A.K. Sawhney | : Electrical and Electric measurement and Instrumentation. |
| 4. C.S. Rangan | : Instrumentation, Devices and Systems. |
| 5. Alan S Morris | : Measurement and instrumentation principles4. |
| 6. H.S. Kalsi | : Electronic Instrumentation |

Course Code:	EEE-3104
Course Title:	Measurements and Instrumentation Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-3103

Rationale of the Course: This course will make familiar to students with the measurement of physical variables as encountered in our everyday life and during scientific experimentation along with their significance of measurements.

Course Objectives:

- To measure physical variables and analyze measured data statistically.
- To measure resistance, inductance and capacitance under different situations.
- To familiar with instruments for measuring magnetic flux and magnetic flux density.
- To measure current, voltage, electrical power and energy.
- To familiar with digital instruments for measuring and visualizing electrical quantities.
- To familiar with instruments for measuring non-electrical quantities.

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Measure physical variables and how to analyze measured data statistically.	Psychomotor (Imitation, Manipulation)
CLO2	Measure resistance, inductance and capacitance under different situations.	Psychomotor (Imitation, Manipulation)
CLO3	Measure magnetic flux and magnetic flux density.	Psychomotor (Imitation, Manipulation)
CLO4	Use analog instruments for the measurements of current, voltage, electrical power and energy.	Psychomotor (Manipulation, Precision)
CLO5	Use digital instruments for measuring and visualizing electrical quantities.	Psychomotor (Manipulation, Precision)
CLO6	Apply instruments for measuring non-electrical quantities	Psychomotor (Precision, Articulation)

Course Contents:

Sl. No.	List of the Experiments
1.	Measurement of medium resistance by ammeter voltmeter method.
2.	Measurement of power a) using 3 ammeter b) using 3 voltmeter
3.	1) Measurement of "Energy meter 2) Range extension of a) ammeter and b) Voltmeter
4.	1) Localization of cable fault by Murray Loop test 2) Measurement of earth resistance by fall of potential method
5.	a) Measurement of medium resistance by Wheatstone bridge method. B) Measurement off lumen of a light source.
6.	Study of characteristics of different transducers: thermocouple, thermistor, LM35D2.
7.	To convert a Galvanometer into a Voltmeter.
8.	To convert a Galvanometer into an Ammeter.
9.	To convert a Galvanometer into an Ohmmeter.
10.	To extend the range of a Voltmeter.
11.	To extend the range of an Ammeter.
12.	To extend the range of an Ohmmeter.
13.	Introduction to an Oscilloscope and measure amplitude, frequency of wave.
14.	To construct a sine wave generator.
15.	To construct a sweep frequency generator.
16.	To construct a square wave generator.
17.	To construct a saw tooth wave generator.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2			√	√								
CLO3			√	√								
CLO4			√									
CLO5		√			√							
CLO6						√						

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Participation
CLO2	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Participation
CLO3	- Demonstration - Experiment, - Report writing	- Lab performance - Semester end final examination - Participation

CLO4	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO5	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO6	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Lab guide/Manuals
2. Design of OP AMP circuit with experiments- Howard M. Berlin

Course Code:	EEE-3105
Course Title:	Electromagnetic Fields and Waves
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	PHY-1103

Rationale of the Course: Engineering electromagnetics plays the most fundamental role over the modern wireless communications. This course brings the basics and application of electromagnetic theory in electrical and electronic engineering.

Course Objectives:

- To know the basic principle of EM transmission and reception
- To be introduced with antenna fundamentals and how EM work with antenna.
- To employ basic and advanced mathematics to materialize the EM.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Understand the basic concepts and application of electromagnetics (EM), various laws of E-fields and H-fields along with relationship with each other and various distributions, antenna fundamentals and EM transmission and reception.	Cognitive (Understand)
CLO2	Employ basic and advanced mathematics to materialize the EM.	Cognitive (Apply)
CLO3	Apply Maxwell's equations in the study of the EM wave.	Cognitive (Apply)
CLO4	Apply EM wave propagation in reflection and refraction.	Cognitive (Apply)
CLO5	Design, apply and evaluate the different antennas.	Cognitive (Analyze, Evaluate)
CLO6	Develop the concepts and applications of waveguides, EM transmission mechanism through transmission line.	Cognitive (Create)

Course Contents:

Sl. No	Course content (as Summary)
1.	Electromagnetics: Field equations on coulomb, Ampere, Lenz, Gauss and Faraday's law, Displacement current, Maxwell's equations, boundary conditions, Plane wave in space, Poynting's radiation vector.
2.	Transmission Lines: Balance, unbalanced lines, Primary Line constants, Characteristics impedance, propagation coefficient, Phase and Group velocity, standing waves, standing wave ratio, losses in transmission line, Smith Chart, Transmission line components: Stub, couplers, Balun, Slotted lines. Telephone lines and Cables, RF lines, Microstrip lines and strip transmission lines

3.	Propagation of Waves: Electronic spectrum, propagation in free space, Tropospheric propagation, Ionospheric propagation,, Ground (surface) wave propagation, space waves, LF-VLF-ELF propagations.
4.	Waveguides: TM, TE, TEM Waves; Rectangular wave guide; modes, Temn modes. Circular and other wave guides, Waveguide Coupling, Matching and Attenuation, Cavity resonator; Directional coupler, Isolator, Circulator, Duplexers
5.	Antenna: Radiation current element; Antenna equivalent circuit, Elementary Doublet, Classification of Antennas, Current-voltage distribution of antenna; Resonant antenna, Radiation pattern, Length of antenna, Non-resonant antenna, Antenna gains, Radiated Power, Antenna resistance, Beamwidth, Bandwidth, Polarization, Effects of ground; Effects of Height, Antenna Coupling; Directional HF, UHV, Microwave an Helical Antenna, Discone antenna, Loop antenna.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3		√	√									
CLO4		√	√									
CLO5			√	√								
CLO6			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Journal Paper	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO6	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | |
|--------------------------------|--|
| 1. Matthew N.O. Sadiku | : Elements of Electromagnetics |
| 2. D. K. Cheng | : Fields and Wave Electromagnetics |
| 3. W. H. Hayt& J. A. Buck | : Engineering Electromagnetics |
| 4. A.B. Brownell and R.E. Beam | : Theory and Application of Microwave. |
| 5. J.D. Kraus | : Antenna |
| 6. Louis E. Frenzel | : Communication Electronics |

Course Code:	EEE-3106
Course Title:	Electromagnetic Fields and Waves Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-3105

Rationale of the Course: This course will make familiar to students with the basics and application of electromagnetic theory in electrical and electronic engineering and during scientific experimentation along with their significance of measurements.

Course Objectives:

- To employ basic and advanced mathematics to materialize the EM.
- To know the basic principle of EM transmission and reception
- To be introduced with antenna fundamentals and how EM work with antenna.

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Know the basic concepts and application of pattern design of antenna	Psychomotor (Imitation, Manipulation)
CLO2	Apply Maxwell's equations to study the EM wave	Psychomotor (Imitation, Manipulation)
CLO3	Determine the impedance, reflection co-efficient and return loss of antenna.	Psychomotor (Manipulation, Precision)
CLO4	Calculate the VSWR of antenna.	Psychomotor (Manipulation, Precision)
CLO5	Design different type antennas	Psychomotor (Manipulation, Precision, Articulation)

Course Contents:

Sl. No.	List of the Experiments
1.	Plot Radiation Pattern of Antenna
2.	Plot Azimuth and Elevation Pattern of Antenna
3.	Calculate the Directivity of Antenna
4.	Calculate the EH fields of Antenna
5.	Plot Different Polarizations of Antenna
6.	Calculate the Axial Ratio of Antenna
7.	Calculate the Beam width of Antenna
8.	Calculate the Impedance of Antenna
9.	Calculate the Reflection Coefficient of Antenna
10.	Calculate the Return Loss of Antenna
11.	Calculate the Voltage Standing Wave Ratio (VSWR) of Antenna

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√	√									
CLO2		√	√									
CLO3		√	√									
CLO4		√	√									
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation

CLO2	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO3	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO4	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO5	▪ Demonstration ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-3107
Course Title:	Microprocessor, Microcontrollers and Peripherals
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-2205

Rationale of the Course: In the era of digitalization, the use of microprocessor and microcontroller is very common in each and every field. Microprocessor and microcontroller is specialized and essential field in the everyday functioning of modern industrialized societies where need highly skilled manpower. Hence, this course is designed to cover introduction to basic 8085 & 8086 microprocessors as well as 8-bit microcontrollers in detail with sufficient exposure to design and maintain indigenous real-time and embedded systems.

Course Objectives:

- Grow concept of how a microprocessor operates.
- Explain memory interfacing technique with microprocessor.
- Explain the I/O operation and I/O interfacing
- Discuss how a Microprocessor based system is designed.
- Explain basic architecture of Micro-controllers.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Understand the history of microprocessor & microcontroller, microprocessor based systems & its operation, functional block diagram & internal architecture of 8085, 8086 & 8051, and peripherals including memory.	Cognitive (Understand)
CLO2	Justify the Intel 8085, 8086 & 8051 addressing mode of an instruction.	Cognitive (Analyze)
CLO3	Interpret how the different peripherals including memory are interfaced with the microprocessor and microcontroller.	Cognitive (Analyze)
CLO4	Examine the data transfer information through different ports and the microprocessor/microcontroller based systems.	Cognitive (Evaluate)
CLO5	Develop programming skills in assembly language.	Cognitive (Apply)
CLO6	Design microprocessor/microcontroller-based systems for real world applications.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction: Microprocessors and Microcomputer, Evolution of Microprocessor, Microprocessor instruction set and Computer language, General architecture of microprocessor and its operation, Memory, Input/output (I/O) devices, RISC and CISC processors, Co-processors, Parallel processing.
2.	Intel 8085 Microprocessor: Introduction, Internal architecture, Register structure, Memory addressing, Addressing modes, Instruction set, 8085 programming model, Instruction classification, How to write and execute a simple program, Memory interfacing, Interfacing output displays, Interfacing Input Devices, Memory mapped I/O, Testing and Troubleshooting I/O interface Circuits, 8085-based System Design.
3.	Intel 8086 Microprocessor: Introduction, Architecture, Addressing modes, Instruction set, Programming model, Assemble language programming, How to write and execute a simple program & 8086 input/output system design using 8086.
4.	Programmable Interface Devices: Interrupts, Basic concepts in programmable devices, The 8155/8156 and 8355/8755 etc multipurpose programmable devices, The 8279 programmable keyboard/display interface.
5.	Peripheral Interfacing: Parallel versus serial transmission, Synchronous and asynchronous serial data Transmission, Basic concept of serial I/O, Software controlled asynchronous serial I/O; The 8085-serial I/O Lines: SID and SOD, Hardware controlled serial I/O using programmable chips, DMA controllers
6.	Microcontrollers: Introduction, History, Microcontrollers versus microprocessors, Internal architecture, Memory organization, Special function registers (SFRs), Addressing modes, Instruction set, Assembly language programming of 8051 /PIC24/AVR32, I/O ports & circuits, Interfacing of Keyboard & External memory, & Motor speed control system design using 8051/PIC24/AVR32

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√										
CLO3			√	√								
CLO4				√								
CLO5			√									
CLO6			√		√	√						

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Participation
CLO2	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO6	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation

Learning Materials:

- | | |
|---------------------|---|
| 1. Douglas V. Hall | : Microprocessors and Interfacing |
| 2. M. Rafiquzzaman | : Microprocessor and Microprocessor-based System Design |
| 3. 3. Barry B. Brey | : Microprocessor Hardware Interfacing and Application |
| 4. Morris Manno | : Digital Logic and Computer Design |
| 5. M. Morris Manno | : Computer System and Architecture |
| 6. R. Gaonkar | : Microprocessor Architecture, Programming and Applications |
| 7. John P. Hayes | : Computer Architecture and Organization |
| 8. Ajay V. Deshmukh | : Microcontrollers: Theory and Applications |

Course Code:	EEE-3108
Course Title:	Microprocessor, Microcontrollers and Peripherals Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-3107

Rationale of the Course: In this course students will perform experiments to practically verify the theories learned in the theory course EEE- 3107.

Course Objectives:

- To show how assembly language programming is done in 8085 microprocessor
- To show how assembly language programming is done in 8086 microprocessor
- To show how a microcontroller is programmed for using it in a particular application
- Explain basic architecture of Micro-controllers.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Identify 8085 and 8086 ICs	Psychomotor (Imitation, Manipulation)
CLO2	Program 8085 using assembly language program	Psychomotor (Manipulation, Precision)
CLO3	Program 8086 using assembly language or C language	Psychomotor (Manipulation, Precision)
CLO4	Design an input output system using microcontroller	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	(a) Introduction to 8085 Microprocessor architecture and 8085 Assembly Language Programming (b) Familiarization with 8085 Microprocessor Training Kit Objectives
2.	a) Study on 8085 Microprocessor Instruction set b) Performing different arithmetic operation using 8085 Microprocessor.
3.	(a) Introduction to 8255 programmable peripheral interface (PPI) b) Data transfer using PPI for simple I/O mode of operation Mode 0 (c) Bit set/reset (BSR) mode of operation of 8255 PPI
4.	Design and implementation of traffic light controller using 8255 PPI in 8085 microprocessor environment
5.	Demonstrate a counter program of up counter and down counter with time delays by 8085 microprocessor with 8255 peripherals
6.	(a) Introduction to 8086 Microprocessor Internal Architecture, Addressing modes and Hex code (b) Constructing the Machine code for 8086 instruction (c) Familiarization with MDA-Win 8086 Microprocessor Trainer Kit.
7.	a) Introduction to 8086 assembly language programming b) Introduction to serial monitor operation of MDAwin8086 trainer kit.
8.	Dot Matrix LED Interface Using 8255 PPI in 8086 Microprocessor Environment
9.	(a) Introduction to Arduino board and Arduino IDE (b) Design and implementation of blinking LED, analog voltage measurement, string operation and traffic controller projects in Arduino environment
10.	Design and implementation of DC motor speed control drive in Arduino environment
11.	Design and implementation of systems using programmable microcontroller.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√										
CLO3		√										
CLO4			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Participation
CLO2	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Participation
CLO3	- Demonstration - Experiment, - Report writing	- Lab performance - Semester end final examination - Participation
CLO3	- Demonstration - Experiment, - Report writing	- Lab performance - Semester end final examination - Participation
CLO4	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-3110
Course Title:	Electrical and Electronic Shop Practice
Course Credit:	1.0
Course Type:	Core Course
Pre-requisite:	N/A

Rationale of the Course: This course intends to develop the skills to assemble and troubleshoot the electrical and electronic system.

Course Objectives:

- Acquire knowledge about electrical and electronic components, measuring instruments, bread board assembling, etc.
- Acquire knowledge and practice soldering, de-soldering and PCB design.
- Able to analyze the circuit and troubleshoot.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Identify and test the various active and passive components.	Psychomotor (Precision, Manipulation)
CLO2	Solder and de-solder of electronic component in electronic circuits	Psychomotor (Manipulation, Precision)
CLO3	Design the PCB Layout and fabricate PCB.	Psychomotor (Precision, Articulation)
CLO4	Troubleshoot and repair the common electronic system	Psychomotor (Manipulation, Precision)

Course Content:

Sl. No.	List of the Experiments
1.	Familiarization with a) Conventional symbols for electrical circuit installation. b) Electrical Switches and tools.
2.	Performing different types of wire splicing and terminations.
3.	Experiments on a) Tube light connection. b) Staircase lamp circuit by two way switch.
4.	Experiments on a) Traffic signal lighting. b) Holiday & Smart lighting.
5.	Experiments on a) transformers different testing. b) Working principle of ceiling fan.
6.	Testing of a transformer a) Megger Test. b) Dielectric Strength of transformer oil.
7.	Experiments on a) Installation of wiring system b)Electrical earthing & Safety instruction

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√					√						
CLO2			√									
CLO3				√		√						
CLO4				√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Online video demonstration for high voltage safety requirements	- Quiz - Lab performance - Participation
CLO2	Practical implementation of electrical wiring using wires and light, Fans and switch	- Quiz - Lab performance - Participation
CLO3	- Discussion, - Experiment, - Report writing	- Lab performance - Semester end final examination - Participation
CLO4	- Discussion, - Experiment, - Report writing	- Lab performance - Semester end final examination - Participation

Learning Materials:

- | | |
|--------------------------------------|------------------------------------|
| 1. Keith Mobley | : Maintenance Engineering Handbook |
| 2. Lindley Higgins and Darrin Wikoff | |
| 3. Tim Williams | : The Circuit Designer's Companion |
| 4. Marcus and Lavy | : Elements of Radio Servicing |
| 5. Mark I. Monstrose | : A Handbook for Designers |

Course Code:	EEE-3112
Course Title:	Electrical and Electronic Circuit Simulation
Course Credit:	1.0
Course Type:	Core Course
Pre-requisite:	N/A

Rationale of the Course: In this course students will verify theories and concepts learned in electrical and electronic circuit theory courses using simulation software(s). PSPICE, MATLAB, PROTEUS and other simulation software will be introduced.

Course Objectives:

- To familiarize the students with software used for simulating electrical and electronic circuits.
- To discuss how to verify a circuit after it is designed, and before implementing it in a circuit board.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Solve the parameters of a circuit consisting of resistors, capacitors, inductors, diodes and transistors using PSPICE and MATLAB	Psychomotor (Manipulation, Precision)
CLO2	Draw a circuit in Proteus software and simulate its operation.	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Familiarization with some basic function of MATLAB
2.	Basic of M-Files, Control loops, 3 Structures and cell array, 3D line draw and problem solving.
3.	Familiarization with plotting various functions and circuit simulation on MATLAB
4.	Experiment on symbolic variables, numbers and expressions and to calculate differentiation, integration, sum and limiting value of a function
5.	Study and demonstration of Laplace Transform and circuits-2, random numbers
6.	Introduction to PSPICE and different circuit analysis by using Capture and OrCAD
7.	Introduction to dependent sources, transient, variable parameter, user defined sources and solving various circuits on that basis
8.	Introduction to AC analysis and simulating them with determining resonant frequency
9.	Study on Power Supply and various kinds of amplifier circuits

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO3			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Participation
CLO2	- Demonstration - Experiment, - Report writing	- Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

B.Sc. Engineering Third Year, Second Semester Examination 2023

Course Code:	EEE-3201
Course Title:	Power System Analysis I
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-1101, EEE-1201

Rationale of the Course: Power system analyses are an essential part of electrical power system design to determine the necessary generating capacity to satisfy the system demand in the presence of scheduled and unscheduled outages and unforeseen variations in the system load.

Course Objectives:

- To model transmission lines
- To use of PU quantities in power system analysis
- To obtain steady state solution of power system using G-S Load Flow
- To study Effect of faults

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Understand the basic concept of power system, inductance and capacitance of transmission lines also faults on power system	Cognitive (Understand)
CLO2	Analyze different methods for power system analysis in steady state operation.	Cognitive (Evaluate)
CLO3	Analyze different problems using symmetrical components transformation, symmetrical faults and unsymmetrical faults.	Cognitive (Apply)
CLO4	Develop the simulation tools to perform comprehensive short circuit studies and load flow studies use instruments and equipment in the laboratory.	Cognitive (Create)

Course Contents:

Sl. No.	Course Content (as Summary)
1.	Introduction to Transmission Lines: Flux linkages, inductance due to external flux, inductance of single-phase two-wire line, composite conductor lines, G.M.D, 3-phase line with equilateral and with unsymmetrical spacing, parallel circuit of 3-phase line, and use of tables.
2.	Capacitance of Transmission Lines: Electric field, capacitance of two wire line, three-phase lines with symmetrical & with equilateral spacing, effect of earth, parallel circuits lines, representation of lines: short, medium and long transmission lines, T and π representation, exact solutions, equivalent circuit of long transmission line.
3.	Generalized Line Constants: General line equations in terms of ABCD constants, relations between constants, charts of line constants, constants of combined networks measurements of line constants.
4.	Power Flow Solutions: Power flow problem, Gauss-Seidel method, Newton-Raphson method, Power flow studies in system design and operation, regulating transformers.
5.	Symmetrical Faults: Transients in RL series circuits, internal voltages of loaded machines under fault conditions, Fault calculations using Z_{bus} . Circuit breakers.
6.	Unsymmetrical Faults: Unsymmetrical faults on power system, Single line-to-ground faults, Double line-to-ground faults, Open-conductor faults.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3		√	√	√								
CLO4			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Web material /Videos	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration ▪ Power plant visiting ▪ Web material /Videos	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion ▪ Journal Paper ▪ Web material /Videos	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | |
|--------------------------------|--|
| 1. William D. Stevenson | : Elements of power system analysis Fourth Edition |
| 2. V.K. Mehta and Rohit Mehta | : Principles of Power System |
| 3. D P Kothari and I J Nagrath | : Power System Engineering Second Edition |
| 4. Ashfaq Hussain | : Electrical Power System Fourth Edition |
| 5. Hadi Saadat | : Power system analysis |
| 6. M.N. Bandyopadhyay | : Electrical Power Systems: Theory and Practice |

Course Code:	EEE-3203
Course Title:	Control System Engineering
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-2205

Rationale of the Course: Control system engineering ensures that there is a strategic method to improve productivity and enhance the best practices of a company. We want to eliminate the redundant manual controls and reduce human errors that could cost us a significant amount of money and time. Hence, this course is designed to be introduced to different aspects of a control system, various manual and automatic control systems, and programmable control systems.

Course Objectives:

- To introduce the fundamentals of manual and automatic control systems and their advantages, disadvantages along with their aspects.
- To understand the concepts of the mathematical modeling, feedback control, and stability analysis in time and frequency domains along with exhibiting an understanding of the fundamentals of feedback control systems, complete the basic design of a PLC system.
- To demonstrate different ways to program PLC and their applications with proper analysis.
- To design and develop different manual and automatic controlled systems for industrial purposes and as daily appliances.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the fundamentals of feedback control for dynamic system.	Cognitive (Understand)
CLO2	Determine and use models of physical systems in forms suitable for use in the analysis and design of control systems.	Cognitive (Analyze)
CLO3	Determine the time and frequency-domain responses of first and second-order systems to step and sinusoidal (and to some extent, ramp) inputs.	Cognitive (Analyze)
CLO4	Determine the (absolute) stability of a closed-loop control system and Apply root-locus technique to analyze and design control systems	Cognitive (Apply, Analyze)
CLO5	Apply software tools (PLC) for designing and analyzing of control systems	Cognitive (Apply, Analyze, Evaluate)

Course Contents:

Sl. No.	Course Content (as Summary)
1.	Introduction: Introduction to Control System: Conventional control system, steady state response to step, ramp, and parabolic inputs, transient response, poles and zeros, frequency response from pole-zero diagram, Routh's stability criterion; block diagrams, canonical forms, transfer functions and signal flow graph, root locus, frequency response, Nyquist's stability criterion.
2.	Modern Control System: Introduction, state variable analysis, controllability and observability, application of Eigen value, linear control system design by state feedback, pole placement.
3.	Introduction to Industrial Automation and Control: Definitions of Automation and Control, Relation between Automation and Information Technology, Role of Automation in Industry, Functional elements of Industrial Automation Systems that perform a variety of functions related to Instrumentation, Control, Supervision and Operations Management, Automation using PLC and SCADA.
4.	Introduction to PLC: Definitions & History of PLC, PLC systems, I/O modules, CPU, Memory, Power supplies, Programmer, Advantages & limitations, Selection criteria for PLC. Programming of PLC: IEC Programming Standard, Ladder Diagram, Function Block Diagram, Instruction List and Structured Text, Construction of PLC ladder diagram, Basic components & their symbols in ladder diagram, Fundamentals of ladder diagram, Boolean logic & relay logic, Analysis of rungs, Input ON/OFF switching devices, Input analog devices, Output ON/OFF devices, Output analog devices

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√	√									
CLO3		√	√									
CLO4		√	√									
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Participation
CLO2	- Lecture - Presentation - Discussion/Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation

CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial ▪ Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Benjamin C Kuo, : Automatic Control Systems”, Prentice Hall of India.
2. M. Gopal, : Control Systems- Principle of Design
3. I. J. Nagrath, : Control Systems Engineering”

Course Code:	EEE-3204
Course Title:	Control System Engineering Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-3203

Rationale of the Course: This lab course provides strong understanding of control system analysis and design with software simulation and PLC trainer board

Course Objectives:

- To gain a strong knowledge on MATLAB software.
- To get the basic knowledge on practical control system and PLC applications.
- To get the knowledge on applications of machines & electronic devices with control systems.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Formulate transfer function for given control system problems.	Psychomotor (Imitation, Manipulation)
CLO2	Plot Root Locus and Bode plots for given control system model	Psychomotor (Manipulation, Precision)
CLO3	Use PLC controllers for given control system model	Psychomotor (Manipulation, Precision)
CLO4	Design Lead, Lag, Lead-Lag systems in control systems	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Introduction to system designing by MATLAB and simulate by analyzing time response PID controller system modeling and root locus
2.	Control system design and analysis using MATLAB and Simulink (Bode diagram, Nyquist diagram, state diagram)
3.	Study of PID controller for speed control of DC motor
4.	Introduction to Programmable Logic Controller (PLC)
5.	System implementation in PLC

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Participation
CLO2	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO3	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO4	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-3205
Course Title:	Communication Engineering I
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	N/A

Rationale of the Course: Communication engineering has several potential and presents many job opportunities so this course is offered to gain necessary knowledge within the field and prepare potential students to work within this industry. Course is focused to enable the learners to be familiar with basics of a communication system and different noises, analog modulation and demodulation techniques, radio systems, basics of digital communications including different waveform coding techniques. Television systems also have been introduced.

Course Objectives:

- Introduce with the fundamental concepts in communication engineering.
- Understand different modulation and demodulation techniques and related problem solving.
- Gain basic idea about radio and TV systems including transmitter and receiver circuits design.
- Make students capable enough to manage small and large size communication systems.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Understand basics of communication systems, modulation and demodulation, radio system, waveform coding techniques and TV systems.	Cognitive (Understand)
CLO2	Solve different problems related to noise, modulation, demodulation, transmitter and receiver circuits of radio and TV.	Cognitive (Apply, Analyze)
CLO3	Sketch and resolve problems related to Analog to digital conversion and different coding in digital communication.	Cognitive (Apply, Analyze)
CLO4	Investigation of quantization noise, channel noise, SNR in digital communication, noise limiter and cross modulation, transmitting and receiving antenna in radio system in radio and TV.	Cognitive (Evaluate)
CLO5	Justify among modulation techniques and waveform coding techniques.	Cognitive (Evaluate)
CLO6	Design of transmitter and receiver circuits of radio and TV systems.	Cognitive (Create)

Course Contents:

Sl. No.	Course Content (as Summary)
1.	Overview of Communication Systems and Introduction to Noise: Basic principles, fundamental elements, system limitations, message source, bandwidth requirements, Transmission types- base-band transmission, carrier transmission; transmission media types, bandwidth and transmission capacity. Sources of noise, characteristics of various types of noise and signal to noise ratio.
2.	Analog Modulation and Demodulation: Amplitude modulation (AM)- introduction, DSB, SSB, VSB, quadrature; spectral analysis of each type, envelope and synchronous detection; angle modulation instantaneous frequency, frequency modulation (FM) and phase modulation (PM), spectral analysis, demodulation of FM and PM.
3.	Radio System: Radio Transmitter- classification, elements of AM, FM and SSB transmitter, master oscillator, mixer, RF power amplifier, pre-emphasis circuits, Radio Receiver- classification, elements of AM, FM and SSB receiver, AGC, AFC, de-emphasis circuits, noise limiter, cross modulation, Design of radio transmitter and receiver circuits.
4.	Introduction to digital Communication: Baseband digital transmission, Limitations, Channels for digital communication, AWGN channel model, bit error rate of a baseband transmission system, channel capacity theorem, channel coding theorem.
5.	Waveform Coding Techniques: Sampling- sampling theorem, Nyquist criterion, aliasing, instantaneous and natural sampling, flattopped sampling; message reconstruction from its samples, PAM, PCM, quantization noise, channel noise, SNR, robust quantization, differential PCM, delta modulation (DM) principle, adaptive DM; line coding- formats and bandwidths.
6.	TV System: Elements of TV system, principle of operation, TV signals generation, composite video signal, TV transmitter and receiver, transmitting and receiving antenna, picture tube; Introduction to color TV-compatibility, three color theory, Grassman's law, color display tube; VCR, CCTV, CATV, MATV, TV Booster

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√	√									
CLO3		√	√									
CLO4			√	√								
CLO5			√	√								
CLO6			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion/Online Resource	- Quiz - Tutorial/Assignment - Participation
CLO2	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO6	- Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination

Learning Materials:

- | | |
|----------------------|--|
| 1. B. P. Lathi | : Modern Digital and Analog Communication System |
| 2. Simon Haykin | : Digital Communication Systems |
| 3. Kennedy and Davis | : Electronic Communication Systems |
| 4. Roddy and Coolen | : Electronic Communications |
| 5. G. K. Mathur | : Radio Engineering |
| 6. B. Grob | : Basic TV |
| 7. Gulati | : Monochrome and Color TV |

Course Code:	EEE-3206
Course Title:	Communication Engineering I Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-3205

Rationale of the Course: In this course students will perform experiments to practically verify the theories learned in the theory course EEE-3205.

Course Objectives:

- To show how AM works- SSB and DSB.
- To provide hands on experience of demodulating AM, FM and PM using practical circuits.
- To show the process of designing and constructing radio transmitter and receiver circuits.
- To provide knowledge about pulse code modulation (PCM).

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Understand the operation of practical modulation circuit for AM, FM and PM.	Psychomotor (Imitation, Manipulation)
CLO2	Understand the basic elements of TV system.	Psychomotor (Imitation, Manipulation)
CLO3	Design and construct a simple PCM circuits.	Psychomotor (Precision, Articulation)
CLO4	Design practical radio transmitter and receiver circuits.	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Design, construction and testing of amplitude modulator (AM)
2.	Design, construction and testing of frequency modulator (FM)
3.	Design, construction and testing of Pulse modulator (PM)
4.	Design, construction and testing of a FM transmitter.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2	√	√										
CLO3			√	√	√							
CLO4			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO2	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO3	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO4	- Demonstration - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-3207
Course Title:	Electrical Properties of Materials
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	N/A

Rationale of the Course: In the present modernization, materials have contributed to the advancement of a number of technologies, including medicine & health, information & communication, national security & space, transportation, structural materials, textiles, personal hygiene, agriculture & food science and the environment. Material is the root of all sciences and engineering. This course is intended to build up strong base in structures and properties of materials. Hence, this course is designed to help in understanding different types of materials, their structure, different macroscopic and microscopic properties.

Course Objectives:

- To understand the basic building blocks of solid, band theory of solids, dielectric properties and dielectric parameters of materials.
- To explain the materials properties using classical theory.
- To introduce quantum theory for better understanding of properties of solid.
- To know various magnetic materials and their properties.
- To explain superconductivity property of solid and their models.

Course Learning Outcomes (CLOs): On successful completion of the course, the student should be able to-

CLOs	Description	Learning Domain with Level
CLO1	Explain the basic concepts that are used to describe the structure and physical properties of crystalline substances to formulate the theory of X-ray diffraction by crystal and apply this knowledge to determine the crystal structure.	Cognitive (Understand)
CLO2	Understand a wide spectrum on the established theoretical models to describe the electrical properties of materials and to formulate the problem of electrons in a periodic potential and develop a framework of the origin of energy band that explains the physical properties of solids.	Cognitive (Understand)
CLO3	Understand and analyze the properties of dielectric materials to understand the relation between the electron structure of crystalline solids and their dielectric properties.	Cognitive (Understand, Analyze)

CLO4	Acquire the knowledge on the theories of different types of magnetic materials underlying the electronic and magnetic behavior of materials and make use of its applications.	Cognitive (Understand, Analyze)
CLO5	Get the basic theoretical knowledge on superconducting materials and to learn the properties of superconductors in the frame of BCS theory.	Cognitive (Apply)
CLO6	Explain the fundamental principles of nanotechnology and their applications and to explain the importance of reduction in material dimensionality and its relationship with materials properties.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Elementary Materials Concepts: Crystalline and amorphous solids, Bravais and reciprocal lattice, crystal defects and their significance, X-rays and their generation, Bragg's law and Laue equations, Experimental methods of X-ray diffraction, Bonding and types of solids.
2.	Electrical Properties: Electrical and thermal conduction in solids, Free electron gas, classical-Drude theory of metal, failure of classical theory, Sommerfeld theory, Fermi-Dirac distribution, density of energy states, Electrons in periodic solids, band theory of solids, Bloch theorem and Kronig-Penny model, E-K diagram & its physical meaning, electron effective mass, direct and indirect band gap of semiconductor.
3.	Dielectric properties: Polarization and relative permittivity, electronic and ionic polarizations, orientational polarization, polar and nonpolar dielectrics, Langevin's theory of polarization, Clausius-Mossotti equation, interaction between field and matter, frequency dependent dielectric constant and dielectric loss, dielectric strength and insulation breakdown, piezo-electricity.
4.	Magnetic Properties: Magnetization, dipole and atomic moment, magnetization vector, field intensity, permeability and susceptibility, magnetic moments, origin of different magnetism, classical and quantum theory of diamagnetic and paramagnetic, ferromagnetism: Molecular field theory, Ferromagnetic domain, ferrimagnetic and antiferromagnetic materials and their applications, permanent magnetic materials, soft and hard magnetic materials, magnetic recording materials.
5.	Superconductivity: Basic properties of superconductors, Zero resistance and Meissner effect, Type-1 and Type-2 superconductors, thermodynamic of superconductors, London equations, Penetration depth, and Coherence length, BCS theory of superconductivity, Quantum theory of superconductivity, Josephson Effect, Superconducting magnets.
6.	Nanotechnology: Low dimension structures, Quantum well, Quantum wires, Quantum dots, Metal, Semiconductor and organic multilayer's, Fullerenes, Nanotubes, Porous silicon and other special nanomaterials. Electronic states in 2-D, 1-D and 0-D solids. Strong and weak confinement.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2	√	√										
CLO3		√	√									
CLO4			√	√								
CLO5		√	√									
CLO6		√	√		√	√						

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion - Journal Paper	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation

CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO6	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | |
|---------------------------------|--|
| 1. Safa O. Kasap | : Principles of Electronic Materials and Devices |
| 2. J. Dekker | : Solid State Physics |
| 3. C. Kittel | : Introduction to Solid State Physics |
| 4. J. Mckelvy | : Solid State and Semiconductor Physics |
| 5. L.J. Azaroff and J.J. Brophy | : Electronic Process in Materials |
| 6. C.A. Wert and R.M. Thomson | : Physics of Solids |
| 7. Rogers, Pennathur, Adams | : Nanotechnology: Understanding Small Systems |

Course Code:	EEE-3209
Course Title:	Signals and Linear Systems
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	MATH-2107

Rationale of the Course: Signals and Linear Systems is an introduction to analog signal processing. Signal processing forms an integral part of engineering systems in many diverse areas, including seismic data processing, communications, speech processing, image processing, defense electronics, consumer electronics, and consumer products. The course presents and integrates the basic concepts for signals and systems. Signal and system representations are developed for both time and frequency domains. These representations are related through the Fourier transform and its generalizations as well as Laplace transform, those are explored in detail.

Course Objectives:

- To introduce the students with different types of signals and systems, properties of systems.
- To understand differential equation, the meaning and practical relevance of LTI system properties such as linearity, time invariance, stability, causality, impulse response, transfer function, and frequency function.
- To introduce the students with Fourier series, Fourier transformation, Laplace transformation, Properties of Fourier and Laplace transformation.
- To describe the student about the application of Fourier and Laplace transformation.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the basic concepts of signals and systems.	Cognitive (Understand)
CLO2	Explain the operation of the random variables and processes.	Cognitive (Understand Apply)
CLO3	Apply LTI systems to calculate their output signal for impulse response, convolution, and transfer function and frequency response.	Cognitive (Apply, Analyze)
CLO4	Apply the Fourier and Laplace transforms in problem solving.	Cognitive (Apply, Analyze)
CLO5	Design and analyze systems using Fourier transform and Laplace transform.	Cognitive (Apply, Analyze, Evaluate, Create)

Course Content

Sl. No.	Course Content (as Summary)
1.	Introduction: Signals, Systems and Signal Processing, Classification of Signals, Analog-to-Digital and Digital-to-Analog Conversion, Operations on signals, Properties of signals.
2.	Random Variables and Processes: Probability, Cumulative distribution function, Probability Density Function, Tchebycheff's Inequality, The Gaussian Probability Density, The Error Function, The Rayleigh Probability Density, Correlation between random variables, the central-limit theorem, Random Processes, Autocorrelation, Power Spectral Density of a Sequence of Random Pulses, Power Spectral Density of Digital Data, Effect of rudimentary filters of digital data, the complimentary error function.
3.	Discrete Time Signals and Systems: Discrete Time Signals, Discrete Time Systems, Analysis of Discrete Time Linear Time-Invariant Systems, Discrete Time Systems Described by Difference Equations, Correlation of Discrete-Time Signals, Time Domain Representations for Linear Time Invariant Systems, Convolution Properties of convolution Difference and differential equations - characterizing solutions.
4.	Transformation: Fourier Representations of Signals, Discrete time periodic signals - the discrete time Fourier series Continuous time periodic signals - the Fourier series, Discrete time non-periodic signals - the discrete time Fourier transform Continuous time non-periodic signals - the Fourier transform Properties of Fourier representations. Applications of Fourier Representations, Frequency response from time-domain system, descriptions Fourier transform representations for periodic signals, Convolution and modulation revisited - mixing periodic and non-periodic signals,
5.	The Laplace Transformation: Definition, Convergence Properties, Inversion Solving, Differential Equations, Transform Analysis of Systems.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√								
CLO5			√	√	√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Participation
CLO2	- Lecture - Presentation - Problem Solving - Discussion/Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Problem Solving - Discussion	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Problem Solving - Discussion	- Quiz - Tutorial/Assignment - Semester end final examination - Participation

Learning Materials:

- | | |
|--|------------------------|
| 1. Oppenheim A.V., Willsky A. S. and Young I. T | : Signals and Systems |
| 2. Simon Haykin and Barry Van Veen | : Signals and Systems |
| 3. R. F. Ziemer, W. H. Tranter and J. D. R. Fannin | : Signals and Systems |
| 4. M. J. Roberts | : Signals and Systems. |

Course Code:	EEE-3212
Course Title:	Electrical Services Design
Course Credit:	1.0
Course Type:	Core Course
Pre-requisite:	N/A

Rationale of the Course: This course provides the student a practical experience about the electrical layout, wiring, protections and wattage rating of wire in a building. Students will also know about the electrical wiring, symbols and regulations used in a building. Building securities like fire alarm, smoke alarm and CCTV systems implementation will be taught in the courses. Implementation of various utility services like water supply and gas supply will be covered in the course. Execution of communication services like intercom, internet and telephone systems in a building.

Course Objectives:

- To understand domestic wiring procedures practically.
- To implement the electrical wiring and to select the wattage rating of wires in a building.
- To provide hands on experience of three phase connection and electrical protections.
- To show how the demand of security systems in a building is fulfilled.
- To execute various utility systems need in a building design.
- To know how to install air conditioning, lift and heating systems.

Course Learning Outcomes (CLOs): On successful completion of the course, the student should be able to-

CLOs	Description	Learning Domain with Level
CLO1	Choose the proper electrical wires, its ratings, fuses, bus-bars, circuit breakers, switches and rating of a power transformer for a building.	Psychomotor (Imitation, Manipulation)
CLO2	Inspect the electrical faults and problems in a building.	Psychomotor (Imitation, Manipulation)
CLO3	Solve the security and alarm systems in a building.	Psychomotor (Precision, Articulation)
CLO4	Examine the power requirement for a building.	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	To design and implement a stair case wiring
2.	To design and implement electrical wiring of a multi-storied building
3.	To assemble a residential house wiring using fuse, switch, indicator, lamp and energy meter
4.	To implement and test of Fire and Smoke alarm systems in an office building
5.	To calculate and measure the maximum power requirements in a building
6.	To install and test a security system in an office

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√					√						
CLO2			√									
CLO3				√		√						
CLO4				√	√	√						

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Online video demonstration for high voltage safety requirements	- Quiz/Lab performance - Semester end final examination - Participation
CLO2	Practical implementation of electrical wiring using wires and light, Fans and switch	- Quiz/Lab performance - Semester end final examination - Participation
CLO3	- Discussion, - Experiment, - Report writing	- Quiz/Lab performance - Semester end final examination - Participation
CLO4	Online video demonstration for high voltage safety requirements	- Quiz/Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-3200
Course Title:	Viva-voce
Course Credit:	1.0
Course Type:	Core Course
Pre-requisite:	Knowledge of 3 rd Year courses

Rationale of the Course: Communication plays very vital role in aspect of life. Viva-voce is such an opportunity where a learner is judged how he/she can communicate with others from his/her gained knowledge. The objective of this course is to verify the students how well they can express their knowledge orally.

Course Objective:

- To assess the students ability to communicate with another person.
- To supplement the information obtained through other evaluation techniques.
- To identify and analyze the students presence of mind.
- To evaluate the students spontaneity and mannerism.

Course Learning Outcomes (CLOs): By the end of the all courses in First Year, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Convince the audience orally about their knowledge from the courses in 3 rd Year.	Affective(Receiving, Responding, Valuing)

Course Content:

Sl. No.	Course Contents
1.	3 rd Year Course Curriculum

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√								√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation/Discussion	- Participation; - Presentation

Learning Materials: Knowledge from 3rd Year Curriculum

B.Sc. Engineering Fourth Year, First Semester Examination 2024

Course Code:	EEE-4000
Course Title:	Thesis/Project
Course Credit:	1.0
Course Type	Capstone Course

Rationale of the Course: The undergraduate thesis project should provide the student with knowledge of how to seek scientific facts and how to plan, carry out and present scientific work as well as theoretical and practical specialization within the area of Electrical and Electronics Engineering.

Course Learning Outcomes (CLOs): on successful completion of the course, the student should be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Show basic knowledge on how to pursue scientific fact.	Cognitive (Understand)
CLO2	Have theoretical and practical professional specialization within the EEE discipline including understanding of the current research questions.	Cognitive (Understand)
CLO3	Show basic knowledge on planning and performance of a scientific work.	Cognitive (Apply, Analyze)
CLO4	Show basic knowledge on analysis of scientific data.	Cognitive (Apply)

Course Content:

The undergraduate thesis project within the field of Electrical and Electronics research is an individual study that should include hypothesis testing that will substantiate new data.

The undergraduate thesis project includes search, studies and summary of scientific literature, practical work in close relation to ongoing research of the corresponding project, compilation and critical analysis of the results, and oral and written presentation.

The undergraduate thesis project is mastered under individual supervision. The supervision includes how to perform a scientific study and how to orally and in writing present gathered data in good scientific manner. The supervisor must have documented scientific experience.

The studies take place under individual supervision and include literature studies, seminar exercises and practical work.

Project Dissertation/Report must be submitted by the end of the second semester and make an oral defense of the project. Project Evaluation will be made in the second semester.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2	√	√										
CLO3			√	√								
CLO4			√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation/Problem Solving ▪ Discussion	▪ Quiz ▪ Participation
CLO2	▪ Presentation ▪ Problem Solving ▪ Discussion/Demonstration	▪ Quiz ▪ Participation
CLO3	▪ Presentation ▪ Problem Solving/Discussion/Group Discussion/Demonstration	▪ Quiz ▪ Participation
CLO4	▪ Presentation ▪ Problem Solving/Discussion	▪ Quiz ▪ Participation

Learning Materials:

1. As per the guideline of the corresponding supervisor.

Course Code:	EEE-4100
Course Title:	Industrial Attachment Training/Internship
Course Credit:	1.0
Course Type	Capstone Course

Rationale of the Course: Students will be attached with the Electrical and Electronic Engineering related industries/service agencies for two weeks to take Professional/Industrial/In-Plant training. This training is to be organized after completion of their third year odd semester or during any vacation in Third year second semester to gain practical knowledge.

Course Objectives:

- To give students learn from practical environment and important problems in the industries or companies.
- To learn from the working engineers about the common problems and solutions in the attached industries.
- To give them an opportunity to get along with staff of the industries and learn how to develop cordial relationship among the co-workers

Course Learning Outcomes (CLOs): On successful completion of the course, the student should be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Learn the practical knowledge about industrial equipment.	Affective (Receiving, Responding)
CLO2	Learn from the working engineers how to operate this equipment.	Affective (Receiving, Responding)

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√							√	√		
CLO2	√	√							√	√		

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Discussion	▪ Participation
CLO2	▪ Presentation ▪ Discussion/Demonstration	▪ Participation

Learning Materials:

1. As per the guideline of the corresponding supervisor.

Course Code:	EEE-4101
Course Title:	Power System Analysis II
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-3201

Rationale of the Course: This course will teach fundamentals on power system modeling, dynamics, stability and control, introduce the methods and tools for planning and operating a modern power grid to meet reliability criteria under disturbances. Students are required to review literature on recommended topics to gain deeper insight on emerging techniques for power systems analysis.

Course Objectives:

- To understand the concepts of transmission lines characteristics and transmission cost and tariff.
- To learn about underground and overhead transmission lines.
- To obtain stability and economic operation of power system

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the modern power systems, interconnections and load flow.	Cognitive (Understand)
CLO2	Perform interconnected power systems.	Cognitive (Apply)
CLO3	Analyze load flow in power system analysis.	Cognitive (Analyze)
CLO4	Analyze and calculate system faults and stability in power systems.	Cognitive (Analyze, Evaluate)
CLO5	Model and analyze of power system control.	Cognitive (Apply, Analyze, Evaluate)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction of Modern Power Systems: Power and energy resources, Overview of modern power system, Operations and planning, Classification and definitions of power system stability
2.	Interconnected Power System: Introduction, Real power, Frequency balance, Reactive power, Voltage balance, Node equations, Bus admittance matrixes, Applications of Bus admittance matrixes in network analysis, Basic concept of Bus impedance matrixes
3.	Load Flow Analysis: Basic complex power flow equations for a power system networks, Data for Load flow studies, Iterative approaches for solving power flow equations-Gauss-Seidal method, Newton- Raphson-Methods
4.	Power System Fault Calculation: Definition and purpose of fault calculation, Types of faults in power system, Symmetrical fault calculations, Calculation of short circuit MVA
5.	Power System Control: Modeling on frequency regulation and control, Modeling on voltage regulation and control
6.	Power System Stability: Small-signal stability, Power oscillation in interconnected systems, Power system stabilizer, Transient stability, Direct methods, Numerical methods, Voltage stability, Voltage collapse, Analysis methods and mitigation measures

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2			√									
CLO3			√	√								
CLO4		√	√	√								
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Problem Solving - Discussion	- Quiz - Tutorial/Assignment - Participation
CLO2	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Participation
CLO3	- Lecture - Problem Solving - Discussion/Group Discussion	- Quiz - Tutorial/Assignment/group Assignment - Participation
CLO4	- Lecture - Presentation - Discussion	- Quiz - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Problem Solving/Discussion	- Tutorial/Assignment - Semester end final examination - Participation

Learning Materials

- | | |
|--|--------------------------------------|
| 1. Hadi Saadat | : Power System Analysis |
| 2. Prabha Kundur | : Power System Stability and Control |
| 3. John Grainger and William Stevenson | : Power System Analysis |

Course Code:	EEE-4102
Course Title:	Power System Analysis II Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-4101

Rationale of the Course: In this course students will get hands-on experience on performing experiments for verifying the theories and concepts learned from the course EEE-4101.

Course Objectives:

- To show how the inductance and capacitance of transmission line are calculated.
- To perform power flow analysis effectively with Iterative Newton-Raphson algorithm
- To provide hands on experience for both symmetrical and unsymmetrical fault analysis
- To perform fault current analysis for determining ratings of relays and circuit breakers.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the concept of MATLAB programming.	Psychomotor (Imitation,)
CLO2	Determine ratings of relays, circuit breakers, transmission line parameters, power flow analysis using MATLAB programming.	Psychomotor (Imitation, Manipulation)
CLO3	Analyze Line to Ground (L-G) Fault analysis of a Single Phase and three phase Transmission Line.	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Determination of inductance and capacitance of transmission line using MATLAB
2.	Analysis of power flow with Iterative Newton-Raphson method
3.	Measurement of Line to Ground (L-G) Fault current of a Single Phase Transmission Line
4.	Measurement of Single Line to Ground Fault (L-G) current of a Three Phase Transmission Line
5.	Determination of ratings of relays and circuit breakers using MATLAB

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											√
CLO2		√	√									
CLO3		√	√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Using MATLAB programming tool	▪ Quiz/Lab performance ▪ Semester end final examination ▪ Participation
CLO2	Using MATLAB programming tool	▪ Quiz/Lab performance ▪ Semester end final examination ▪ Participation
CLO3	Using MATLAB programming tool	▪ Quiz/Lab performance ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-4103
Course Title:	Communication Engineering II
Course Credit:	3.0
Course Type:	Core
Pre-requisite:	EEE-3205

Rationale of the Course: Digital communications has become a fundamental industry with stable long-term job growth that connects the world. In this digital communication era learner need to have basic knowledge in digital communication systems. Course is intended to enable the learners to familiar with digital modulation techniques, switching systems, detection, estimation and error correction techniques. Also introduce with different multiplexing, multiple access techniques along with different switching systems.

Course Objective:

- To introduce the students with different digital modulation and demodulation techniques.
- To present the details of error correction techniques
- To differentiate and compare different multiplexing and multiple access techniques.
- To introduce different coding schemes digital system and the modern TV system
- To explain the mechanism of different analog and digital switching system used in telecommunication networks.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand and explain digital modulation techniques, detection, estimation and error correction coding, multiplexing techniques and multiple access techniques, analog and digital switching system, traffic engineering.	Cognitive (Understand)
CLO2	Analyze and solve problems related to detection, estimation and error correction, multiplexing and multiple access techniques.	Cognitive (Analyze)
CLO3	Compare different multiplexing and multiple access techniques.	Cognitive (Analyze)
CLO4	Investigate analog and digital switching systems for traffic engineering.	Cognitive (Evaluate)
CLO5	Justify different multiple access techniques in mobile communication systems.	Cognitive (Evaluate)
CLO6	Design of different filters and trafficking systems in digital communication.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Digital Modulation Techniques: Binary modulation techniques: ASK, PSK, and FSK, Detection of ASK, PSK, and FSK, Quadrature modulation techniques, M-ary modulation techniques, power spectra, effect of inter-symbol interference.
2.	Detection, Estimation and Error Correction Coding: Model of digital communication system, detection of signals in noise, probability of error, correlation receiver, matched filter receiver. Estimation: MLE, Weiner filters, Adaptive filters, linear prediction. Bit error rate calculation of a digital link, digital link design. Block codes, cyclic codes, systematic and nonsystematic cyclic codes, convolutional codes, Trellis codes, decoding techniques.
3.	Multiplexing Techniques and Multiple Access Techniques: FDM, TDM, SDH, PDH, SONET, WDM, SONET over WDM, FDMA, TDMA, CDMA and SDMA. Introduction to 2G and 3G mobile communication systems, DSL, VDSL, HDSL, Fiber Access Network, FTTX
4.	Introduction to Analog and digital Switching System: Stronger and, Crossbar switching systems, Telephone Networks: Subscriber loop systems, 79 switching hierarchy and routing, Transmission plan, Transmission systems, numbering plan, charging plan, signaling techniques, In channel signaling, Common channel signaling, Stored program control, Software architecture, Application software, Enhanced services, Space division switching, time division switching, blocking probability and multistage switching, and digital memory switch.
5.	Traffic Engineering: Traffic characterization, Grade of services and blocking probability, Modeling switching systems, Blocking models and loss estimates, delay system and queuing.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√	√									
CLO3			√	√								
CLO4			√	√	√							
CLO5				√	√							
CLO6			√	√		√						

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Problem Solving ▪ Discussion	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Problem Solving ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Problem Solving ▪ Discussion/Group Discussion	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Problem Solving ▪ Discussion	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Problem Solving ▪ Discussion	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO6	▪ Lecture ▪ Presentation ▪ Problem Solving ▪ Discussion	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

1. B. P. Lathi : Modern Digital and Analog Communication System
2. Simon Haykin : Digital Communication Systems
3. Roddy and Coolen : Electronic Communications
4. George Kennedy : Electronic Communication System
5. Herbert Taub and Donald L. Schilling : Principles of Communication Systems

Course Code:	EEE-4104
Course Title:	Communication Engineering II Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-4103

Rationale of the Course: In this course students will perform experiments to practically verify the theories learned in the theory course EEE 4103.

Course Objectives:

- To show how ASK, FSK and PSK.
- To provide practical experience of detecting ASK, FSK and PSK using practical circuits.
- To show the process of designing and constructing multiplexing techniques.
- To provide knowledge about telecommunication switching networks.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the operation of practical modulation circuit for ASK, FSK and PSK.	Psychomotor (Imitation, Manipulation)
CLO2	Understand the basic elements of telecommunication switching system.	Psychomotor (Precision, Articulation)
CLO3	Design practical circuits for ASK, FSK and PSK detection.	Psychomotor (Precision, Articulation)
CLO4	Design and construct a simple FDM and TDM systems.	Psychomotor (Precision, Articulation)

Course content:

Sl. No.	List of the Experiments
1.	Design, construction and testing of pulse amplitude modulator (PAM)
2.	Design, construction and testing of pulse width modulator (PWM)
3.	Design, construction and testing of pulse position modulator (PPM)
4.	Design, construction and testing of amplitude shift keying (ASK) modulator
5.	Design, construction and testing of frequency shift keying (FSK) modulator
6.	Design, construction and testing of pulse shift keying (PSK) modulator
7.	Design, construction and testing of pulse code modulator (PCM)
8.	Design and construct a simple FDM and TDM systems

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√			√							
CLO3			√		√							
CLO4			√		√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO2	- Discussion, - Experiment, - Report writing	- Lab performance - Semester end final examination - Participation
CLO3	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO4	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-4105
Course Title:	Digital Signal Processing
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-3209

Rationale of the Course: This course presents the concept of signals and systems in the time and frequency domain and their processing. Students will learn about Digital Signal Processing in regards to the common mathematical tools such as- impulse response, solution of difference equation, z-transform, discrete time harmonic analysis, discrete Fourier transform. Techniques for processing signals are introduced including FIR and IIR filters.

Course Objectives:

- To classify signals and systems and their mathematical representation.
- To realize the impulse response of different discrete time systems.
- To apply different transformation techniques, such as z- and Fourier transforms and their computations.
- To study about filters and their design for digital implementation.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand different signals and systems and their processing.	Cognitive (Understand)
CLO2	Find impulse response of discrete-time systems.	Cognitive (Analyze)
CLO3	Convert signals and systems from time domain to z or frequency domain and vice-versa.	Cognitive (Analyze)
CLO4	Design and implement digital filters for filtering and prediction of signals.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction to Digital Signal Processing (DSP): Digital signals and systems: Operations in digital signal processing, the scope of DSP, analog to digital conversion, frequency Domain Effects of Sampling: Periodic repetitions in frequency domain due to sampling in time domain, recovery of continuous-time signal from its samples (reconstruction), role of anti- aliasing and reconstruction filters, examples of aliased signals (show how waveform is distorted), impulse response, finite impulse response (FIR) and infinite impulse response (IIR) of discrete-time systems, difference equation.
2.	Discrete Transformations: Discrete Fourier series, the Discrete-Time Fourier Transform, discrete Fourier transform (DFT) and fast Fourier transform (FFT): Forward and inverse transforms; coefficient ordering; time and frequency resolution; periodic extension, zero padding and modulo-M reduction; properties of the DFT, circular convolution; Cooley- Tukey decomposition, recursive application, radix-2 FFTs , time and frequency decimation, computational complexity.
3.	Z-Transformation: Basic Theory: Background idea behind the z-transform (solution to LTI discrete-time diff. eq.), calculation of z-transform and its inverse (briefly), regions of convergence, Properties of z-transforms: role in solution of discrete-time LTI systems, convolution property and graphical interpretation of the convolution operation, z-transforms of cascaded systems, stability and causality, Realization and frequency Response: Frequency response (Magnitude and Phase), representation of LTI systems with rational polynomials, block-form implementations of a rational polynomial transfer function
4.	Digital Filters: FIR filters- linear phase filters, specifications, design using window, optimal and frequency sampling methods; IIR filters- specifications, design using impulse invariant, bi-linear z-transformation, least-square methods, linear phase, Butterworth, Chebychev , Inverse Chebychev , Bessel and elliptic filters, finite precision effects in implementing digital filters.
5.	Implementing Digital Filters: Block-diagram representations; direct forms, cascade forms, first and second-order factors, parallel forms; feedback loops transposed forms; linear-phase FIR structures.
6.	Wavelets: Short time Fourier transform; fundamentals of wavelets, wavelet transform (continuous and discrete), time - frequency density and orthogonal bases.
7.	Speech Processing: Speech production model, Speech analysis and Synthesis system, Vocoder.
8.	Image Processing: Image acquisition, Storage and processing, Structure of the human eye, A simple in model, Introduction to image transforms, Restorations and compressions, Colour image process, Morphological image processors, DSP processors.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√	√									
CLO3		√	√									
CLO4			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Participation
CLO2	- Lecture - Presentation - Problem Solving - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Problem Solving - Discussion/Group Discussion	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Problem Solving - Discussion	- Quiz - Tutorial/Assignment - Semester end final examination - Participation

Learning Materials:

1. J. G. Proakis and D.G. Manolakis : Digital Signal Processing: Principles, Algorithms and Applications
2. A.V. Oppenheim, R. W. Schafer and J.R. Buck : Discrete-Time Signal Processing
3. Dag Stranneby and William Walker : Digital Signal Processing and Applications

Course Code:	EEE-4106
Course Title:	Digital Signal Processing Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-4105

Rationale of the Course: This course intended to verify different methods to analyze digital signals in time and frequency domain using computer programming and simulation software.

Course Objectives:

- Introduce basic concepts in MATLAB programming and use of built-in and user-defined functions for DSP.
- Develop mathematical and analytical skills necessary to analyze digital signals by computer.
- Introduce application of DSP including spectral analysis, digital filters etc.

Course Learning Outcomes (CLOs): By the end of the course student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the different types of continuous and discrete time signals and ADC process.	Psychomotor (Imitation, Manipulation)
CLO2	Calculate Fourier transform of digital sequences.	Psychomotor (Manipulation, Precision)
CLO3	Simulate various applications of DSP using MATLAB tools	Psychomotor (Precision, Articulation)
CLO4	Design and implement different types of filter use in digital signal processing.	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Getting started with MATLAB for digital signal processing
2.	Represent/plot continuous and discrete time signals
3.	Calculate correlation and convolution between two DT sequences
4.	Convert analog signal to digital signal and analyze the effect of sampling rate
5.	Calculate Fourier transform of different DT sequences and functions and plot spectrum
6.	Design and simulate FIR and IIR filters using MATLAB tools

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√	√									
CLO3			√	√	√							
CLO4			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Participation
CLO2	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO3	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO4	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Lab guide/Manuals

B.Sc. Engineering Fourth Year, Second Semester Examination 2024

Course Code:	EEE-4000
Course Title:	Thesis/Project
Course Credit:	3.0
Course Type:	Capstone Course
Pre-requisite:	N/A

Rationale of the Course: The undergraduate thesis project should provide the student with knowledge of how to seek scientific facts and how to plan, carry out and present scientific work as well as theoretical and practical specialization within the area of Electrical and Electronics Engineering.

Course Learning Outcomes (CLOs): On successful completion of the course, the student should be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Show basic knowledge on how to present scientific work.	Cognitive (Apply)
CLO2	Be able to search scientific literature.	Cognitive (Analyze)
CLO3	Be able to summaries scientific literature.	Cognitive (Evaluate)
CLO4	Be able to discuss scientific data related to the question at hand.	Cognitive (Evaluate)
CLO5	Be able to present scientific data and conclusions in written and oral form addressed to different groups.	Cognitive (Analyze, Evaluate, Create)
CLO6	Complete a supervised research project and submit a dissertation as the requirement of partial fulfillment of B. Sc. Engineering Degree to the department.	Affective (Receiving, Responding, Valuing)

Course Content:

The undergraduate thesis project within the field of Electrical and Electronics research is an individual study that should include hypothesis testing that will substantiate new data.

The undergraduate thesis project includes search, studies and summary of scientific literature, practical work in close relation to ongoing research of the corresponding project, compilation and critical analysis of the results, and oral and written presentation.

The undergraduate thesis project is mastered under individual supervision. The supervision includes how to perform a scientific study and how to orally and in writing present gathered data in good scientific manner. The supervisor must have documented scientific experience.

The studies take place under individual supervision and include literature studies, seminar exercises and practical work.

Project Dissertation/Report must be submitted by the end of the Second semester and make an oral defense of the project. Project Evaluation will be made in the Second semester.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1			√	√								
CLO2			√	√								
CLO3				√	√							
CLO4				√	√							
CLO5			√	√	√				√	√		
CLO6									√	√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation/Problem Solving ▪ Discussion	▪ Quiz ▪ Participation
CLO2	▪ Presentation ▪ Problem Solving ▪ Discussion/Demonstration	▪ Quiz ▪ Participation
CLO3	▪ Presentation ▪ Problem Solving/Discussion/Group Discussion/Demonstration	▪ Quiz ▪ Participation

CLO4	▪ Presentation ▪ Problem Solving/Discussion	▪ Quiz ▪ Participation
CLO5	▪ Presentation ▪ Problem Solving/Discussion	▪ Quiz ▪ Participation
CLO6	▪ Presentation ▪ Problem Solving/Discussion/Group	▪ Quiz ▪ Participation

Learning Materials:

1. As per the guideline of the corresponding supervisor

Course Code:	EEE-4201
Course Title:	Switchgear and Protection
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	

Rationale of the Course: Safety is an important issue in all engineering sectors. This course is designed to cover the safety techniques of electrical line and systems along with their practical applications in industries, house-buildings and house hold appliances.

Course Objectives:

- To finalize the students with different protective measures taken against power faults.
- To explain the operation of different protective devices, such as connectors, relays, circuit breakers, etc.
- To explain the performance metrics and ratings of switchgear so that students can appropriately select necessary devices in order to use them in a protective scheme.
- To familiarize the students about basic protective measures for generator, transmission line and motor.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the fundamentals of protective devices.	Cognitive (Understand)
CLO2	Distinguish between different type of relays and circuit breakers according to their construction and uses.	Cognitive (Apply, Analyze)
CLO3	Evaluate the performance of different protective devices.	Cognitive (Evaluate)
CLO4	Design protective scheme for transformer, generator, motor, and transmission line.	Cognitive (Evaluate, Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction to Switchgear: Purpose of power system protection, Introduction to Switchgear, circuit interruption and protection. Criteria for detecting faults and requirements of protective devices, Terminologies and general characteristics of relays and circuit breaker.
2.	Fuse & Relay: Fuse and its types, Relays: over-current, differential, directional, distance. Electromechanical relay.
3.	Circuit Breakers: control systems, Trip circuit, arc extinction methods, Types of circuit breaker, Different types of protective devices used in switchgear.
4.	Circuit Breaker Ratings: circuit breaker ratings, recovery voltage, TRV, Switching in a capacitive circuit, Current chopping. Air, Oil, air blast, SF6, vacuum and high voltage DC circuit breaker, Selection criteria, testing of circuit breakers.
5.	Transformer Protection: Different types of faults in Transformer, different types of protection scheme in transformer, Buchholz Relay etc. Integrated HV transmission line protection, Combined Transformer and Bus bar protection.
6.	Generator and Motor protection: Introduction, Different types of faults in Generator and motor, different types of protection scheme.
7.	Bus and Transmission Line Protection: Bus bar arrangement, Pilot-wire and carrier current protection, different types of Bus and Transmission line protection scheme, Over voltage protection, lightning and lightning arresters, Grounding
8.	Static and Digital/Numerical Relay: Definition, features, Operation, application, Block diagram and types, Microcontroller and Microprocessor based protection.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√	√									
CLO3			√	√								
CLO4			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion	▪ Quiz ▪ Tutorial/Assignment ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Problem Solving ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Problem Solving ▪ Discussion/Group Discussion	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Problem Solving ▪ Discussion	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

- | | | |
|---------------------------------|---|---|
| 1. Sunil S. Rao | : | Switchgear protection and power systems |
| 2. T. S. MadhavaRao | : | Power System Protection Static Relays |
| 3. Badri Ram and D. Vishwakarma | : | Power System Protection and Switchgear |
| 4. Paul M. Anderson | : | Power System Protection |

Course Code:	EEE-4202
Course Title:	Switchgear and Protection Sessional
Course Credit:	1.0
Course Type:	Core Course
Co-requisite:	EEE-4201

Rationale of the Course: In this course student will learn practically how the device of power protection system works. They will have working experience with relays, circuit breakers, switches, fuses etc. in the laboratory.

Course Objectives:

- To familiarize the students with different protective measures taken against power faults.
- To introduce the students with switches, protective relays, circuit breakers, fuses, bus bar etc.
- To introduce students with LT and HT switchgear.
- To introduce students with Generator protection system.

Course Learning Outcomes (CLOs): By the end of the course student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Identify the circuit breakers, switches, fuses, protective relays, LT, CT, PT and HT switchgear.	Psychomotor (Imitation, Manipulation)
CLO2	Explain and figure out common faults occur in the transmission line, generator and substation.	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Introduction protective devices: relays, switches, circuit breakers, fuses, bus bar, surge suppressor, lighting arrestor, earthing etc.
2.	LT and HT switchgear: Introduction to LT and HT switchgear and demonstration of their working principle.
3.	Substation protection: Demonstrate all sections of a substation and show common faults occur in the system and their protective measures
4.	Transmission line faults: Visiting a High voltage transmission line and show common faults occur and their protective measures in the system.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√	√									

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	Discussion and Demonstration	- Quiz/Lab performance - Participation
CLO2	Discussion and Demonstration	- Quiz/Lab performance - Semester end final examination

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-4200
Course Title:	Technical Seminar
Course Credit:	1.0
Course Type:	Core course

Rationale of the Course: The objective of this course is to verify the students how well they can express their knowledge through presentation and explain it orally. A student has to face a board of members and answer the questions from the board members.

Course Objective:

- To assess the students ability to communicate with another person.
- To supplement the information obtained through other evaluation techniques.
- To identify and analyze the students presence of mind.
- To evaluate the students spontaneity and mannerism.

Course Learning Outcomes (CLOs): By the end of the course student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Convince the audience orally about their knowledge on the courses in Fourth Year.	Affective(Receiving, Responding, Valuing)

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√								√		√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Presentation - Discussion	- Participation - Presentation

Learning Materials: Knowledge from 1st Year to 4th Year Curriculum

Elective Courses

Elective I

Course Code:	EEE-4111
Course Title:	Power Electronics
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-2101, EEE-2201, EEE-3101

Rationale of the Course: The systems and machines of our world depend on power electronics for the ability to run efficiently and sustainably. Power electronics is the application of solid-state electronics for the control and conversion of electric power. It applies to both the systems and products involved in converting and controlling the flow of electrical energy, allowing the electricity needed for everyday products to be delivered with maximum efficiency in the smallest and lightest package. Power engineering and electronics, while maybe not as popular as other forms of engineering, is of great importance to the world around us and has been important for a long time. Power electronics is an essential part of many devices and systems in one or more of their phases. Hence, this course is designed to be introduced with different power electronics components, devices, systems and their detailed way of working in industrial electronics.

Course Objectives:

- To introduce with the fundamentals of Power Electronics such as introduction to solid-state devices, Thyristors, Schottky rectifier, Zener diode, SCR, DIAC, and TRIAC.
- To understand the working of different power electronics devices and automated industrial systems.
- To facilitate the demonstration of different industrial systems like heating, welding, inverter, magnetic amplifiers, etc., with proper analysis.
- To design and develop different manual and automatic controlled systems for industrial purposes and as daily appliances.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the fundamentals of power electronics.	Cognitive (Understand)
CLO2	Explain the techniques of the motor control process.	Cognitive (Apply)
CLO3	Utilize the simulation tool to check stability test of motor's speed control process of poly-phase induction motor, synchronous motor and PMSM motor control.	Cognitive (Analyze)
CLO4	Design & analyze the inverters	Cognitive (Evaluate)
CLO5	Demonstrate the findings of DC – DC converters and AC –AC power converters.	Cognitive (Evaluate)
CLO6	Design & analyze satiable reactors, magnetic amplifiers and manual and automatic controlled systems for industrial purpose.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Semiconductor Power Devices: SCRs, Thyristors, TRIACS, BJT, power MOSFET and IGBT. AC to AC Controlled converter: On-off and phase control, single-phase and three-phase power controllers and cyclo-converter.
2.	DC to DC Converter: Chopper, characteristics and operation of step up and step-down choppers, switching converter, and power regulators. AC to DC Controlled converter: Single phase and three phase half wave, full wave and semi-converter, phase control, characteristics, power factor control and converter triggering circuits.
3.	Inverter: Three phase and single-phase voltage source and current source inverters, voltage, frequency and harmonic control, PWM inverters, SVM inverter. Introduction to power electronic control of motors: Scalar and vector control of poly-phase induction motors, rotor power control, synchronous motor, and PMSM motor control.
4.	Industrial Applications: Introduction to resistance welding, satiable reactors and magnetic amplifiers, dielectric heating, induction heating.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√	√							
CLO5			√	√								
CLO6			√	√	√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO6	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials

1. M. Rashid, "Power Electronics", Pearson Education India, 2004.
2. Ned Mohan, "Power Electronics", Third Edition, 2012, John Wiley.
3. M.D. Singh, K.B. Khanchandani, "Power Electronics", Tata McGraw Hill.
4. C.Y. Lander., "Power Electronics", Third Edition, 1993, McGraw Hill International.
5. B.K .Bose, "Modern Power electronics", 2003, Pearson Education India.
6. Joseph Vithyathil, "Principles of Power Electronics", First Edition, 2010, Tata McGraw Hill

Course Code:	EEE-4112
Course Title:	Power Electronics Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-4111

Rational of the Course: This lab course provides strong understanding of power electronics course with experimental verification of switching device characteristics, experimental study of power converter and controller circuits.

Course Objectives:

- To verify of power semiconductor devices.
- To study of power converters and controllers

Course Learning Outcomes (CLOs): By the end of the course student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the behavior of power semiconductor devices	Psychomotor (Imitation, Manipulation)
CLO2	Able to design and construct power converters and controllers	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Design, construction and testing of power control circuit using Diac.
2.	Design, construction and testing of power control circuit using Triac.
3.	Design, construction and testing of power control circuit using SCR
4.	To find the characteristics of UJT and hence design a waveform generation circuit using UJT.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√			√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO2	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-4121
Course Title:	Cellular Mobile Communication
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-3205, EEE-4103

Rationale of the Course: In this modern age of global communication, the use of cell phone is common in each and every person all-over the world. Cellular Mobile Communication is essential field in the modern science and technology where need highly skilled manpower. Hence, this course is designed to cover introduction to Cellular Mobile Communication in detail.

Course Objectives:

- To learn the fundamental ideas on the mobile communication.
- To understanding in details networks components and data base networks

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to

CLOs	Description	Learning Domain with Level
CLO1	Understand the fundamental concept on networks, network components, cellular systems and data services.	Cognitive(Understand)
CLO2	Design various types of channels.	Cognitive(Apply)
CLO3	Design, construct and test network components.	Cognitive(Apply)
CLO4	Produce and apply different networks.	Cognitive(Apply, Analyze)
CLO5	Analyze and develop mobile station.	Cognitive(Analyze)
CLO6	Apply, analyze and evaluate different data services and networks.	Cognitive(Apply, Analyze and Evaluate)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Mobile Communication: Introduction, concept, evolution and fundamentals, analog and digital cellular systems, cellular radio system, frequency reuse, co-channel interference, cell splitting and components, Mobile radio propagation, propagation characteristics, models for radio propagation, antenna at cell site and mobile antenna, frequency management and channel assignment, fundamentals, spectrum utilization, fundamentals of channel assignment, fixed channel assignment, non-fixed channel assignment, traffic and channel assignment, handoffs and dropped calls, reasons and types, forced handoffs, mobile assisted handoffs and dropped call rate.
2.	Network Components: Switching, TRC, BCS, MSC, GMSC, HLR, VLR, ILR, AUC, EIR, Base station system, RBS, Cell antennas and Mobile antennas.
3.	Introduction to Networks: ISDN, B-ISDN, LAN, MAN, WAN, BLUETOOTH, ATM, and multimedia communication, Unicast, Multicast, and Broadcast.
4.	Mobile Station: Function, Idle mode, Types, SIM, MS features, Evolution of MS, MS in Idle mode, Access to network, Roaming in cells, Detaching from network, MS in active move, Call set up from MS, Call set up and call termination.
5.	Data Services and Networks: CDPD, HSCSD, GPRS, D-AMPS+, CDMA Two, WAP, GPRS architecture, GPRS traffic case, GSM to 3G, EDGE, WCDMA and 4G. Introduction to 5G.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2			√	√								
CLO3			√	√								
CLO4				√	√							
CLO5		√	√									
CLO6			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion	- Quiz - Tutorial/Assignment - Participation
CLO2	- Lecture - Presentation - Discussion - Online Resources	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Online Resources	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Discussion/Group Discussion - Online Resources	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation

CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion	▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO6	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials

1. GSM System Survey : Ericson student text by Ericson Radio System
2. Dharma Prakash Agarwall : Introduction to Wireless and Mobile System
3. C. Y. Lee : Mobile Cellular Telecommunication
4. Saleh Faruque : Cellular Mobile System

Course Code:	EEE-4122
Course Title:	Cellular Mobile Communication Sessional
Course Credit:	1.5
Course Type:	Core Course
Co-requisite:	EEE-4121

Rationale of the Course: This lab course provides strong practical understanding of cellular and mobile communication course with experimental verification of switching device characteristics, experimental study of cellular mobile communication system.

Course Objectives:

- To learn the fundamental practical ideas on the mobile communication.
- To understanding in details networks components and data base networks.
- To verification of switching devices.
- To study the characteristics and management of radio wave propagation

Course Learning Outcomes (CLOs): By the end of study of the course, the student will be able to-

CLOs	Description	Learning Domain with Level
CLO1	Understand the generation of mobile communication system	Psychomotor (Imitation, Manipulation)
CLO2	Design and construct the different multiple access techniques	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Test the different sections of mobile phone
2.	Perform the process of call connection and call release of cellular Mobile System
3.	Design and implementation of Local area network (LAN).
4.	Transfer an image, audio and video file using Bluetooth protocol with varying distance between two devices and analyze the performance.
5.	Configure Wi-Fi setting in mobile devices using mobile tethering.
6.	Apply RFID technology for real life applications using RFID kit.
7.	Establish seamless wireless connectivity using multiple access point
8.	Use AT commands to understand working of 3G network using 3G mobile phone Trainer kit.
9.	Simulate Bluetooth voice transmission to observe effect of AWGN & interference of 802.11b on transmission using MATLAB & Simulink.
10.	Simulate the line coding techniques using MATLAB and Simulink.
11.	Simulate the Binary amplitude shift keying using MATLAB and Simulink.
12.	Simulate the Binary Phase shift keying using MATLAB and Simulink.
13.	Simulate the Delta Modulation using MATLAB and Simulink
14.	Simulate the Direct Sequence Spread Spectrum using MATLAB and Simulink.
15.	Study of MSK modulation process by simulation
16.	Study of MSK demodulation process by simulation.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√			√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation
CLO2	- Discussion, - Experiment, - Report writing	- Quiz - Lab performance - Semester end final examination - Participation

Learning Materials:

1. Lab guide/Manuals

Course Code:	EEE-4131
Course Title:	High Voltage Engineering
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-3201, EEE-4101

Rationale of the Course: This course intends to develop and apply the theory of high voltage generation and measurements to the appropriate places. Students will also be taught of the breakdown of matters and how to test electrical equipment which are supposed to be exposed to high voltages.

Course Objectives:

- To learn the principles of theory of high voltage generation and measurements.
- To understand the operation of high voltage power supplies for dc, ac, and impulse voltages.
- To get familiar with high voltage testing and various applications where high voltage field is used.
- To improve their ability to bring clarity to high voltage breakdown in gas liquid, and solid dielectric materials.
- To gain an understanding about overvoltage phenomenon and insulation coordination in electric power systems.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the theory and operation of high voltage dc, ac and impulse voltage generation.	Cognitive (Understand)
CLO2	Improve their ability to bring clarity to high voltage breakdown in gas liquid, and solid dielectric materials.	Cognitive (Apply)
CLO3	Comprehend breakdown phenomena in insulating materials and overvoltage phenomena in power transmission systems and insulation coordination	Cognitive (Apply)
CLO4	Analyze different methods for high voltage measurements to measure the voltages accurately, ensuring perfect safety to the personnel and equipment.	Cognitive (Analyze)
CLO5	Get familiar with high voltage testing and various applications.	Cognitive(Apply)

Course Content:

Sl. No.	Course Content (as Summary)
1.	High Voltage DC Generation: Rectifier circuits, ripple minimization, voltage multipliers, Van-de-Graf and electrostatic generators; applications.
2.	High Voltage AC Generation: Tesla coils, cascaded transformers and resonance transformers
3.	Impulse Voltage Generation: Shapes, mathematical analysis, codes and standards, single and multi-stage impulse generators, tripping and control of impulse generators.
4.	Breakdown in gas, liquid and solid dielectric materials, applications of gas and solid dielectrics in transformer. Corona.
5.	High Voltage Measurements and Testing: IEC and IEEE standards, sphere gap, electrostatic voltmeter, potential divider, Schering bridge, Mega Ohm meter, HV current and voltage transducers: contact and noncontact.
6.	Overvoltage Phenomenon and insulation coordination. Lightning and switching surges, basic insulation level (EV, EHV and UHV systems), surge diverters and arresters.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2	√	√										
CLO3		√	√									
CLO4			√	√								
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Online Resources	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials

- | | |
|--------------------------------|--|
| 1. M.S. Naidu and V. Kamaraju, | : High Voltage Engineering |
| 2. Kuffeland, W.S. Zaengl, | : High Voltage Engineering Fundamentals' |
| 3. C.L.Wadhaw | : High Voltage |

Course Code:	EEE-4132
Course Title:	High Voltage Engineering Sessional
Credit Value:	1.5
Course Type:	Core Course
Co-requisite:	EEE-4131

Rationale of the Course: This course provides the student a hands on experience of verifying the theorems, measurement and testing phenomena they are taught in EEE-4131 course.

Course Objectives:

- To study the filtration and treatment techniques of transformer oils and determine their dielectric strength.
- To study solid dielectrics used in power apparatus and capable to measure the capacitance and dielectric loss of insulating materials.
- To provide knowledge about applications of insulating materials and analysis of breakdown phenomena in these materials.
- To provide hands on experience of measuring and testing high voltage accurately.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the filtration and treatment techniques of transformer oils and capable to determine their dielectric strength.	Psychomotor (Imitation, Manipulation)
CLO2	Recognize applications of insulating materials and analyze breakdown phenomena in these materials.	Psychomotor (Precision, Articulation)
CLO3	Examine different methods for high voltage measurements and testing to measure the voltages accurately.	Psychomotor (Precision, Articulation)
CLO4	Understand overvoltage phenomena in power transmission system	Psychomotor (Precision, Articulation)

Course Content:

Sl. No.	List of the Experiments
1.	Study filtration and Treatment of transformer oil.
2.	Determine dielectric strength of transformer oil.
3.	Study solid dielectrics used in power apparatus.
4.	Study applications of insulating materials
5.	Study high voltage testing of electrical equipment: (i) Different line insulator (iii) cable (ii) bushing (iv) power capacitor (v) power transformer
6.	Breakdown studies of uniform field and non-uniform field gaps under A.C. excitation:(i) Sphere to sphere (ii) Sphere to plane (iii) Sphere to point (iv) Plane to plane (v) Plane to Point (vi) Point to Point
7.	To study of high voltage ac measurement
8.	Study the capacitance and dielectric loss measurement of an insulating material using Schering Bridge.
9.	Design an EHV transmission line

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs):

CLO1	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO2	√	√	√	√		√						
CLO3	√	√	√	√								
CLO4	√	√	√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO2	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO3	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation
CLO4	▪ Discussion, ▪ Experiment, ▪ Report writing	▪ Quiz ▪ Lab performance ▪ Semester end final examination ▪ Participation

Learning Materials:

3. Lab guide/Manuals

Elective II (Electronics Group)

Course Code:	EEE-4213
Course Title:	Optoelectronics
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-2101

Rationale of the Course: Optoelectronics is quickly becoming a fast emerging technology field that consists of applying electronic devices to sourcing, detection, and control of light. Devices that fall under optoelectronics are vast which requires immense manpower for maintenance provides huge job opening for engineer graduates. This course is introduced to give basic knowledge of optoelectronics and its related fields.

Course Objectives:

- To acquaint students with the laws and phenomena in the area of optical properties of semiconductors.
- To acquire in depth of knowledge in emission and amplification of light.
- To get dated with the design, fabrication and interfacing of LED, LASER, Photo-detectors
- To develop the concept about available devices, current trends and future developments in the application of light into electronics.

Course Learning Outcomes (CLOs): Upon completion of the course, students should be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the optical properties of semiconductor, properties of light devices, constructions and working of LED, LASERs, Photo-detectors and modulation of light.	Cognitive (Understand)
CLO2	Use different properties of light in optoelectronics and select appropriate optoelectronic device for a particular use.	Cognitive (Apply)
CLO3	Design and construct of light sources as LED and LASER circuits with their modulation techniques.	Cognitive (Apply)
CLO4	Solve problems related to circuits or design.	Cognitive (Analyze)
CLO5	Compare between different stimulated emissions.	Cognitive (Evaluate)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Optical Properties in Semiconductor: Direct and indirect band-gap materials, basic transitions in semiconductors, radiative and nonradiative recombination, optical absorption, photo-generated excess carriers, minority carrier life time, luminescence and quantum efficiency in radiation. (CLO1)
2.	Properties of Light: Particle and wave nature of light, polarization, interference, diffraction and blackbody radiation. (CLO2)
3.	Light Emitting Diode (LED): Principles, materials for visible and infrared LED, internal and external efficiency, loss mechanism, structure and coupling to optical fibers. Double-Hetero-structure (DH) LEDs, Characteristics, Surface and Edge emitting LEDs. (CLO3)
4.	Stimulated Emission and Light Amplification: Spontaneous and stimulated emission, Einstein relations, population inversion, absorption of radiation, optical feedback and threshold conditions. (CLO4)
5.	Semiconductor Lasers: Population inversion in degenerate semiconductors, laser cavity, operating wavelength, threshold current density, power output, elementary laser diode characteristics, hetero-junction lasers, optical and electrical confinement. single frequency solid state lasers-distributed Bragg reflector (DBR), distributed feedback (DFB) laser.(CLO5)
6.	Introduction to Quantum Well Lasers: Introduction to quantum well lasers, Vertical Cavity Surface Emitting Lasers (VCSELs), optical laser amplifiers.(CLO6)
7.	Photo-detectors: Photoconductors, junction photo-detectors, PIN detectors, avalanche photodiodes, hetero-junction photodiodes, Schottky photo-diodes and phototransistors. Noise in photo-detectors. PIN and APD. Photo-detector design issues. (CLO1)
8.	Modulation of Light: Phase and amplitude modulation, electro-optic effect, acousto-optic effect and magneto-optic devices. Introduction to integrated optics.(CLO7)

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3		√	√									
CLO4			√	√								
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO3	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO4	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO5	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam

Learning Materials

1. Wilson and Hawkes : Optoelectronics: An Introduction of Optoelectronics
2. J. Wilson, J.F.B. Hawkes : Optoelectronics
3. Michael A. Parker : Physics of Optoelectronics
4. Pallab Bhattacharya : Semiconductor Optoelectronic Devices
5. S.C. Gupta : Optoelectronic Devices and Systems
6. Joachim Piprek : Optoelectronic Devices

Course Code:	EEE-4215
Course Title:	VLSI Design
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-2101, EEE-2201, EEE-3101, EEE-3107

Rationale of the Course: The course introduces the concepts and physical procedures involved in the design, integration and manufacture of VLSI circuits. The course teaches fundamental design principles and simulations of hands-on experience in fabricating integrated circuits based on silicon technology. By the end of the course, the students should understand the broad aspects of CMOS, BiCMOS technology for integrated circuits including testing and evaluation, concepts of yield, lab procedures including safety, assembly and packaging.

Course Objectives:

- To introduce the fundamental ideas on the MOS devices, CMOS technology.
- To perform the programmable logic devices, CMOS chip design and VLSI design

Course Learning Outcomes (CLOs): After successful completion of this course, the students should be able to

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the MOS, CMOS, P -Well process, N -Well process, twin - tub process.	Cognitive (Understand)
CLO2	Explain the concepts of MOS transistors operations and their characteristics	Cognitive (Understand)
CLO3	Analysis of MOS, CMOS and BiCOM technology	Cognitive (Analysis)
CLO4	Evaluate power estimation and delay calculations in CMOS circuits	Cognitive (Evaluate)
CLO5	Design layout and schematic and analysis digital logic gates	Cognitive (Apply)
CLO6	Implement, P -Well process, N -Well process, twin - tub process, Digital Integrated Circuits and System Building Blocks in VLSI design.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	MOS Technology And Circuits: MOS Technology and VLSI, Process parameters and consideration for BJT, MOS and CMOS, CMOS logic, MOS transistor theory, Enhancement mode transistor action, Ideal I-V characteristics, DC transfer characteristics, Threshold voltage, Body effect, Design Equations, Second order effects. MOS models and small signal, AC characteristics, Simple MOS capacitance Models, Detailed MOS gate capacitance model, Detailed MOS Diffusion capacitance model.
2.	CMOS Circuits Design Process: CMOS fabrication, P -Well process, N -Well process, twin - tub process, MOS layers stick diagrams and Layout diagram, Layout design rules, Latch up in CMOS circuits, CMOS logic structures, Transmission gates, Static CMOS design, dynamic CMOS design.
3.	Circuit Characterization and Performance Estimation: Resistance estimation, Capacitance estimation, Inductance, switching characteristics, transistor sizing, power dissipation and design margining, Charge sharing. Scaling.
4.	VLSI System Components Circuits And System Level Physical Design: Multiplexers, Decoders, comparators, priority encoders, Shift registers, Arithmetic circuits, Ripple carry adders, Carry look ahead adders, High-speed adders, Multipliers. Physical design – Delay modelling, cross talk, floor planning, power distribution, Clock Distribution
5.	Specification Using Verilog HDL: Basic concepts- identifiers- gate primitives, gate delays, operators, timing controls, procedural assignments conditional statements, Data flow and RTL, structural gate level switch level modeling, Design hierarchies, Behavioral and RTL modeling, Test benches, Design of decoder, equality detector, comparator, priority encoder, half adder, full adder, Ripple carry adder, D latch and D flip flop.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√										
CLO3				√								
CLO4				√								
CLO5			√									
CLO6					√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion - Online Materials	- Quiz - Participation - Tutorial/ Assignment - Semester final exam
CLO2	- Lecture - Presentation - Discussion - Online Materials	- Quiz - Participation - Tutorial/ Assignment - Semester final exam

CLO3	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO4	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO5	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam
CLO6	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online Materials	▪ Quiz ▪ Participation ▪ Tutorial/ Assignment ▪ Semester final exam

Learning Materials

1. Douglas A. Pucknell and Kamran Eshraghian,: Basic VLSI Design Systems and Circuits
2. Wayne Wolf, : Modern VLSI Design
3. Amar Mukherjee, : Introduction to NMOS and CMOS VLSI System Design

Course Code:	EEE-4217
Course Title:	Micro-Electro-Mechanical Systems (MEMS)
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-3207

Rationale of the Course: This course aims the exploration of micro-electromechanical devices and systems

Course objective:

- To be familiar with the basic terminologies of MEMS
- To know the operational principles of MEMS Devices
- To understand the different micromachining techniques to fabricate MEMS devices
- To become familiar with a wide variety of MEMS applications.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the basic terminologies of MEMS	Cognitive (Understand)
CLO2	Explain the concepts of MEMS Devices operations and their characteristics	Cognitive (Understand)
CLO3	Analysis the different micromachining techniques to fabricate MEMS devices	Cognitive (Analysis)
CLO4	Evaluate power estimation and delay calculations in MEMS devices	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction to Micro/Nano Engineering: MEMS technology today, MEMS and nanotechnology, micromachining techniques, fabrication process used in MEMS, basic terminologies of MEMS, commercial applications of MEMS. Market overview of MEMS
2.	Materials for MEMS: Single-crystal silicon, polysilicon stress, characteristics silicon dioxide, silicon nitride and silicon carbide, germanium-based materials, metallic thin film materials, piezoelectric material, polymer materials in MEMS.
3.	MEMS Fabrication: Introduction to wet bulk micromachining, bulk vs. surface micromachining, surface micromachining process, silicon as a mechanical element in MEMS, wet isotropic and anisotropic etching, etch-stop techniques, surface micromachining

4.	LIGA and Micro-Molding: LIGA-background, LIGA and LIGA-Like process steps
5.	LIGA and Micro-Molding: LIGA-background, LIGA and LIGA-Like process steps
6.	Packaging of MEMS Devices: Material requirements, wafer-level packaging, high-temperature electrical interconnection systems, and high-temperature ceramic packaging systems.
7.	Molecular Self-Assembly: Molecule-molecule interactions, molecule-substrate interactions, applications of functionalized surfaces.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2					√							
CLO3			√	√								
CLO4		√		√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials

- | | |
|-------------------------|--|
| 1. Traylor and Francis. | : MEMS design and fabrication |
| 2. Stephen D. Senturia | : Microsystem Design |
| 3. Chang Liu | : Foundations of MEMS |
| 4. Marc J. Madou | : Fundamentals of Microfabrication |
| 5. Minhang Bao | : Analysis and Design Principles of MEMS Devices |

Course Code:	EEE-4219
Course Title:	Bioelectronics
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-2101, EEE-2201

Rationale of the Course: This course aims the basic concepts analog and digital electronics and its application to bioengineering.

Course objective:

- To provide the background to design electronic instrumentation to assess physiological and molecular functions from bioelectrical measurements

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the basic concepts analog and digital electronics	Cognitive (Understand)
CLO2	Explain the concepts of bioelectronics Devices operations and their characteristics	Cognitive (Understand)
CLO3	Analysis the different micromachining techniques to fabricate bioelectronics Devices	Cognitive (Analyze)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction of Bioelectronics: Introduction to analog and digital electronics for bioengineering, Market overview of bioelectronics devices.
2.	Biological Materials: Analogy between semiconductor and biological materials, water and electrolyte solutions; biological molecules - Proteins, Nucleic acids, Phospholipids; cell membrane; Eucaryotic cell.
3.	Motion in Solution and Chemical Reaction: Diffusion, Brownian motion, electrophoresis, enzyme kinetics; Solid electrolyte junctions: electrode-electrolyte interfaces, Poisson-Boltzmann equation, Membrane transport, Nernst-Planck equation and solution
4.	Electrolyte-Insulator-Semiconductor (EIS): EIS Structure, Site binding Theory, Electrical double layer theory
5.	MOSFET Based Bioelectronics Devices : Biosensor overview, Ion Sensitive Field Effect Transistor (ISFET), Enzyme Field Effect Transistor (ENFET), Chemical Field Effect Transistor (CHEMFET), Reference Field Effect Transistor (REFET), Immune Field Effect Transistor (IMFET), Organic Thin Film Transistor (TFT), Cell-Based Biosensors & Sensors of Cell Metabolism, Light Addressable Potentiometric Sensors (LAPS); Interfacing of Biological Systems with electronic systems, non-conventional bioelectronics devices, conducting polymer based ISFET.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2	√	√										
CLO3			√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion - Online resources	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation

Learning Materials:

- | | |
|-------------------------|--|
| 1. Trailor and Francis. | : MEMS design and fabrication |
| 2. Stephen D. Senturia | : Microsystem Design |
| 3. Chang Liu | : Foundations of MEMS |
| 4. Marc J. Madou | : Fundamentals of Microfabrication |
| 5. Minhang Bao | : Analysis and Design Principles of MEMS Devices |

Elective III (Power and Energy Group)

Course Code:	EEE-4223
Course Title:	Renewable Energy for Sustainable Development
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	

Rationale of the Course: The course provides the structure and operation of renewable energy systems such as; wind energy, water energy, solar energy, bioenergy, etc., and determine their potential use and efficiency for sustainable development.

Course Objectives:

- To understand basic aspects of renewable energy sources.
- To gather knowledge about renewable energy conversion technique to useful energy source
- To introduce about energy coupling between turbine and electric generator

Course Learning Outcomes (CLOs): After successful completion of this course, the students should be able to

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand in detail about the energy science and technology, differences between conventional and renewable energy, the role of renewable energy for sustainable development	Cognitive(Understand)
CLO2	Explain in detail the main components of these different renewable energy systems	Cognitive(Apply)
CLO3	Explain why renewable energy i.e. wind, water, solar, and bioenergy is important for sustainable development	Cognitive(Apply)
CLO4	Recognize the effects that current energy systems based on fossil fuels have over the environment and the society	Cognitive(Apply)
CLO5	Compare different renewable energy technologies and choose the most appropriate based on local conditions	Cognitive(Analyze)
CLO6	Design and dimension technological solutions based on wind energy, bioenergy or solar energy that meet specific energy demands, are economically feasible and have a minimal impact on the environment	Cognitive(Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction To Energy Studies: Introduction, Energy science and Technology, Forms of Energy, Importance of Energy Consumption as Measure of Prosperity, Per Capita Energy Consumption, Roles and responsibility of Ministry of power energy and mineral resources, Classification of Energy Resources, Conventional Energy Resources, Non-Conventional Energy Resources, World Energy Scenario, Bangladesh Energy Scenario.
2.	Wind Energy: Introduction, History of Wind Energy, Wind Energy Scenario of World and Bangladesh, Basic principles of Wind Energy Conversion Systems (WECS), Types and Classification of WECS, Parts of WECS, Power, torque and speed characteristics, Electrical Power Output and Capacity Factor of WECS, Stand alone, grid connected and hybrid applications of WECS, Economics of wind energy utilization, Site selection criteria, Wind farm, Wind rose diagram. Challenges, Wind Energy Industry, Sustainability
3.	Water Energy (Hydro, Tidal & Wave): Hydro Resources, Hydro Project Types, Hydro Conversion Technologies, Tidal Resource, Tidal Power Conversion, Wave Resource, Wave Energy Conversion, Challenges to Sustainability
4.	Solar Energy: Introduction, Solar Radiation, Sun path diagram, Basic Sun-Earth Angles, Solar Radiation, Geometry and its relation, Measurement of Solar Radiation on horizontal and tilted surfaces, Principle of Conversion of Solar Radiation into Heat, Collectors, Collector efficiency, Solar Greenhouse. Solar Photovoltaic, Solar Cell fundamentals, Characteristics, Classification, Construction of module, panel and array, Solar PV Systems (stand-alone and grid connected), Solar PV Applications. Government schemes and policies, Sustainability
5.	Bioenergy: Introduction, Sources of Feed stocks, Bioenergy, Photosynthesis process, Conversion Technology, Diesel and Ethanol, Biogas, Electricity Production, Transportation, Challenges, Types of biogas plants, Factors affecting biogas generation, Energy plantation, Environmental impacts and benefits, Future role of biomass, Sustainability & Economics

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√					√	√					
CLO2		√	√									
CLO3			√	√			√					
CLO4				√		√	√					
CLO5			√	√								
CLO6			√	√	√							√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO6	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials

- | | |
|----------------------------------|---|
| 1. Sukhumi. S.P. | : Solar Energy |
| 2. B. H. Khan, | : Non-Conventional Energy Resources |
| 3. Twidell, J.W. & Weir, | : A. Renewable Energy Sources. |
| 4. S. P. Sukhumi and J.K. Nyack, | : Solar Energy – Principles of Thermal Collection and Storage |
| 5. Garg, Prakash, | : Solar Energy, Fundamentals and Applications. |
| 6. G.D. Rai, | : Non-Conventional Energy Sources, |
| 7. Godfrey Boyle, | : Renewable Energy, Power for a Sustainable Future |

Course Code:	EEE-4225
Course Title:	Power Plant Engineering
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	ME-1205

Rationale of the Course:

Upon completion of this course, students can able to understand different types of power plants, its functions, flow lines, fuels, and different issues related to them. Analyze economic and safety-related issues in power sectors.

Course Objectives:

- To introduce the fundamental ideas on the manual and automatic power plants.
- To understanding in details the generation of power with control

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the steam power plants, steam generator and turbine functioning.	Cognitive(Understand)
CLO2	Explain the construction and working of hydroelectric power plants, gas turbine power plants and nuclear power plants.	Cognitive(Apply, Analyze)
CLO3	Create awareness about various power plants.	Affective (Responding)

Course Content:

Sl. No.	Course Content (as Summery)
1.	Fundamental of Power Plant: Concept of power plant, Classification of power plant, Energy, Major source of energy, Power development in Bangladesh, Present power position in Bangladesh.
2.	Steam Power Plants: High steam pressure and temperature, Super heating of steam, Reheat cycle, Regenerative cycle, Binary vapour cycle, Coal Classification, Use of high ash coal, Coal supply, storage and handling, Ash handling and dust collectors.
3.	Steam Generators and Turbine: Fire tube and water tube boilers, Modern boilers, Economizer and air preheater, Condenser, Supply of cooling water to condenser, Cooling towers, Steam prime movers, Impulse and reaction types Turbine, Heat balance and efficiency, Modern development in steam power plants.
4.	Hydro Electric Power Plants: Classification and basic schemes, Types of turbines, Capacity calculation, Selection of site, Pump storage projects, Hydroelectric power plant in Bangladesh
5.	Gas Turbine Power Plants: Operation of gas turbine power plant, Open cycle plant, Closed cycle plant, Combined gas turbine and steam turbine cycle.
6.	Nuclear Power Plant: Types of fuels. Classification of reactors, Methods of cooling, Moderators, Methods of control, Safety measures, Cost of nuclear energy, Nuclear power plant and prospects in Bangladesh.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3						√	√				√	√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion - Online resources	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO2	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation

CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
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Learning Materials:

1. P. K. Nag: Power Plant Engineering
2. M. M. El-Wakil: Power Plant Technology
3. Janet Wood: Nuclear Power

Course Code:	EEE-4227
Course Title:	Nuclear Power Engineering
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	N/A

Rationale of the Course: Modern life is meaningless without proper supply of electrical power. Nuclear energy is prime source of energy worldwide. Professionals in this field are very low in number. This course is intended to enable learner to get basic knowledge and understanding of nuclear energy and nuclear power plant to become power plant engineer or researcher.

Course Objectives:

- To understand basic concepts of nuclear energy and nuclear processes
- To gathering the knowledge about different types of nuclear power reactor, instrumentation and control.
- To introduce and highlights the safety issues and grid interconnections issues.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the basic concept of nuclear energy, nuclear processes & nuclear fission-fusion reactions. In addition, the competences covering the design principles of nuclear reactors and nuclear power plants that will help to understand the nuclear fuel cycle from cradle to grave.	Cognitive (Understand)
CLO2	Apply & develop mathematical models to solve energy technology problems; which lead to design energy technology related equipment, plants, processes and systems.	Cognitive (Apply)
CLO3	Explain different types of nuclear power reactors, nuclear power plants processes relevant to power generation & distribution.	Cognitive (Analyze)
CLO4	Specify & investigate about energy economics, savings, environmental impacts, emissions, safety, security and grid interconnection issues.	Cognitive (Analyze)
CLO5	Utilize different numerical methods of reactor physics and thermal hydraulic safety analysis with introducing the radiation protection and reliability engineering. Managerial skills also manage and organize both national and international projects.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Basic Concepts: Nuclear energy, atoms and nuclei, radioactivity, nuclear processes, fission, fusion.
2.	Nuclear Systems: Particle accelerator, isotope separators, neutron chain reaction, reactor types, power generation. Layout of nuclear power plant (NPP).
3.	Nuclear Power Plant Reactors: Pressurized water reactor, boiling water reactor, CANDU reactor, gas cooled reactor, liquid metal cooled reactor, and breeder reactor, auxiliaries, instrumentation and control.
4.	Grid Interconnection Issues: Effects of frequency and voltage changes on NPP operation, Advanced and next generation nuclear plants; very high temperature reactors, Biological effects, reactor safety and security; Three Mile island case; Chernobyl case; Fukushima case, Fuel cycle; radioactive waste disposal.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√											
CLO2		√	√									
CLO3			√	√								
CLO4			√	√								
CLO5			√	√	√						√	√

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	- Lecture - Presentation - Discussion/Online resources	- Quiz - Tutorial/Assignment - Participation
CLO2	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO3	- Lecture - Presentation - Discussion/Group Discussion - Demonstration	- Quiz - Tutorial/Assignment/group Assignment - Semester end final examination - Participation
CLO4	- Lecture - Presentation - Discussion - Online resources	- Quiz - Tutorial/Assignment - Semester end final examination - Participation
CLO5	- Lecture - Presentation - Discussion - Demonstration	- Quiz - Tutorial/Assignment - Semester end final examination - Participation

Learning Materials

- | | |
|---|---|
| 1. M M El-Wakil | : Nuclear Power Engineering |
| 2. Julian Meyer, Jürgen Schnell | : Design & Construction of Nuclear Power Plants |
| 3. Janet Wood | : Nuclear Power |
| 4. John R. Lamarsh and Anthony J. Baratta | : Introduction to Nuclear Engineering |

Course Code:	EEE-4229
Course Title:	Power System Operation and Control
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-3201, EEE-4101

Rationale of the Course: Modern power system is a complex system, spread over a large geographically area. Hence, this course is designed to cover modern power system as well as reactive power and active power control, angle stability and voltage stability, inter-area power transfer, power quality, automatic generation and frequency control for multi-machine system, reliability evaluation operation in competitive environment are important factors in operation and control of the power system.

Course Objectives:

- To understand basic concepts of power system monitoring and control
- To gather knowledge about different types various on line functions and network topology.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the operation, control, deregulation and restructuring of power systems	Cognitive (Understand)

CLO2	Explain and apply the concept of computer control of power systems and data acquisition.	Cognitive (Understand, Apply)
CLO3	Analyze various functions of Energy Management System (EMS).	Cognitive (Analyze)
CLO4	Design the computer control of power systems in stable/unstable position.	Cognitive (Analyze)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Load Frequency Control: Introduction, load frequency problem-Megawatt frequency (or P-f) control channel, MVAR voltages (or Q-V) control channel-Dynamic interaction between P-f and Q-V loops. Mathematical model of speed governing system-Turbine models, division of power system into control areas, P-f control of single control area (the uncontrolled and controlled cases)-P-f control of two area systems (the uncontrolled cases and controlled cases)
2.	Economic Operation of Power Systems: Distribution of load between units within a plant-Transmission loss as a function of plant generation, Calculation of loss coefficients-Distribution of load between plants.
3.	Power System Stability: The stability problem-Steady state stability, transient stability and Dynamic stability-Swing equation, Equal area criterion of stability-Applications of Equal area criterion, Step by step solution of swing equation-Factors affecting transient stability, Methods to improve steady state and Transient stability, Introduction to voltage stability
4.	Computer Control of Power Systems: Need of computer control of power systems. Concept of energy control center (or) load dispatch center and the functions - system monitoring - data acquisition and control. System hardware configuration – SCADA and EMS functions. Network topology – Importance of Load Forecasting and simple techniques of forecasting.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation

Learning Materials

- | | |
|------------------------------------|---------------------------------|
| 1. C. L. Wadhwa | : Electrical Power Systems, |
| 2. D. P. Kothari and I. J. Nagrath | : Modern Power System Analysis |
| 3 D. P. Kothari | : Modern Power System Analysis- |
| 4 Hadi Sadat | : Power System Analysis . |

Elective IV (Communication Group)

Course Code:	EEE- 4233
Course Title	Optical Fiber Communication
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite	EEE-2101, EEE-2201, EEE-3207

Rationale of the Course: This is basic course of optical fiber communication. Fiber-optic communication is a method of transmitting information from one place to another by sending pulses of infrared light through an optical fiber. The light is a form of carrier wave that is modulated to carry information. Fiber is preferred over electrical cabling when high bandwidth, long distance, or immunity to electromagnetic interference is required. This type of communication can transmit voice, video, and telemetry through local area networks or across long distances

Course Objectives:

- To understand the basics of optical fiber communications.
- To understand the physics of optical fiber and its characteristics.
- To describe the principle of operation of optical sources and detectors.
- To describe different type dispersions and limitations which are inherent to the system.
- To learn the coherent optical communication and multi-channel optical systems.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the basics characteristics of optical fiber communications.	Cognitive (Understand)
CLO2	Explain the coherent optical communication and multi-channel optical systems.	Cognitive (Apply)
CLO3	Demonstrate the principle of operation of optical sources and detectors.	Cognitive (Analyze)
CLO4	Develop the different type dispersions and limitations which are inherent to the system.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Introduction: Optical fibers; Structure, Step-index and graded-index fibers, Modes of propagation, modal theory for circular waveguide, Modal equations, Waveguide equations, Power flow in optical fibers, Signal degradation in optical fibers, Fiber attenuation, Distortion in optical guides, Dispersions, Mode coupling.
2.	Optical Sources and Detectors: Light emitting diode (LED) and semiconductor laser diode (SLD), Structures, Modulation capability, Transient response, Power bandwidth product, Modal noise, Temperature Effects and reliability; p-i-n and avalanche photo detectors, Structures, Principles of operations.
3.	Optical Modulation and Amplifier: Optical modulation and detection schemes, Direct and coherent detection receivers: configuration, operation, noise sources, sensitivity calculation, performance curves, Optical Amplifiers, Design of analog and digital receivers.
4.	Multiplexing and Coding: Wavelength Division Multiplexing (WDM), Dense Wavelength Division Multiplexing (DWDM) and Optical frequency division multiplexing (OFDM) transmission schemes. Optical data coding, links, power budget. Optical data buses, optical networks, Fiber Distributed Data Interface (FDDI) and Synchronous Optical Network (SONET)/SDH.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs):

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√								
CLO4			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategy

CLOs	Teaching –Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online resources	▪ Participation ▪ Class test ▪ Oral exam ▪ Assignment ▪ Final exam
CLO2	▪ Multimedia projector ▪ Guided teaching ▪ Text book ▪ Lecture note ▪ E-learning	▪ Participation ▪ Class test ▪ Oral exam ▪ Assignment ▪ Final exam
CLO3	▪ Multimedia projector ▪ Guided teaching ▪ Text book ▪ Lecture note ▪ E-learning	▪ Participation ▪ Class test ▪ Oral exam ▪ Assignment ▪ Final exam
CLO4	▪ Multimedia projector ▪ Guided teaching ▪ Text book ▪ Lecture note ▪ E-learning	▪ Participation ▪ Class test ▪ Oral exam ▪ Assignment ▪ Final exam

Learning Materials:

- | | |
|------------------|---|
| 1. John Senior | : Optical Fiber Communications |
| 2. G. P. Agrawal | : Optical Fiber Communication System |
| 3. Chrin | : An Introduction to Optical Fiber |
| 4. Barnoski | : Fundamentals of Optical Fiber Communication |

Course Code:	EEE-4235
Course Title:	Microwave and Satellite Communication
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-3205, EEE-4103

Rationale of the Course: With the increasingly application of microwave technologies in applications like radio astronomy, long-distance communications, space navigation, radar systems, medical equipment students need to understand the theoretical and experimental design and analysis of microwave devices and circuits. This course provides a through coverage of fundamental principles of microwave engineering on microwave production, amplification, analysis and applications.

Course Objectives:

- To be familiar with microwave tubes solid-State devices.
- To understand characteristics of microwave generated by tube and solid state devices.
- To design microwave components such as Directional Couplers, Microwave Wave-guides and Components.
- To analyze and study rectangular and circular wave guides using field theory.
- To design impedance matching and tuning using lumped and distributed elements for network.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs):

CLOs	Descriptions	Learning Domain with Level
CLO1	Understanding the basic theories, models and design methods applied in microwave communication systems.	Cognitive (Understand)
CLO2	Apply this knowledge to microwave communication systems.	Cognitive (Apply)

Course Content:

Sl. No.	Course Content (as Summary)
1.	Microwave Generation: Klystrons, Reflex Klystrons, Magnetrons and other devices, Microwave tubes, Microwave components, Microwave measurements, power, frequency, wavelength, microwave transmission media, Anisotropy media, impedance, Hollow, surface wave, dielectric waveguide, waves on a transmission line, standing wave ration, quarter wave transformers.
2.	Micro-strip Lines: wave propagation and micro-strip lines, dielectric constants, characteristic impedance, attenuation factors.
3.	Waveguide: Microwave components using waveguide; Circulator, Isolator, T-junction, Magic T, Horn antenna.
4.	Microwave Circuit Design: Low frequency parameters, s-, y-, z-, h-parameter, ABCD parameters, transmission matrix, passive circuit design, the Smith Chart, mixer design, microwave control circuit design, wireless microwave systems, noise in microwave circuits, Microwave IC, HMICs, MMICs.
5.	Microwave Engineering for Wireless System: Microwave networks, Active networks, Design of matching networks, Noise, Microwave link, Rectifier and Detector Design, Frequency and Modulation Techniques, Transmitting and Receiving Equipment.
6.	Satellite Communication: Introduction, Orbits, station keeping, satellite altitude, transmission path, path loss, noise consideration, satellite system, saturation flux density, effective isotropic radiated power, multiple access methods, earth station antenna, satellite link design, frequency plan, satellite communication for Internet, VSAT network, One-way, two-way and open-sky satellite communication, GNSS-GPS and Galileo systems and GIS, Satellite Navigation, DBS-TV.
7.	Radar: Basic principles, Radar equation, factors influencing maximum range, effect of noise, power and frequencies used in Radar, types of Radar, Basic pulsed Radar system, Modulators, receivers, Bandwidth requirements, factors governing pulse characteristics, Duplexer, moving target indicator (MTI), tracking Radar systems and search systems.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2			√	√								

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation

Learning Materials

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|---------------------|----------------------------------|
| 1. Samuel Y. Liao | : Microwave Devices and Circuits |
| 2. Herbert J. Reich | : Microwave Principles |
| 3. David M. Pozar | : Microwave Engineering |

Course Code:	EEE-4237
Course Title:	Data Communication and Networking
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-3205, EEE-4103

Rationale of the course: This course serves as a general introduction for students seeking to acquire a foundation in current computer network technologies for data communication. The course provides basic idea about hardware, software, terminology, components, design, and connections of a network.

Course Objectives:

- To study the basics of data communication and computer network.
- To study the functions of the layers of internet model.
- To study the protocols of TCP/IP model and addressing used in different layers.
- To study the coding techniques, digital modulations techniques, multiplexing techniques and access control techniques.
- To study the structure and functions of networking devices.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs):

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand of the basic concepts of data communications and networking, and the purpose of the layers of internet model.	Cognitive (Understand)
CLO2	Understand how networking devices performs and make communication among heterogeneous networks possible.	Cognitive (Understand)
CLO3	Explain how binary data are converted to signals for transmission, how transmission media limits the data transmission rate, and how consistency of the message sent is ensured (error control, flow control).	Cognitive (Understand, Apply)
CLO4	Acquire the knowledge on multiple access control, utilization of single channel to carry data from multiple sources, utilization of addressing in different layers and signal switching techniques.	Cognitive (Apply, Analyze)
CLO5	Demonstrate in detail how point-to-point, host-to-host and process-to-process communication is happened.	Cognitive (Create)

Course Content

Sl. No.	Course Content (as Summary)
1.	Overview of Data Communications and Networking: Introduction to data communication, Data communications networks, Protocols and standards, Layered tasks, OSI model, TCP/IP model.
2.	Physical Layer: Signals, Analog and digital, Data rate limits, Transmission impairment, Digital transmission, Line coding, Block coding, Sampling, Transmission mode, Modulation of digital data, Modulation of analog signals, Multiplexing FDM, TDM, WDM, Transmission media, Guided and unguided, Circuit switching, Packet switching, Virtual circuit switching.
3.	Data Link Layer: Error detection and correction, Types of error detection, Error correction, Flow and error control, HDLC, Point to point protocol, Multiple access control, Random access, Controlled access, Channelization, Local area networks, Ethernet, Virtual LANs.
4.	Network Layer: Host-to-host delivery, Internetworking, Addressing routing, Network layer protocol, Unicast and multicast routing protocol.
5.	Transport Layer: Process-to-process delivery, UDP and TCP, Congestion control and quality of service
6.	Application Layer: Client-Server Model, Domain Name System, Electronic Mail, File transfer, HTTP, WWW, Multimedia.
7.	Queuing system and Congestion Control: Queuing models, Single server and multiserver queue, Effects of congestion, Congestion control-backpressure, Choke packet, Implicit and explicit congestion signaling, Traffic management, TCP traffic control.

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2	√	√										
CLO3		√	√									
CLO4			√	√								
CLO5			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Presentation ▪ Online resources	▪ Quiz ▪ Tutorial/Assignment ▪ Participation

CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion/ ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation
CLO4	▪ Lecture ▪ Presentation ▪ Discussion ▪ Online resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO5	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

1. Behrouz A. Forouzan : Data Communications and Networking
2. A.S. Tannenbaum : Computer Networks
3. W. Stallings : Data and Computer Communication

Course Code:	EEE-4239
Course Title:	Telecommunication Engineering
Course Credit:	3.0
Course Type:	Core Course
Pre-requisite:	EEE-3205, EEE-4103

Rationale of the Course: This course provides a through coverage of fundamental principles of Telecommunication engineering and applications.

Course Objectives:

- To describe and relate fundamentals of telecommunication networks and associated technologies.
- To apply the principles of queuing theory in evaluating the performance of telecommunication networks.
- To solve problems and design simple systems related to telecommunications networks.
- To appreciate the reasons for switching, and the relative merits of the possible switching modes.
- To understand the principles of the internal design and operation of communication switches, and the essence of the key protocols that are used with switched networks.

Course Learning Outcomes (CLOs): By the end of the course, the student will be able to-

CLOs	Descriptions	Learning Domain with Level
CLO1	Understand the principles of the internal design and operation of communication switches	Cognitive (Understand)
CLO2	Explain the switching and the queuing theory with their applications.	Cognitive (Understand, Apply)
CLO3	Design simple telecommunications network systems.	Cognitive (Create)

Course Content:

Sl. No.	Course Content (as summary)
1.	Introduction: Principle, evolution and telecommunication networks. National and International regulatory bodies, Telephone apparatus, telephone Exchanges, subscriber loop, supervisory tones, PSTN.
2.	Introduction to Analog Switching Systems: Stronger and Crossbar switching systems, Telephone Networks: Subscriber loop systems, switching hierarchy and routing, Transmission plan, Transmission systems, numbering plan, charging plan, signaling techniques, In-channel signaling, Common channel signaling.

3.	Introduction to Digital Switching Systems: Stored program control, Software architecture, Application software, Enhanced services. Space division switching, time division switching, blocking probability and multistage switching and digital memory switch trucking Efficiency
4.	Traffic Analysis: Traffic characterization, Grade of services, trucking Efficiency, and network blocking probabilities, Modeling switching systems, Blocking models and loss estimates, delay system and queuing.
5.	Telephone Networks: Analog termination requirements, BORSCHT configuration, digital termination requirements, signaling tones, touch tone dial generation, design consideration, touch tone detection, switching hierarchy and routing, transmission plan, numbering plan- CCITT No. 7 Signaling systems.
6.	Network Hierarchy: Network hierarchy in the telephone network, Network hierarchy in other networks; Network Intelligence
7.	Access Network Technology: Digital subscriber loop (DSL), Wireless local loop (WLL), FTTx, SONET/SDH, WDM Network, IP telephony and VoIP, ATM network and Next Generation Network (NGN).

Mapping of Course Learning Outcomes (CLOs) with Program Learning Outcomes (PLOs)

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9	PLO10	PLO11	PLO12
CLO1	√	√										
CLO2		√	√									
CLO3			√	√	√							

Mapping Course Learning Outcomes (CLOs) with the Teaching-Learning and Assessment Strategies

CLOs	Teaching-Learning Strategy	Assessment Strategy
CLO1	▪ Lecture ▪ Prese ▪ ntation ▪ Discussion ▪ Online resources	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO2	▪ Lecture ▪ Presentation ▪ Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment ▪ Semester end final examination ▪ Participation
CLO3	▪ Lecture ▪ Presentation ▪ Discussion/Group Discussion ▪ Demonstration	▪ Quiz ▪ Tutorial/Assignment/group Assignment ▪ Semester end final examination ▪ Participation

Learning Materials:

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| 1. | John Dunlop and D. Geoffrey Smith | : | Telecommunication Engineering |
| 2. | Roger L. Freeman | : | Telecommunication System Engineering |
| 3. | T. Viswanathan | : | Telecommunication Switching Systems and Networks |
| 4. | John Bellamy | : | Digital Telephony |